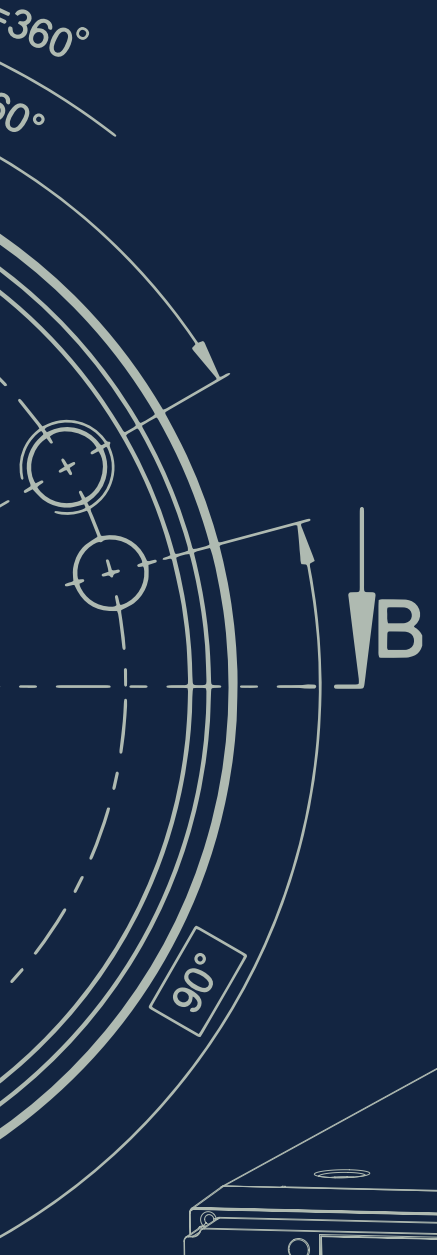
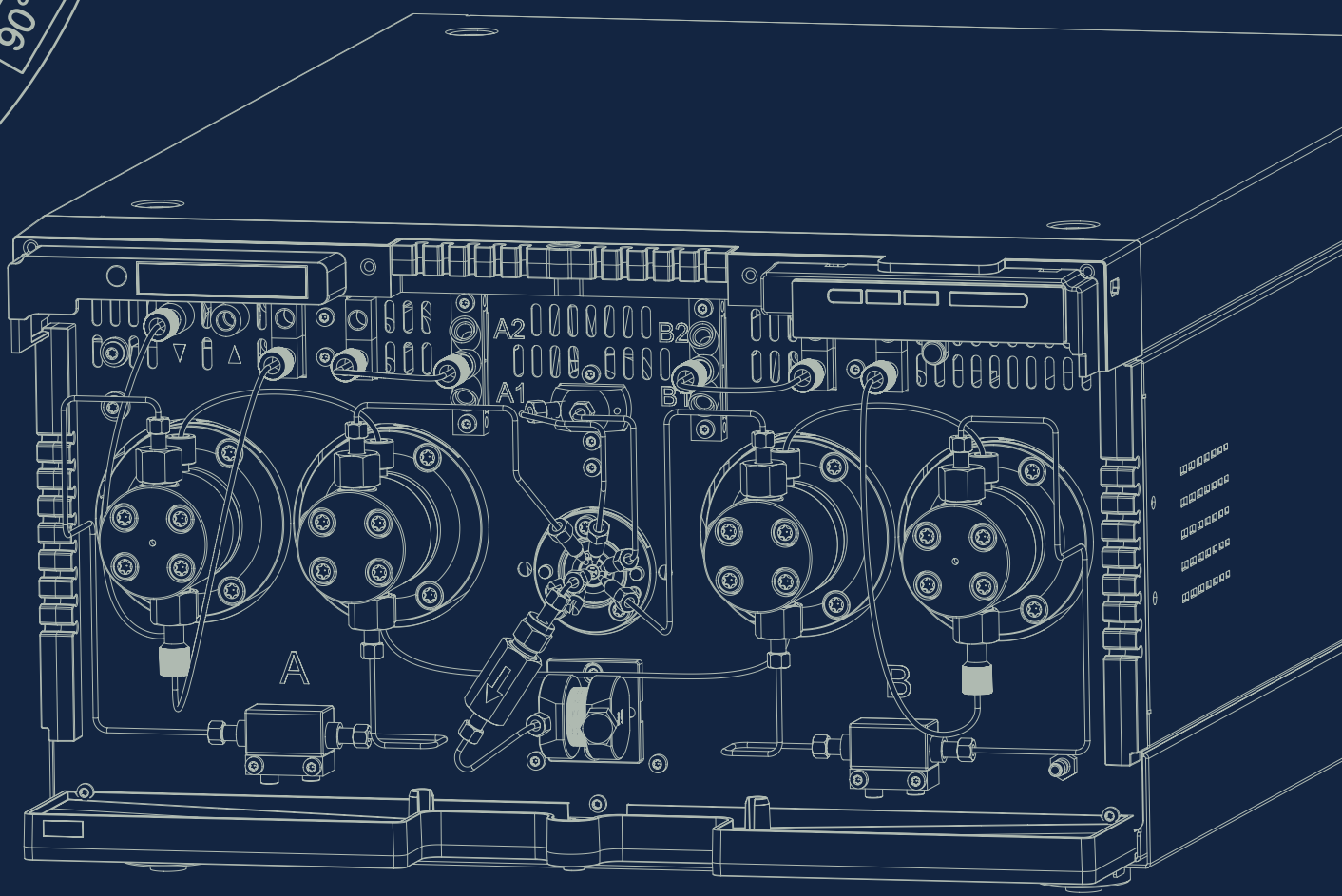
Science with Passion

KNAUER

Product Selection Guide

2025/2026





Get in touch

Sales

If you want to learn more about our products and services or get a quote, the experts from our sales team are happy to assist you with your request.

Phone: +49 30 809727-0 (workdays 9-17h CET)
Fax: +49 30 8015010
E-mail: **sales@knauer.net**

Support

Do you have questions about the installation or the operation of your device or software?

International Support:

Contact your local KNAUER partner for support:
www.knauer.net/local-distributors

Phone: +49 30 809727-111 (workdays 9-17h CET)
Fax: +49 30 8015010
E-mail: **support@knauer.net**



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Welcome to KNAUER



About KNAUER

Based in Berlin, KNAUER is a medium-sized, owner-managed company that has been serving the sciences since 1962. We develop and manufacture scientific instruments of superior quality for liquid chromatography. The range includes systems and components for analytical HPLC/UHPLC, preparative HPLC, fast protein liquid chromatography (FPLC), multi-column chromatography/simulated moving bed (SMB), gel permeation chromatography/size exclusion chromatography (GPC/SEC), osmometry and Skids for the production of lipid nanoparticles (LNP).

Sustainability & ecological commitment

We are committed to protect the environment for ourselves and our children. KNAUER contributes to the conservation of a healthy environment by basing our work on an environmental management system according to DIN EN ISO 14001. The KNAUER quality management system according to DIN EN ISO 9001 and EN ISO 13485:2016 makes sure that we continuously manufacture products in the best quality possible. As a family business with about 190 employees, KNAUER focuses on sustainability and takes responsibility for our future.

Some of our ecological activities:

- The regular creation of an input and output balance for the determination and evaluation of energy and resource flows
- Environmentally friendly product development, energy-efficient production, and shipping with biodegradable packaging materials and re-usable packaging with local suppliers
- Fixed specifications for the development of new products according to ecological aspects such as low solvent consumption, repairability, and longevity of the products
- Complete modernization of the company building included thermal insulation, new windows, electric blinds, and a green rooftop, which resulted in a 50 % heating energy saving
- 100 % green electricity and generation of solar power with our photovoltaic system on the roof
- Guidelines for business travel from an environmental, economic, and social perspective
- Tips and instructions for clients to reduce solvent consumption during instrument use
- Environmentally compatible working and manufacturing of HPLC instruments and accessories, e.g. by using energy-efficient working equipment and reducing the use of solvents and harmful substances
- A life cycle assessment to optimize the manufacturing process and concentrate on electricity saving components

Sustainability: #KNAUERforFuture

Many KNAUER employees have good ideas for sustainability, and so we all get better together every year. We would like to inspire YOU to implement sustainability in many areas of your company, too. May these short videos keep you entertained and invite you to act!
www.knauer.net/corporate-social-responsibility sustainable.

KNAUER

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Analytical HPLC/UHPLC systems

Efficient and adaptable – with ULDC option

KNAUER AZURA® liquid chromatography instruments are designed to support and facilitate your work. Whether doing routine analysis or demanding separation tasks, AZURA® systems are the right tool to overcome your analytical challenges. Choose between different gradient forming technologies and maximum flow rates to find the best configuration for your task. A large variety of detectors is available.

AZURA® Analytical ULDC/UHPLC systems brochure



GPC/SEC systems

GPC/SEC from analytical to preparative scale

KNAUER AZURA® SEC is a line of GPC and SEC systems that offer solutions for different applications and labs. KNAUER AZURA® SEC is a product line for size exclusion and gel permeations chromatography. Depending on customer requirements, the systems are fully biocompatible, without metal in the wetted parts or resistant to common organic GPC solvents. The systems come in various options, each tailored to specific lab needs, such as the AZURA® SEC Compact for budget-friendly solutions or the AZURA® SEC Lab for highest analytical performance. Different detectors, fractionation or sample preparation options can be modularly combined.

AZURA® GPC/SEC systems brochure



Preparative HPLC systems

Customized purification

AZURA® preparative systems are the perfect solution for frequently changing separation tasks – from milligram to kilogram scale. These prep HPLC systems combine flexibility and reliability. The systems can be configured freely choosing different materials, flow rates, valves, and detectors. Due to the flexible design of the devices, parts like pump heads or flow cells can be easily exchanged. All components of the compact system e.g. can be integrated into the pilot-scale system.

AZURA® Purification solutions brochure



FPLC systems (Bio purification)

The flexible FPLC platform for protein purification

The AZURA® FPLC with its biocompatible/metal-free components is the perfect choice for any protein purification task. Multiple functionalities such as automatic sample injection via autosampler, column switching, buffer and sample selection, as well as fraction collection enable the user to automate purification processes.
A large range of different detectors make your target molecules visible. Different flow rates and compatibility to columns from all vendors offer maximum flexibility.

AZURA® Purification solutions brochure



Multi-column chromatography (SMB)

Continuous separation for higher productivity and purity

Simulated moving bed chromatography is increasingly applied as a separation technique in the pharmaceutical industry, production of fine chemicals and in the field of bioengineering. SMB is a method in process chromatography that enables substance mixtures to be continuously separated and extracted in two fractions. By repeated use of the SMB process each partial fraction can be separated into a further fraction – down to binary substance mixtures. It´s efficiency is significantly higher than batch chromatography, through better utilization of the column stationary phase.

AZURA® SMB systems brochure



Chromatography data systems

Choose your software drivers and control software

KNAUER modules can be controlled via various CDS. Above that, Mobile Control is the right solution as control software if you need a basic, easy-to-use, and cost-effective software for your LC system.

System configurator

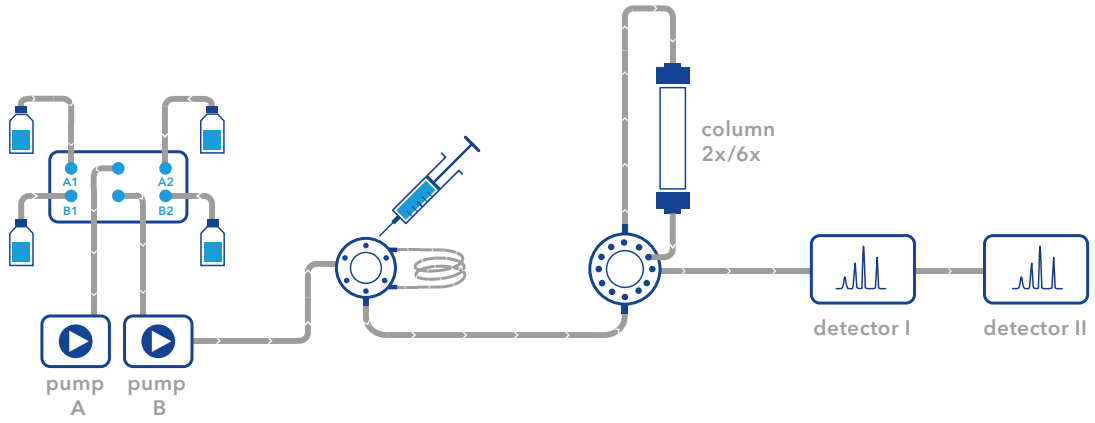
HPLC/UHPLC by KNAUER

MAKE YOUR PRESELECTION

☐ **UHPLC**
(SST, max. 1 240 bar)

☐ **HPLC**
(SST, max. 862 bar)

☐ **Bio-Inert**
(metal-free, max. 400 bar)



ELUENT SELECTION & DELIVERY

- ☐ 5 ml/min binary gradient pump P 8.1L (UHPLC), max. delivery pressure 1 240 bar
- ☐ 10 ml/min binary gradient pump P 6.1L, max. delivery pressure 862 bar
- ☐ 10 ml/min quaternary gradient pump P 6.1L, max. delivery pressure 862 bar
- ..... × Solvent selection valve (6 further inlets)

SAMPLE INJECTION

- ☐ Injection valve
- ☐ Autosampler AS 6.1L
- ☐ Liquid Handler LH 8.1
- ☐ Autosampler AS 8.1L
- ☐ Sample cooling/ heating

COLUMN SELECTION & THERMOSTAT

- ☐ 2 columns
- ☐ 4 columns
- ☐ 8 columns
- ☐ Column thermostat
- ☐ Column kit HPLC
- ☐ Column kit UHPLC
- ☐ Eluent pre-heating cartridge 0.1 mm ID UHPLC
- ☐ Eluent pre-heating cartridge 0.18 mm ID HPLC

DETECTION

- ☐ UV/VIS single wavelength
- ☐ UV/VIS multiple wavelength
- ☐ Conductivity
- ☐ Refractive index
- ☐ Light Scattering
- ☐ A/D-converter (integration of further detectors)
- ☐ DAD 2.1L
- ☐ DAD 6.1L
- ☐ Fluorescence Detector RF-20 A
- ☐ Fluorescence Detector RF-20 Axs

ACCESSORIES

- ☐ Tubing kit UHPLC & ULDC
- ☐ Tubing kit HPLC
- ☐ PEEK tubing
- ..... × Back pressure regulator
- ☐ Workstation (Windows)

FLOW CELLS FOR UV-DETECTOR

- ☐ 10 mm/10 µl Pressure proof
- ☐ 10 mm/2 µl LightGuide®
- ☐ 50 mm/6 µl LightGuide®
- ☐ 3 mm/2 µl (up to 100 ml/min) Pressure proof

SOFTWARE

- ☐ ClarityChrom®
- ☐ Mobile Control
- ☐ knauerOS®
- ☐ OpenLAB EZChrom/CDS
- ☐ Chromeleon™

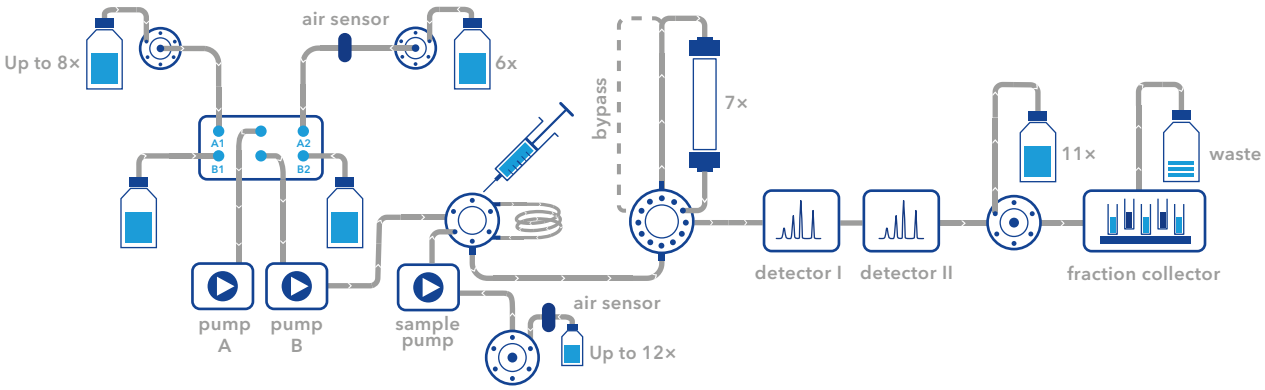
COMMON APPLICATIONS

- ☐ Reversed phase
- ☐ Normal phase
- ☐ other...
- ☐ System Qualification

System configurator

Purification by KNAUER

MAKE YOUR PRESELECTION ☐ **Stainless steel** ☐ **Biocompatible**



SOLVENT SELECTION & DELIVERY

- ☐ 50 ml/min binary gradient pump P 6.1L
- ..... × 100 ml/min pump P 2.1L
- ..... × 250 ml/min pump P 2.1L
- ..... × 500 ml/min pump P 2.1L
- ..... × 1000 ml/min pump P 2.1L
- ☐ Ternary gradient module for pump P 2.1L
- ☐ Binary gradient module for pump P 2.1L
- ..... × solvent selection valve VU 4.1

SAMPLE INJECTION

- ☐ Multi-injection valve
- ☐ Injection valve
- ☐ Sample pump module
- ☐ Sample selection valve: ..... × inlets
- ☐ Autosampler AS 6.1L
- ☐ Liquid Handler LH 2.1
- ☐ VariLoop

COLUMN SELECTION & THERMOSTAT

- ☐ Column selection (two columns or one bypass)
- ☐ Column selection high flow (7 columns, one bypass, reverse flow)
- ☐ Eluent heater ELH 2.1L for one eluent
- ☐ Two eluents
- ☐ Heating sleeve for HPLC columns

DETECTION

- ☐ UV/VIS UVD 2.1S (single wavelength)
- ☐ UV/VIS MWD 2.1L (multiwave length)
- ☐ DAD 2.1L
- ☐ UV/VIS VWD 2.1L incl.
  - ☐ Conductivity
  - ☐ pH
- ☐ Conductivity Mikron 81
- ☐ Conductivity & pH CM 2.1S
- ☐ Refractive index RID 2.1L
- ☐ Light Scattering ELSD
- ☐ A/D-converter IFU 2.1 (integration of further detectors)

FRACTION COLLECTION

- ☐ FC 6.1 Fraction Collector
- ☐ LABOCOL fraction collector
- ☐ Vario 4000 Plus
- ☐ Liquid Handler 2.1
- ☐ Foxy R1 fraction collector
- ☐ Foxy R2 fraction collector
- ☐ Fractionation valve VU 4.1

ACCESSORIES

- ..... × Airsensor main pump
- ☐ Flow splitter
- ☐ Mass flow controller
- ☐ AZURA SmartClick
- ☐ Pressure Control (2 pressure sensors)
- ..... × Tubing 1/16"
- ..... × Tubing 1/8"
- ..... × Tubing 1/4"
- ☐ Backpressure Regulator
- ☐ Workstation (Windows)

SOFTWARE

- ☐ ClarityChrom®
- ☐ Chromeleon™
- ☐ OpenLAB®
- ☐ Mobile Control
- ☐ PurityChrom® 6

COMMON APPLICATIONS

- ☐ Reversed phase
- ☐ Affinity Chromatography (AC)
- ☐ Size Exclusion Chromatography (SEC)
- ☐ Hydrophobic Interaction Chromatography (HIC)
- ☐ Normal phase
- ☐ Ion Exchange Chromatography (IEX)

SERVICES

- ☐ Operation Qualification
- ☐ Material Documentation
- ☐ Factory Acceptance Test (FAT/ Site Acceptance Test (SAT)
- ☐ Software Compliance (GAMP 5/21 CFR Part 11)

AZURA® Pump P 8.1L

The new AZURA® P 8.1L UHPLC pump offers a maximum delivery pressure of 124 MPa/1240 bar and a flow rate range up to 5 ml/min for ultra-fast and high-resolution applications.

The adaptive pulsation compensation of the pump is independent of flow rate, backpressure, and eluent type through real-time eluent compressibility monitoring and variable piston stroke. Together with the ultra-precise piston movement thanks to KNAUER's proprietary advanced piston drive technology, this results in outstanding flow reproducibility at any working conditions.

Developed with innovative technology and decades of continuous improvements the AZURA® P 8.1L UHPLC pump enters a new level of performance and durability. With a high level of in-house component production KNAUER achieves an industry-leading level of manufacturing precision. This results in unmatched piston seal life and system uptime.



KNAUER offers various software control options:
www.knauer.net/software

For pump accessories, **see p. 74**

Further information:
www.knauer.net/Pump P8.1L

Specifications

Solvent delivery

Pump type	Analytical UHPLC pump
Delivery system	Dual Serial Piston Pump
Pulsation compensation	Adaptive pulsation compensation
Piston seal washing	Active Wash
Flow rate accuracy	± 0.25 % (water, 1 ml/min, 1 200 bar)
Flow rate precision	≤ 0.04 % RSD or 0.008 min SD whichever is greater (water, 1 ml/min, 1 200 bar)
System protection	Soft start, P_{min} und P_{max} are programmable
Gradient range	0 – 100 % in 0.1 % increments
Solvent selection valve	2 × 2 channels
Gradient formation	HPG
Liquid temperature range	4 – 60 °C (39.2 – 140 °F)
HPG: gradient accuracy	± 0.3 %
HPG: gradient precision	< 0.1 % RSD at 1 ml/min, based on retention time at constant room temperature
Pump head inlet (standard)	UNF 1/4-28 Thread (for 1/8" tubing)
Pump head outlet (standard)	UNF 10-32 Thread (for 1/16" capillary)

Degasser module

Degasser channels	4 channels
Max. flow rate/channel	5 ml/min
Degassing method	Gas Permeation through Teflon(R) AF amorphous fluoropolymer membrane
Degassing efficiency	< 0.5 ppm dissolved O_2 at 1 ml/min
Degassing chamber volume	280 µl volume per channel
Solvent applicability	Universal, with exception of hydrochloric acid and halogenated hydrocarbons
Wetted materials	PEEK, Tefzel® (ETFC), Systec AF™

Communication

Display	Mobile Control (optional)
Inputs	LAN, Pin header connectors (Analog IN, Start In, Error IN)
Analog inputs	Flow rate, 0 - 10 V via pin header connectors
Analog control input	Flow Rate
Level/event outputs	8 event outputs (TTL, OC, Relais) and 24 V
Control	LAN, Analog and event control, Mobile Control

Technical parameters

Leak sensor	Yes
Special features	Pump Head is detected automatically using Radio frequency identification (RFID)
Ambient conditions	4 – 40 °C (39.2 – 104 °F) Air humidity below 90 %, non-condensing

General

Power supply	100 – 240 V; 50 – 60 Hz; Maximum power consumption 310 Watt
Dimensions	361 mm × 208.2 mm × 523 mm (W × H × D)
Weight	26.7 kg

AZURA® Pump P 8.1L with 5 ml pump head

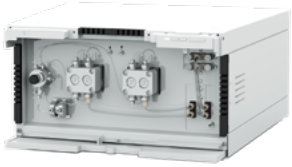
Pump specifications	
Pump head	5 ml
Continuous working conditions	0.1 – 4 ml/min
Best working conditions	0.02 – 5 ml/min
Flow rate increment	0.001 ml/min
Mixing volume	50 µl (HPG)
Wetted materials	Stainless steel, sapphire, ruby, PEEK, zirconium oxide, nickel-cobalt-chromium-molybdenum alloy (MP35N®), diamond-like carbon (DLC), polyimide (Vespel®), polyethylene
Maximum delivery pressure	18 000 psi / 1 240 bar / 124 MPa
Flow rate range	0.001 – 5 ml/min
Pump head material	Stainless steel
Purge valve	Automated

Ordering details

Device	
APF45PA	AZURA® P 8.1L UHPLC pump with a maximum delivery pressure of up to 124 MPa/1240 bar for ultra-fast and high-resolution applications

AZURA® Pump P 6.1L

The AZURA® Pump P 6.1L uses technology to overcome the challenges of pumping LC solvents at high pressure with low pulsation. This pump is designed to fulfill the needs for high pressure and low pressure mixing tasks. The pump can deliver flow in the range of 0.001–50 ml/min at pressures up to 1 000 bar (depending on model and flow rate). The AZURA® binary pump contains two identical high-pressure pumps, a 2 × 2-channel solvent selection valve and the new developed AZURA® mixer, a low-volume microfluidic mixing device. The AZURA® quaternary pump contains one high-pressure pump and an integrated LPG mixing block with a 4 channel valve and mixer. The integrated degasser and AZURA® inline filter are completing the analytical AZURA® HPLC pump and turn this pump into a working horse in the lab. This pump is also available with wetted materials made from ceramic, PEEK and titanium for biocompatible applications.



KNAUER offers various software control options:
www.knauer.net/software

For pump accessories, **see p. 74**

Further information:
www.knauer.net/Pump P 6.1L

Specifications

Solvent delivery

Pump type	Analytical HPLC pump
Delivery system	Dual Serial Piston Pump
Pulsation compensation	Active Pulsation Compensation
Piston seal washing	Active Wash
Flow rate accuracy	± 0.25 %, measured at 5–80 % of flow range, using ethanol
Flow rate precision	≤ 0.04 % RSD or 0.008 min SD whichever is greater
System protection	Soft start, P_{min} and P_{max} are programmable
Gradient range	0–100 % in 0.1 % increments
Solvent selection valve	2 × 2 channels (HPG only)
Gradient formation	LPG / HPG
Liquid temperature range	4–60 °C (39.2–140 °F)
HPG: gradient accuracy	± 0.3 % at 1 ml/min, 150 bar (ethanol/caffeine tracer) ± 1 % (5–95 %, measured at 0.1–10 ml/min, water/caffeine tracer)
HPG: gradient precision	< 0.1 % RSD at 1 ml/min, 0.3 % RSD overall, based on retention time at constant room temperature
LPG: gradient accuracy	± 0.3 % at 1 ml/min, 150 bar (ethanol/caffeine tracer) ± 2 % (1–99 %, measured at 5–50 % of the flow range, water/caffeine tracer)
LPG: gradient precision	< 0.1 % RSD at 1 ml/min, 0.5 % RSD overall, based on retention time at constant room temperature

Degasser module

Degasser channels	4 channels (LPG versions), 2/4 channels (HPG versions); optional
Max. flow rate/channel	10 ml/min
Degassing method	Gas permeation through amorphous fluoropolymer membrane
Degassing efficiency	< 0.5 ppm dissolved O_2 at 1 ml/min
Degassing chamber volume	280 µl volume per channel
Solvent applicability	Universal, with exception of hydrochloric acid and halogenated hydrocarbons
Wetted materials	PEEK, Tefzel® (ETFC), Systec AF™

Communication

Display	Mobile Control (optional)
Inputs	LAN, Pin header connectors (Analog IN, Start IN, Error IN)
Analog inputs	Flow rate, 0–10 V via pin header connectors
Analog control input	Flow rate
Level/event outputs	8 event outputs (TTL, OC, Relais) and 24 V
Control	LAN, Analog and event control, Mobile Control

Technical parameters

Leak sensor	Yes
Special features	Pump head is detected automatically using Radio frequency identification (RFID)
Ambient conditions	4–40 °C (39.2–104 °F) Air humidity below 90 %, non-condensing

General

Power supply	100–240 V; 50–60 Hz; Maximum power consumption 100 W
Dimensions	361 mm × 208.2 mm × 523 mm (W × H × D)
Weight	14.1 kg

AZURA® Pump P 6.1L with 5 ml pump head

Pump specifications	
Pump head	5 ml
Continuous working conditions	0.1 – 4 ml/min
Best working conditions	0.02 – 5 ml/min
Flow rate increment	0.001 ml/min
Mixing volume	100 µl (HPG, LPG)
Wetted materials	GFP, stainless steel, FKM, PEEK, sapphire, aluminum oxide, ruby, zirconium oxide
Maximum delivery pressure	14 500 psi / 1 000 bar / 100 MPa up to 2 ml/min, 10150 psi / 700 bar / 70 MPa up to 5 ml/min
Flow rate range	0.001 – 5 ml/min
Pump head material	Stainless steel

Ordering details

Device	
APH34GA	AZURA® Pump P 6.1L (LPG), with 5 ml pump head (stainless steel), degasser and mixer (100 µl)
APH35GA	AZURA® Pump P 6.1L (HPG), with 5 ml pump head (stainless steel), degasser and mixer (100 µl)

AZURA® Pump P 6.1L with 10 ml pump head

Pump specifications	
Pump head	10 ml
Continuous working conditions	0.1 – 4.0 ml/min
Best working conditions	0.1 – 8.0 ml/min
Flow rate increment	0.001 ml/min
Mixing volume	400 µl (HPG), 600 µl (LPG)
Wetted materials	GFP, stainless steel, FKM, PEEK, sapphire, aluminum oxide, ruby, zirconium oxide
Maximum delivery pressure	12 500 psi / 862 bar / 86 MPa up to 2 ml/min; 5 800 psi / 400 bar / 40 MPa up to 10 ml/min
Flow rate range	0.001 – 10 ml/min
Pump head material	Stainless steel

Ordering details

Device	
APH30EA	AZURA® Pump P 6.1L isocratic, without degasser, with 10 ml pump head (stainless steel)
APH31EA	AZURA® Pump P 6.1L isocratic, with degasser, with 10 ml pump head (stainless steel) and solvent selection valve
APH30ED	AZURA® Pump P 6.1L isocratic, without degasser, with 10 ml NP pump head (stainless steel)
APH34EA	AZURA® Pump P 6.1L (LPG), with 10 ml pump head (stainless steel), degasser and mixer (600 µl)
APH35EA	AZURA® Pump P 6.1L (HPG), with 10 ml pump head (stainless steel), degasser and mixer (400 µl)
APH35ED	AZURA® Pump P 6.1L (HPG), with 10 ml NP pump head (stainless steel), degasser and mixer (400 µl)
APH38EA	AZURA® Pump P 6.1L (HPG), without degasser, with 10 ml pump head and mixer (400 µl)
APH38ED	AZURA® Pump P 6.1L (HPG), without degasser, with 10 ml NP pump head and mixer (400 µl)
APH39EA	AZURA® Pump P 6.1L (LPG), without degasser, with 10 ml pump head (stainless steel) and mixer (600 µl)

AZURA® Pump P 6.1L with 50 ml pump head

Pump specifications	
Pump head	50 ml
Continuous working conditions	0.1 – 20 ml/min
Best working conditions	0.1 – 40 ml/min
Flow rate increment	0.001 ml/min
Mixing volume	400 µl (HPG)
Wetted materials	GFP, FKM, PEEK, sapphire, aluminum oxide, ruby, zirconium oxide
Maximum delivery pressure	4 350 psi / 300 bar / 30 MPa up to 10 ml/min; 2900 psi / 200 bar / 20 MPa up to 50 ml/min
Flow rate range	0.01 – 50 ml/min
Pump head material	Stainless steel

Ordering details

Device	
APH30FA	AZURA® Pump P 6.1L isocratic, without degasser, with 50 ml pump head (stainless steel)
APH30FD	AZURA® Pump P 6.1L isocratic, without degasser, with 50 ml normal phase pump head (stainless steel)
APH38FA	AZURA® Pump P 6.1L (HPG), without degasser, with 50 ml pump head (stainless steel) and mixer (400 µl)

AZURA® Pump P 6.1L biocompatible

Pump specifications	
Pump head	10 ml / 50 ml
Continuous working conditions	For 10 ml pump heads: 0.1 – 4.0 ml/min; for 50 ml pump heads: 0.1 – 20 ml/min
Best working conditions	For 10 ml pump heads: 0.1 – 8.0 ml/min; for 50 ml pump heads: 0.1 – 40 ml/min
Flow rate increment	0.001 ml/min
Mixing volume	250 µl
Wetted materials	UHMW PE, PEEK, sapphire, aluminum oxide, ruby, titanium
Maximum delivery pressure	5 800 psi / 400 bar / 40 MPa for 10 ml head, 2900 psi / 200 bar / 20 MPa for 50 ml head
Flow rate range	0.001 – 10 ml/min / 0.01 – 50 ml/min
Pump head material	Ceramic

Ordering details

Device	
APH60EB	AZURA® Pump P 6.1L, isocratic, without degasser, with 10 ml pump head (ceramic)
APH60FB	AZURA® Pump P 6.1L, isocratic, without degasser, with 50 ml pump head (ceramic)
APH61EB	AZURA® Pump P 6.1L, isocratic, with degasser and solvent selection valve, with 10 ml pump head (ceramic)
APH64EB	AZURA® Pump P 6.1L (LPG), with 10 ml pump head (ceramic), degasser and mixer (250 µl)
APH69EB	AZURA® Pump P 6.1L (LPG), without degasser, with 10 ml pump head (ceramic) and mixer (250 µl)
APH65EB	AZURA® Pump P 6.1L (HPG), with degasser, with 10 ml pump head (ceramic) and mixer (250 µl)
APH68EB	AZURA® Pump P 6.1L (HPG), without degasser, with 10 ml pump head (ceramic) and mixer (250 µl)
APH68FB	AZURA® Pump P 6.1L (HPG), without degasser, with 50 ml pump head (ceramic) and mixer (250 µl)

AZURA® Pump P 2.1L

AZURA® preparative HPLC pump P 2.1L covers wide flow rate range and pressure capabilities. It has been designed for purification of mg and gram samples. The pump can deliver flow in the range of 0.01 – 1 000 ml/min at pressures up to 400 bar (depending on model). The integrated automatic recognition of the pump head with RFID technology allows fast adaptions of the pump for various applications.



KNAUER offers various software control options:
www.knauer.net/software

For pump accessories, **see p. 74**

Further information:
www.knauer.net/Pump P 2.1L

Specifications

Solvent delivery

Pump type	Preparative HPLC pump
Delivery system	Dual Piston Pump with pistons parallel
Pulsation compensation	Yes, with compressibility factor
Piston seal washing	Active Wash
Flow rate accuracy	± 2 %, measured at 5 – 50 % of flow range using ethanol/water 10:90
Flow rate precision	< 0.1% RSD
System protection	Soft start, P_{min} and P_{max} are programmable
Gradient range	0 – 100 %
Gradient formation	LPG / HPG
Liquid temperature range	4 – 60 °C (39.2 – 140 °F)
HPG: gradient accuracy	± 2 % (5 – 95 %, measured at 5 – 50 % of flow range, water/caffeine tracer)
Leak management	Yes
HPG: gradient precision	< 1 % RSD based on retention time at constant room temperature
LPG: gradient accuracy	± 3 % (5 – 95 %, measured at 5 – 50 % of flow range, water/caffeine tracer)
LPG: gradient precision	2 % RSD, based on retention time at constant room temperature

Communication

Display	Mobile Control (optional)
Inputs	LAN, Pin header connectors (Analog IN, Start IN, Error IN)
Analog inputs	Flow rate, 0 – 10 V via pin header connectors
Analog control input	Flow rate
Level/event outputs	8 event outputs (TTL, OC, Relais) and 24 V
Control	LAN, Analog and event control, Mobile Control

Technical parameters

Leak sensor	Yes
Special features	Pump head is detected automatically using radio frequency identification (RFID)
Ambient conditions	10 – 40 °C (50 – 104 °F), air humidity below 90 %, non-condensing

General

Power supply	100 – 240 V; 50 – 60 Hz; maximum power consumption 320 W
Dimensions	361 mm × 208.2 mm × 523 mm (W × H × D)
Weight	19 kg
Optional accessories	Ternary low pressure gradient valve block, 10 – 220 ml/min, binary low pressure gradient valve block, 10 – 800 ml/min, pump head heating and cooling device

AZURA® Pump P 2.1L with 100 ml pump head

Pump specifications	
Pump head	100 ml
Continuous working conditions	1 – 40 ml/min
Best working conditions	1 – 80 ml/min
Flow rate increment	0.01 ml/min
Pump head inlet	4 mm OD, 3 mm ID, PTFE tubing (M8x1 flat bottom)
Pressure sensor outlet	M8x1, coned
Wetted materials	Zirconium oxide (ZrO_2), FFKM, graphite fiber reinforced PTFE, PEEK, sapphire, ruby, stainless steel, titanium
Maximum delivery pressure	5 800 psi / 400 bar / 40 MPa
Flow rate range	0.01 – 100 ml/min
Pump head material	Stainless steel / titanium

Ordering details

Device	
APE20KA	AZURA® Pump P 2.1L with 100 ml pump head (stainless steel)
APE20KB	AZURA® Pump P 2.1L with 100 ml pump head (titanium)

AZURA® Pump P 2.1L with 250 ml pump head

Pump specifications	
Pump head	250 ml
Continuous working conditions	2.5 – 100 ml/min
Best working conditions	2.5 – 200 ml/min
Flow rate increment	0.01 ml/min
Pump head inlet	4 mm OD, 3 mm ID, PTFE tubing (M8x1 flat bottom)
Pressure sensor outlet	M8x1, coned
Wetted materials	Zirconium oxide (ZrO_2), FFKM, graphite fiber reinforced PTFE, PEEK, sapphire, ruby, stainless steel, titanium
Maximum delivery pressure	3 260 psi / 225 bar / 22.5 MPa up to 100 ml/min, 2 900 psi / 200 bar / 20 MPa up to 250 ml/min
Flow rate range	0.01 – 250 ml/min
Pump head material	Stainless steel / titanium

Ordering details

Device	
APE20LA	AZURA® Pump P 2.1L with 250 ml pump head (stainless steel)
APE20LC	AZURA® Pump P 2.1L with 250 ml pump head (titanium)

AZURA® Pump P 2.1L with 500 ml pump head

Specifications

Pump specifications	
Pump head	500 ml
Continuous working conditions	5 – 200 ml/min
Best working conditions	5 – 400 ml/min
Flow rate increment	0.1 ml/min
Pump head inlet	4 mm OD, 3 mm ID, PTFE tubing (M8x1 flat bottom)
Pressure sensor outlet	M8x1, coned
Wetted materials	Zirconium oxide (ZrO_2), FFKM, graphite fiber reinforced PTFE, PEEK, sapphire, ruby, stainless steel, titanium
Maximum delivery pressure	1 450 psi / 100 bar / 10 MPa
Flow rate range	0.01 – 500 ml/min
Pump head material	Stainless steel / titanium

Ordering details

Device	
APE20MA	AZURA® Pump P 2.1L with 500 ml pump head (stainless steel)
APE20MC	AZURA® Pump P 2.1L with 500 ml pump head (titanium)

AZURA® Pump P 2.1L with 1 000 ml pump head

Pump specifications	
Pump head	1 000 ml
Continuous working conditions	10 – 400 ml/min
Best working conditions	10 – 800 ml/min
Flow rate increment	0.1 ml/min
Pump head inlet	9 mm OD, 7 mm ID, PTFE tubing, piped
Pressure sensor outlet	M8x1, coned
Wetted materials	Zirconium oxide (ZrO_2), FFKM, graphite fiber reinforced PTFE, PEEK, sapphire, ruby, stainless steel, titanium
Maximum delivery pressure	1 080 psi / 75 bar / 7.5 MPa up to 350 ml/min, 720 psi / 50 bar / 5 MPa up to 1 000 ml/min,
Flow rate range	1 – 1 000 ml/min
Pump head material	Stainless steel / titanium

Ordering details

Device	
APE20NA	AZURA® Pump P 2.1L with 1 000 ml pump head (stainless steel)
APE20NB	AZURA® Pump P 2.1L with 1 000 ml pump head (titanium)

LPG Modules

Ordering details

Device	
AZZ00AA	LPG module for Pump P 2.1L binary up to 800 ml/min (stainless steel)
AZZ00AB	LPG module for Pump P 2.1L ternary up to 220 ml/min (stainless steel)
AZZ10AB	LPG module for Pump P 2.1L ternary up to 220 ml/min (PEEK)

AZURA® Pump P 4.2S with 10 ml pump head

Pump specifications	
Pump head	10 ml
Flow rate range	0.001 – 10 ml/min
Maximum delivery pressure	For Stainless Steel, Hastelloy, Ti connections: 5 800 psi / 400 bar / 40 MPa up to 10 ml/min For PEEK connections: 5080 psi / 350 bar / 35 MPa up to 10 ml/min
Wetted materials	Graphite fiber reinforced PTFE, FKM, FFKM, sapphire, ruby, zirconium oxide, titanium and pump head material
Maximum viscosity	100 mPa s (at reduced max. flow)
Flow rate increment	0.001 ml/min
Best working conditions	0.1 – 8.0 ml/min
Continuous working conditions	0.1 – 4.0 ml/min
Pump head material	Stainless steel / ceramic / Hastelloy® C

Ordering details

Device	
APK20EA	AZURA® Pump P 4.2S with pressure sensor and 10 ml/min stainless steel pump head, stainless steel connections
APK20EB	AZURA® Pump P 4.2S with pressure sensor and 10 ml/min ceramic pump head, PEEK connections*)
APK20EC	AZURA® Pump P 4.2S with pressure sensor and 10 ml/min Hastelloy® C pump head, Hastelloy® C connections
APK20EF	AZURA® Pump P 4.2S with pressure sensor and 10 ml/min ceramic pump head, Ti connections
APK20EG	AZURA® Pump P 4.2S with pressure sensor and 10 ml/min stainless steel pump head, stainless steel connections, recommended for aqueous solutions
APK20EH	AZURA® Pump P 4.2S with pressure sensor and 10 ml/min ceramic pump head, Titanium connections, recommended for aqueous solutions
APK90EA	AZURA® Pump P 4.2S without pressure sensor and 10 ml/min stainless steel pump head
APK90EB	AZURA® Pump P 4.2S without pressure sensor and 10 ml/min ceramic pump head, PEEK connections*)
APK90EC	AZURA® Pump P 4.2S without pressure sensor and 10 ml/min Hastelloy® C pump head
APK90EG	AZURA® Pump P 4.2S without pressure sensor and 10 ml/min stainless steel pump head, recommended for aqueous solutions

*) Max. delivery pressure: 5 080 psi / 350 bar / 35 MPa

AZURA® Pump P 4.2S with 50 ml pump head

Pump specifications	
Pump head	50 ml
Flow rate range	0.01 – 50 ml/min
Maximum delivery pressure	2 180 psi / 150 bar / 15 MPa up to 50 ml/min
Wetted materials	Graphite fiber reinforced PTFE, FKM FFKM, sapphire, ruby, zirconium oxide, titanium and pump head material
Maximum viscosity	100 mPa s (at reduced max. flow)
Flow rate increment	0.01 ml/min
Best working conditions	0.5 – 40.0 ml/min
Continuous working conditions	0.5 – 20 ml/min
Pump head material	Stainless steel / ceramic / Hastelloy® C

AZURA® Pump P 4.2S

With its small footprint, the compact dual piston pump AZURA® P 4.2S can be used for a wide range of laboratory tasks. It features an integrated digital pressure sensor, which ensures pressure-independent flow rate accuracy. Moreover, for system protection, the pump drive switches off when reaching a pre-defined pressure limit. Pump heads with maximum flow rates of 10 and 50 ml/min are available.

A pressure rating of up to 400 bar and chemical resistance to a wide range of eluents make it the perfect choice for LC and dosing applications.

Pump head and pressure sensor can easily be exchanged, which allows an adaptation of the pump for delivery of aggressive media and bioinert applications. The pump head can be heated or cooled with optional accessories.



KNAUER offers various software control options:
www.knauer.net/software

For pump accessories, **see p. 74**

Further information:
www.knauer.net/dosing pumps

Specifications

Solvent delivery

Pump type	Ultra-compact high pressure pump
Delivery system	Dual piston pump with one working piston, one auxiliary piston
Pulsation compensation	No
Piston seal washing	Passive Wash
Flow rate accuracy	With pressure sensor: ±2 %, measured at 5–50 % of flow range using ethanol/water 10:90 With purge valve: ±5 %, measured at 5–50 % of flow range using ethanol/water 10:90 ±2 % at calibration point (one point calibration), measured at 5–50 % of flow range using ethanol/water 10:90
Flow rate precision	≤0.5 % RSD, measured at 1/5 ml/min using ethanol/water 10:90
System protection	P_{min} and P_{max} are programmable
Liquid temperature range	4–60 °C (39.2–140 °F)
Pump head inlet (standard)	1/84" OD, 2.1 mm ID FEP tubing (UNF 1/4-28 thread, flat bottom)
Pump head outlet (standard)	UNF 10-32 Thread (for 1/16" capillary)

Communication

Display	Yes
Inputs	LAN, Pin header connectors (Analog IN, Start IN, Error IN), RS-232
Analog inputs	0–10 V, 4–20 mA
Analog control input	Flow rate
Control	LAN, RS-232, analog, standalone

Technical parameters

Display	Yes
Ambient conditions	10–40 °C (50–104 °F) air humidity below 90 %, non-condensing

General

Power supply	100–240 V, 50–60 Hz, Maximum power consumption 100 W
Dimensions	121 mm × 129 mm × 231 mm (W × H × D)
Weight	2.8 kg

Ordering details

Device	
APK20FA	AZURA® Pump P 4.2S with pressure sensor and 50 ml/min stainless steel pump head, stainless steel connections
APK20FB	AZURA® Pump P 4.2S with pressure sensor and 50 ml/min ceramic pump head, PEEK connections
APK20FC	AZURA® Pump P 4.2S with pressure sensor and 50 ml/min Hastelloy® C pump head, Hastelloy® C connections
APK20FG	AZURA® Pump P 4.2S with pressure sensor and 50 ml/min stainless steel pump head, stainless steel connections, recommended for aqueous solutions
APK20FI	AZURA® Pump P 4.2S with pressure sensor and 50 ml/min ceramic pump head, PEEK connections, recommended for aqueous solutions
APK90FA	AZURA® Pump P 4.2S without pressure sensor and 50 ml/min stainless steel pump head
APK90FB	AZURA® Pump P 4.2S without pressure sensor and 50 ml/min ceramic pump head
APK90FC	AZURA® Pump P 4.2S without pressure sensor and 50 ml/min Hastelloy® C pump head
APK90FG	AZURA® Pump P 4.2S without pressure sensor and 50 ml/min stainless steel pump head, recommended for aqueous solutions

AZURA® Assistant ASM 2.2L

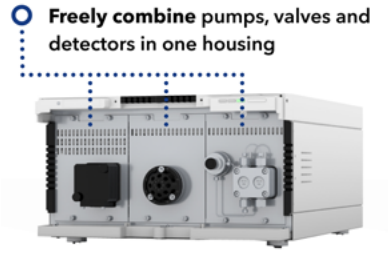
Docking station for pumps, valves and detectors

The Assistant ASM 2.2L is a docking station for up to three compact devices. Valves, pumps and UV detectors can be combined in one housing.

The plug-in modules are removed by loosening four screws allowing the user to exchange modules in case of service within minutes. Likewise, the configuration of the LC system can be adapted to new requirements. Routine maintenance work e.g. replacing the lamp of a detector are easily performed by the user.

Depending on the integrated modules, the assistant fulfills many different tasks like eluent delivery, detection, sample and solvent selection, sample injection, column switching or fraction collection. An assistant including a pump, injection valve, and detector features a complete, compact chromatographic system. As a part of a larger system, the ASM 2.2L is extremely versatile in analytical, preparative and continuous liquid chromatography.

Select your desired plug-in modules for the left, middle and right position in the assistant and you will get your perfect assistant for chromatography and beyond.



KNAUER offers various software control options:
www.knauer.net/software

Further information:
www.knauer.net/assistants

Specifications

General

Power supply	100 – 240 V, 50 – 60 Hz, maximum 130 W
Dimensions	361 × 208 × 523 mm (W × H × D)
Weight	About 17 kg (depending on integrated modules)
Leak sensor	Yes
Ambient conditions	Temperature range: 4 – 40 °C, 39.2 – 104 °F Humidity: 10 – 90 % non-condensing

Communication

Interfaces	LAN
Control	Mobile Control, Software
Inputs	Error (IN), Start (IN), Autozero, 0 – 10 V Analog (IN)
Outputs	Event 1–2, Error (OUT) (OC), + 5 V, + 24 V
Analog inputs	Integrator output (detector signal)

Software functions

Assistant configuration: The ASM 2.2L is supported as complete device. Modules are addressed via the assistant.

	ClarityChrom®	OpenLAB®	Mobile Control (version 6)
Two pumps (independent)	yes	no	yes
Fraction valve	one	one	yes, one valve
Injection module*	no	no	yes, but part of a method

Single device configuration:
The ASM 2.2L is not supported as device. Integrated modules are addressed as separate devices via IP port.

	ClarityChrom®	OpenLAB®	PurityChrom®	Chromeleon®
Two pumps (independent)	no	yes	yes	yes
Fraction valve	no	cascading (Multi valve fraction collector)	one	yes
Injection module*	yes, but part of a method	yes, fully automatic module with trigger for data acquisition	yes, but part of a method	yes, but part of a method

* An injection module is a combination of one pump and one 6 port 2 position valve.

The Assistant ASM 2.2L can be equipped with three plug-in modules. Order the basic device and the modules separately and put together your perfect assistant. Insert the plug-in modules in the desired position of the assistant, tighten the four screws and you're done.

Configuration note

An assistant with following configuration is not allowed:

- more than two pump modules – a high-pressure gradient is not supported
- more than one UV detector
- without a plug-in module



AZURA® Assistant ASM 2.2L – Basic unit

Basic device

ASM 2.2L basic device with two empty modules	AYASM

Plug-in modules

Basic plug-in modules	Article number for ordering individual modules (without the assistant housing)*
Empty module	AG2022
AZURA® Valve Unifier VU 4.1**	AWA04
AZURA® UV Detector UVD 2.1S	ADA03XA
AZURA® UV Detector UVD 2.1S, fiber optics	ADA07XA
Compact pump without pressure sensor	
AZURA® Pump P 2.1S, 10 ml, stainless steel	APG92EA
AZURA® Pump P 2.1S, 10 ml, Hastelloy C	APG92EC
AZURA® Pump P 2.1S, 10 ml, ceramic	APG92EB
AZURA® Pump P 2.1S, 50 ml, stainless steel	APG92FA
AZURA® Pump P 2.1S, 50 ml, Hastelloy C	APG92FC
AZURA® Pump P 2.1S, 50 ml, ceramic	APG92FB
Compact pump with pressure sensor	
AZURA® Pump P 4.1S, 10 ml, stainless steel	APG22EA
AZURA® Pump P 4.1S, 10 ml, stainless steel, normal phase	APG22ED
AZURA® Pump P 4.1S, 10 ml, ceramic	APG22EB
AZURA® Pump P 4.1S, 50 ml, stainless steel	APG22FA
AZURA® Pump P 4.1S, 50 ml, stainless steel, normal phase	APG22FD
AZURA® Pump P 4.1S, 50 ml, ceramic	APG22FB
AZURA® Pump P 4.1S, 50 bar, 10 ml, stainless steel	APG12EA
AZURA® Pump P 4.1S, 50 bar, 10 ml, ceramic	APG12EB
AZURA® Pump P 4.1S, 50 bar, 50 ml, stainless steel	APG12FA
AZURA® Pump P 4.1S, 50 bar, 50 ml, ceramic	APG12FB



AZURA® Valve Unifier VU 4.1**



AZURA® Detector UVD 2.1S



AZURA® Pump P 4.1S

Accessories

* Please consider the configuration notes above.

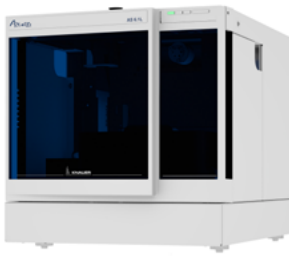
** Note that valves V 4.1 has to be ordered in addition to the valve drive VU 4.1. **For valves, see p. 85.**

Column holder – replacing empty module	AG2022B



Column holder

AZURA® Autosampler AS 6.1L



The AZURA® Autosampler AS 6.1L is available in different versions to meet a wide range of customer requirements (analytical or preparative; see below). Depending on the configuration, the autosampler can inject up to 768 samples when equipped with two microtiter plates, or up to 108 × 1.5 ml vials. The AS 6.1L minimizes carryover through a highly effective interior and exterior needle wash procedure. The system is based on a pulled loop design, in which the sample is aspirated directly into the sample loop. An integrated headspace pressure function prevents bubble formation in the vial during sample uptake. Precolumn derivatization is fully supported.

Three injection modes are available:

- Full loop filling - highest precision and reproducibility
- Partial loop filling - variable injection volumes, e.g. for dilution series
- µl Pickup - loss free injection of extremely small sample volumes

For high-pressure injections, the autosampler is equipped with a so-called ILD™ valve (Intermediate Loop Decompression). This valve includes a central depressurizing port, allowing the sample loop to be depressurized before sample loading. This prevents dilution of the sample with solvent. Extremely fast valve switching (< 100 ms) reduces pressure spikes, resulting in more precise analyses and reduced column wear.

KNAUER offers various software control options:
www.knauer.net/software

For autosampler kits, **see p. 61**

Further information:
www.knauer.net/Autosamplers

Specifications

Sample injection

Autosampler flow path	Analytical
Maximum back pressure	See device versions
Vial/plate dimensions	Dimensions according to ANSI SLAS 4-2004; height: max. 47 mm
Injection volume range	0.1 µl – 10 ml depending on sample loop
Headspace pressure	Built-in compressor, only for sample vials with septum
Switching time inj. valve	< 100 ms
Piercing needle precision	± 0.6 mm
Sample tray cooling	Optional (4 – 40 °C)
Vial detection	Missing vial/well plate detection by sensor
Wash solvents	2 solvents
Leak sensor	No
Wetted materials	ETFE (buffer & needle tubing), stainless steel (sample needle, valve stator), Vespel (rotor seal), Kel-F (syringe valve), glass (syringe), PTFE (tip of syringe plunger)

Analytical performance

Injection design	Pulled loop
Injection modes	Full loop filling, partial loop filling and microliter pickup; PASA™ (pressure-assisted sample aspiration)
Injection precision (valid at 0.2 – 5 cP)	Full loop RSD ≤ 0.3 % Partial loop RSD ≤ 0.5 % µl-pickup RSD ≤ 1.0 %
Linearity	$R^2 \geq 0.9995$
Injection accuracy	0.2 µl for 250 µl injection syringe
Sample carryover	≤ 0.01 % (standard conditions) ≤ 0.0015 % (chlorhexidine) ≤ 0.0003 % (extended needle wash, see Technical Note VTN0004)
Injections per vial	Max. 9 injections
Injection cycle time	Min. 7 s from the same vial, 14 s from different vials; < 60 s for > 100 µl sample injection in all injection modes, incl. 300 µl needle wash
Analysis time	Max. 9 h, 59 min, 59 s

Communication

Inputs	2 programmable TTL inputs (next injection, freeze, stop)
Outputs	1 programmable relay output (inject marker, auxiliary, alarm)
Control	Ethernet (LAN)
Interfaces	LAN, analog

Technical parameters

Ambient conditions	Temperature range: 10 – 40 °C, 50 – 104 °F Air humidity: 20 – 80 %

General

Power supply	95 – 240 V AC
Dimensions	361 × 375 × 570 mm (W × H × D)
Weight	30 kg

Device versions

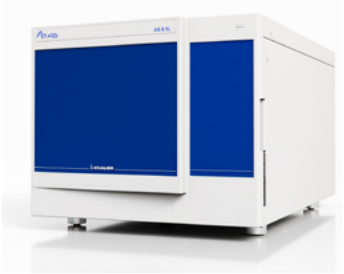
	HPLC+	UHPLC	Bio	Prep
Maximum back pressure	862 bar	1 240 bar	345 bar	350 bar
Sample needle	15 µl	15 µl	15 µl	60 µl
Dispenser syringe	250 µl	250 µl	250 µl	2 500 µl
Buffer tubing	500 µl	500 µl	500 µl	2 000 µl
Sample loop	100 µl, 0.4 mm ID	10 µl, 0.18 mm ID	100 µl, 0.4 mm ID	10 ml
Order number	AAA50AA	AAA10AA	AAA20AA	AAA40AA
Order number (cool/heat option)	AAA51AA	AAA11AA	AAA21AA	AAA41AA*

* also available as biocompatible version: AAA31AA

Ordering details

AAA50AA	AZURA® Autosampler AS 6.1L analytical HPLC autosampler, 862 bar
AAA51AA	AZURA® Autosampler AS 6.1L analytical HPLC autosampler, 862 bar, with sample cooling/heating
AAA10AA	AZURA® Autosampler AS 6.1L analytical HPLC autosampler, 1240 bar
AAA11AA	AZURA® Autosampler AS 6.1L analytical HPLC autosampler, 1240 bar, with sample cooling/heating
AAA20AA	AZURA® Autosampler AS 6.1L analytical HPLC autosampler, 345 bar, with biocompatible flow path
AAA21AA	AZURA® Autosampler AS 6.1L analytical HPLC autosampler, 345 bar, with biocompatible flow path and sample cooling/heating
AAA31AA	AZURA® Autosampler AS 6.1L preparative HPLC autosampler, 350 bar, with biocompatible flow path and sample cooling/heating
AAA40AA	AZURA® Autosampler AS 6.1L preparative HPLC autosampler, 350 bar
AAA41AA	AZURA® Autosampler AS 6.1L preparative HPLC autosampler, 350 bar, with sample cooling/heating

AZURA® Autosampler AS 8.1L



The AZURA® AS 8.1L is designed for analytical UHPLC applications, reliably withstanding pressures of up to 1 379 bar. Flexible sample handling is supported by a range of vial configuration options. The autosampler features two multiwell plate positions, accommodating up to 768 samples in two microtiter plates or 108 × 1.5 ml vials. In addition, two more vial racks can be added with a total capacity of e.g. 24 × 2 ml or 6 × 5 ml. The extension loop enables injection volumes of up to 100 µl.

In the split-loop design, the sample needle aligns precisely with the needle seat, sealing the high-pressure flow path and integrating the sample directly into the mobile phase. This approach minimizes carryover and shortens cycle times, supporting efficient high throughput measurements. Because the needle is part of the system flow path, flushing is obsolete, preventing sample loss while maintaining high precision and excellent injection linearity. Sample usage efficiency can be further enhanced when using spring bottom conical inserts, enabling reproducible injections down to a remaining volume of just 0.5 µl.

KNAUER offers various software control options:
www.knauer.net/software

For autosampler kits, **see p. 61**

Further information:
www.knauer.net/Autosamplers

Specifications

Sample injection	
Application	Analytical
Maximum back pressure	1 379 bar
Vial/plate dimensions	Dimensions according to ANSI SLAS 4-2004; height: max. 47 mm
Injection volume range	0.1 µl – 100 µl
Syringe dispenser	100 µl
Viscosity range	0.2 – 5 cP
Switching time inj. valve	< 100 ms
Sample tray cooling/heating	Optional; 4 – 40 °C (± 2 °C)
Wash solvents	3 solvents
Leak sensor	Yes
Wetted materials in flow path	SST, PTFE, TEFZEL, VESPEL, Glass, Teflon; optional: PEEK

Performance	
Injection design	Split loop
Injection precision	RSD ≤ 0.30 %
Linearity	$R^2 \geq 0.9999$
Sample carryover	≤ 0.0015 % (standard conditions)
Injections accuracy (valid at 0.2 – 5 cP)	≤ 0.2 µl for 10 µl injections
Injection cycle time	≤ 30 s

Communication	
Inputs	2 programmable TTL inputs (next injection, freeze, stop)
Outputs	1 programmable relay output (inject marker, auxiliary, alarm)
Control	Ethernet (LAN)
Interfaces	LAN, analog

Technical parameters	
Ambient conditions	Temperature range: 5 – 40 °C, 41 – 104 °F Air humidity: 20 – 80 %

General	
Power supply	100 – 240 V AC
Dimensions	360 × 300 × 560 mm (W × H × D)
Weight	Standard: 22 kg; cool/heat: 24 kg

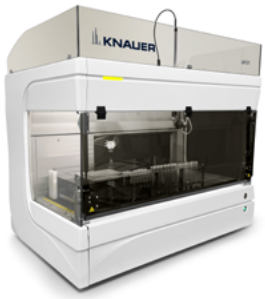
Ordering details

AAB10AA	AZURA® AS 8.1L analytical Autosampler up to 1 379 bar, without cool/heat option
AAB11AA	AZURA® AS 8.1L analytical Autosampler up to 1 379 bar, with cool/heat option

Preparative Liquid Handler LH 2.1

KNAUER's Liquid Handler LH 2.1 allows for the expansion of purification processes with the ability to combine sample injection and fraction collection in one device. A high capacity of sample and fraction vessels meets a flexible arrangement facilitating re-injection of samples to reach new levels of purification. The handler injects samples with minimal loss regardless of their volume – perfect for working with expensive compounds.

- Combine sample injection and fraction collection
- Inject small and large sample volumes with minimal loss
- Expand your vessel capacities
- Flexible arrangement of samples and fractions
- Re-inject collected fractions to reach new levels of purity



KNAUER offers various software control options:
www.knauer.net/software

For LH 2.1 accessories
see p. 70

Further information:
www.knauer.net/Liquid handling

Specifications

Fraction collection

Fraction capacity	Maximum vessel capacity with 5 KNAUER racks • 15 × micro titer well plates • 15 × 24 well plates • 810 × 2 ml tubes • 490 × 15 ml tubes • 160 × 50 ml tubes
Diverter valve	yes
Number of racks	5 KNAUER racks, teaching module for own racks (racks are not included)

Sample injection

Sample injection	Standard and sandwich injection mode
Sample loop	Up to 60 ml; included 10 ml (PEEK)
Injection valve	Valve and valve drive not included, 1/16" or 1/8" V 4.1 injection valves and VU 4.1 supported
Temperature control	No
Needle wash	Single needle wash step after each injection
Wash solvent	4
Wetted materials	Aluminum oxide 99.5 %, Borosilicate Glass, PTFE, FEP, AISI 316L, PEEK

General

Power supply	100 – 240 V, 50 – 60 Hz
Dimensions	96 cm × 104 cm × 70 cm; working area 75 cm × 30 cm
Weight	82 kg
Leak sensor	No
Ambient conditions	10 – 35 °C, 30 – 80 % RH, non-condensing
Note	Valve drive VU 4.1, injection valve V 4.1, sample loop and racks are not included

Communication

Control	LAN, supported by PurityChrom 5 and Chromeleon 7.3
Programming	Loop Volume, Syringe Volume, Syringe Speed, Syringe Delay, Sandwich Volume, Wash Volume, Wash Speed, Dead Volume

Ordering details

Device	
A5080	Preparative Liquid Handler LH 2.1
Accessories	
A50801	LH 2.1 Rack for 3 × microtiter plates
A50802	LH 2.1 Rack for 162 × 2 ml tubes (Eppendorf)
A50803	LH 2.1 Rack for 98 × 15 ml tubes (Falcon)
A50804	LH 2.1 Rack for 32 × 50 ml tubes
A50805	LH 2.1 Rack for 3 × 24 deep well plates

Analytical Liquid Handler LH 8.1

NEW

The KNAUER Liquid Handler LH 8.1 offers finally an alternative in the world of robotic autosamplers. In-house design and development for a new level of simplicity and speed. It comes with a new modularity so that you can customize the autosampler completely to your needs. The "in syringe" sample injection design allows for a special injection method, which combines the precision of partial loop with the zero sample loss of µl pickup. To further develop your automation, the software solutions will also allow sample preparation processes. Like all KNAUER devices, the Liquid Handler LH 8.1 will have full software support from all standard CDS packages.



Specifications

Sample injection

Sample injection modes	Full loop, Sandwich loop, Partial loop
Maximum back pressure	1 240 bar
Sample capacity	Sample racks per robotic cooler. Up to 3 robotic coolers on a standard system. 2 racks for manual rack holder.
Sample rack dimensions	Per rack 60 × 1.5 ml vials, 96 well plates or 284 well plates possible. 130 × 1.5 ml vial rack available (2 rack positions needed)
Sample loop	Possible from 2 µl to 100 µl, 20 µl loop included
Injection valve	Special 6 Port 2-Position Injection Valve with injection port
Switching time inj. valve	< 100 ms
Piercing needle precision	± 0.1 mm
Sample tray cooling/heating	4 – 40 °C possible
Temperature control	Yes with robotic cooler
Vial detection	Yes
Needle wash	Programmable by method
Wash solvent	2 minimum, up to 10
Wetted materials	PTFE, PEEK, Stainless steel, Borosilicate glass

Communication

Interfaces	LAN, analog
Control	Ethernet
Inputs	Programmable TTL inputs
Analog inputs	RS-232

General

Dimensions	Length: 447 mm up to 887 mm in steps of 110 mm Standard models: 557 mm and 887 mm Depth: 520 mm Height: 655 – 681 mm (incl. rail mounted options and supports)
Weight	~ 15.5 kg (~ 27 kg with robotic cooler)
Leak sensor	No
Ambient conditions	10 – 35 °C, 30 – 80 % RH, non-condensing
System type	XYZ autosampler with syringe
Syringe size	25 – 100 µl (other volumes possible)
Software API	Comprehensive support for C#, C++, C and Python on Windows and (embedded) Linux OS
Electrical Interfaces	Solid State Relay Outputs (Ready, Inject, Auxiliary), wide voltage digital inputs (Ready, Auxiliary)
GLP	Yes, valve switches, injections
Display	No

Analytical performance

Injection volumes	1 – 80 µl (higher injection volumes possible on request)
Injection precision	RSD (Relative Standard Deviation): full loop injection: ≤ 0.1 %; sandwich injection at an injection volume > 5 µl: < 0.15 %
Sample carryover	< 0.002 % with caffeine and Fast Wash Station < 0.005 % with chlorhexidin and Fast Wash Station
Linearity	$R^2 \geq 0.999$

Key features

- Inject small and large sample volumes with zero sample loss
- Up to 390 vials or 6 well plates, temperature from 4 to 40 °C with each robotic cooler
- Flexible arrangement of modules
- The standard rail lengths are 557 mm and 887 mm, different dimensions are possible on request

For LH 8.1 accessories, **see p. 70**

Further information:
www.knauer.net/LH 8.1

Ordering details

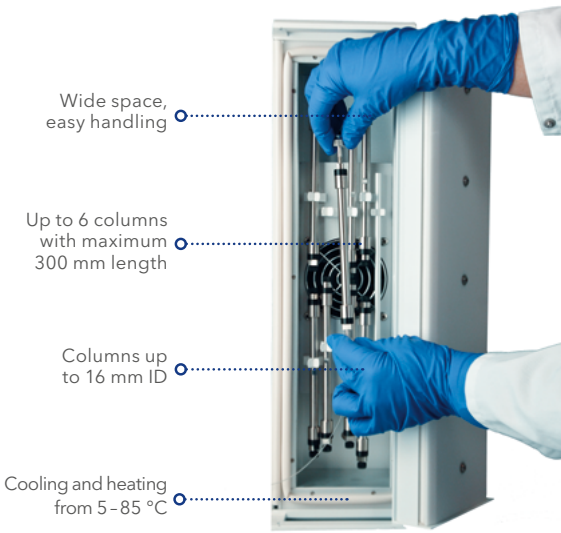
Device

A5100	LH 8.1 Liquid Handler, 557 mm version with manual tray holder, incl. injection valve and Fast Wash Station
A51001	LH 8.1 Liquid Handler, 557 mm version with robotic cooler, incl. injection valve and Fast Wash Station
A5110	LH 8.1 Liquid Handler, 887 mm version with manual tray holder, incl. injection valve and Fast Wash Station
A51101	LH 8.1 Liquid Handler, 887 mm version with robotic cooler, incl. injection valve and Fast Wash Station

AZURA® Column Thermostat CT 2.1

The AZURA® CT 2.1 is a price attractive basic column thermostat. It allows temperature control in the range of 5 °C and 85 °C and thus is appropriate for most HPLC applications. For advanced purification and analysis purposes, the oven can optionally be equipped with an eluent pre-heating cartridge. This ensures even more constant separation conditions leading to higher selectivity and an improved peak shape.

The instrument operates with a microprocessor controlled Peltier element for precise temperature settings. In combination with its high temperature stability, this allows programming of linear as well as non-linear temperature gradients.



KNAUER offers various software control options:
www.knauer.net/software

Further information:
www.knauer.net/CT 2.1

Specifications

Thermostatting

Heating and cooling system	Microprocessor controlled Peltier element for heating and cooling, fan supported 2-way air circulation
Temperature range	5 – 85 °C
Heating/cooling rate	2 °C/min
Temperature accuracy	± 0.2 °C
Temperature stability	± 0.1 °C

Column compartment

Column dimensions

max. number	max. length*	max. outer diameter*	matching column
8	160 mm	12 mm	125 mm × 4.6 mm ID with precolumn
4	325 mm	12 mm	300 mm × 4.6 mm ID
1	325 mm	35 mm	300 mm × 16 mm ID

* total outer dimensions of the column including screw caps

Dimensions, internal	90 × 390 × 47 mm (W × H × D)
Safety	Self-check and auto-calibration at power-on, selectable turn-off temperature
Leak sensor	Gas sensor, adjustable sensitivity, acoustic signal, turn-off switch

Communication

Control	Optional for stand-alone functionality: Mobile Control
Interfaces	LAN Interface

General

Power supply	90 – 230 V, 50 – 60 Hz, 100 W
Dimensions	150 × 470 × 310 mm (W × H × D)
Weight	8.4 kg

Other

Optional accessories	Cartridge for eluent pre-heating for capillary with an ID of 0.1 or 0.18 mm

Ordering details

Device

ATC00	Column Thermostat AZURA® CT 2.1 for constant temperatures and reproducible results

Accessories

A05853	Cartridge for eluent pre-heating, ID 0.18 mm, ~ 8 µl
A05854	Cartridge for eluent pre-heating, ID 0.1 mm, ~ 2.5 µl



A05853

AZURA® Detector DAD 6.1L

The AZURA® DAD 6.1L is a high-end diode array detector (DAD) which combines outstanding performance with easy handling.

A wide range of easily exchangeable flow cells make this device the right choice for fast, standard analytical, semi-preparative and preparative separations with bio-compatible or stainless steel wetted parts.

State-of-the-art total reflection flow cells (LightGuide technology) are available for this detector providing maximum light throughput (due to total internal reflection) with minimal peak dispersion (due to small cell volume) to guarantee an optimized signal to noise (S/N) ratio.

An optional fiber optics adapter offers the possibility to separate the flow cell spatially from the device and thus provides enhanced security for hazardous, explosive or toxic work processes, as well as protecting the device from leakages at high flow rates.

The optical unit with KNAUER Polka-Dot technology and intelligent temperature management ensure maximum sensitivity combined with minimal baseline drift over the whole spectrum.

Furthermore, easy frontal access and improved safety features enable effortless lamp replacement. This eases maintenance and guarantees short downtimes.

The AZURA® DAD 6.1L comes installed with a high brightness deuterium and tungsten halogen lamps, which cover a wavelength range from 190 to 1000 nm.



Key features

- Wide application range
- Large choice of flow cells
- Fiber optics adapter available
- Attractively priced
- Made in Germany

KNAUER offers various software control options:
www.knauer.net/software

For detector accessories and flow cells **see p. 79**

Further information:
www.knauer.net/Detector DAD 6.1L

Specifications

Detection	
Detector type	Diode array detector
Number of diodes	1 024
Pixel pitch	0.8 nm/diode
Detection channels	8 (Digital)/4 (Analog)
Light source	High brightness deuterium (D_2) lamp and halogen lamp with integrated GLP chip
Wavelength range	190 – 1 000 nm
Spectral resolution	< 3.5 nm at H_α line (FWHM) /Note: digital bandwidth 1 – 32 nm
Slit width	2.5 nm
Wavelength accuracy	± 1 nm
Wavelength repeatability	± 0.1 nm
Noise	± 3.5 µAU at 254 nm (ASTM E1657-98)
Drift	300 µAU/h at 254 nm (ASTM E1657-98)
Linearity	> 2.5 AU at 274 nm (ASTM E1657-98)
Maximum data rate	100 Hz (LAN)/12.5 Hz (analog)
Flow cell	Flow cells are not included and need to be ordered separately (see “Accessories”).
Time constants	0.00 / 0.01 / 0.02 / 0.05 / 0.1 / 0.2 / 0.5 / 1.0 / 2.0 / 5.0 / 10.0
Integration time	Automatic
Wavelength verification	Internal holmium filter and deuterium lines
Leak sensor	Yes
Communication	
Inputs	Error (IN), Start (IN), Autozero
Outputs	Events 1 – 2 (Relay and TTL compatible, respectively), Error (OUT), + 5 V, Valve + 24 V, Valve (OUT)
Analog outputs	4 × 0 – 5 V, 20 bit, offset adjustable
Control	Mobile Control, software, event control, analog, terminal protocol
Interfaces	LAN (RJ-45), USB (service only), multi-pin connector, analog (RCA cinch connector)
Technical parameters	
GLP	Detailed report including lamp recognition, operating hours, lamp operating hours, number of lamp ignitions
Display	Mobile Control (optional)
Ambient conditions	Temperature range: 4 – 40 °C, 39.2 – 104 °F Humidity: below 90 %, non condensing
General	
Power supply	100 – 240 V, 50 – 60 Hz, 75 W
Dimensions	361 × 158 × 523 mm (W × H × D)
Weight	13.8 kg

Ordering details

Device

ADC11	AZURA® Detector DAD 6.1L Diode array detector DAD 6.1L without flow cell 190 – 1 000 nm, incl. test cell

Accessories

AMC19XA	10 mm path length, 2 µl, 1/16", 50 bar, LightGuide Flow cell cartridge for AZURA® Detector DAD/MWD
AMD59XA	50 mm path length, 6 µl, 1/16", 50 bar, High Sensitivity LightGuide Flow cell cartridge for AZURA® Detector DAD/MWD
AMC37	10 mm path length, 10 µl, 1/16'', 300 bar, 200 ml/min, PressureProof Flow cell cartridge for AZURA® Detector DAD/MWD
AMC38	10 mm path length, 10 µl, 1/16", 300 bar, PressureProof Flow cell cartridge for AZURA® Detector DAD/MWD
AMB18	3 mm path length, 2 µl, 1/16", 300 bar, PressureProof Flow cell cartridge for AZURA® Detector DAD/MWD
AMKX8KIT	Fiber optics adapter kit for AZURA® Detector DAD/MWD, with fiber optic cables (1 × 400 mm and 1 × 750 mm) and mounting bracket
AMLX8	Test cell for AZURA® Detector DAD/MWD
AZZ00OC	AZURA® Heat Exchanger for analytical PressureProof flow cells, 14 µl

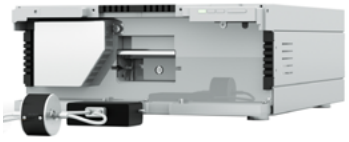
AZURA® Detector DAD 2.1L & MWD 2.1L

The AZURA® DAD 2.1L and MWD 2.1L is a highly competitive diode array detector and a sensitive, 8-channel multiwavelength detector, respectively. Both combine high performance with easy handling at an affordable price.

State-of-the-art total reflection flow cells (LightGuide technology) are available for this device providing maximum light throughput (due to total internal reflection) with minimal peak dispersion (due to the small cell volume) to guarantee an optimized S/N ratio. An optional fiber optics adapter offers the possibility to separate the flow cell spatially from the device and thus provides enhanced security for hazardous, explosive or toxic work processes, as well as protecting the device from leakages at high flow rates.

The newly developed optical unit and intelligent temperature management ensure maximum sensitivity combined with minimal baseline drift. Furthermore, easy frontal access and improved safety features enable effortless lamp replacement. This simplifies maintenance and guarantees short downtimes.

The DAD 2.1L and MWD 2.1L come installed with a deuterium lamp which covers a wavelength range from 190 to 700 nm.



Key features

- Wide application range
- Large choice of flow cells
- Fiber optics adapter available
- Leak management
- Made in Germany

For detector accessories and flow cells, **see p. 79**

Further information:
www.knauer.net/Diode array detector

Specifications

Detection

Detector type	Diode array detector or variable multi wavelength detector
Number of diodes (for DAD 2.1L)	256
Pixel pitch (for DAD 2.1L)	2 nm/diode
Detection channels	8 (Digital)/4 (Analog)
Light source	Deuterium (D_2) lamp with integrated GLP chip
Wavelength range	190 – 700 nm
Spectral resolution	< 10 nm at H_α line (FWHM) /Note: digital bandwidth 1 – 32 nm
Slit width	7.0 nm
Wavelength accuracy	± 1 nm
Wavelength repeatability	± 0.1 nm
Noise	± 5 µAU at 254 nm (ASTM E1657-98)
Drift	400 µAU/h at 254 nm (ASTM E1657-98)
Linearity	> 2.0 AU at 274 nm (ASTM E1657-98)
Maximum data rate	100 Hz (LAN)/12.5 Hz (analog)
Flow cell	Flow cells are not included and need to be ordered separately (see “Accessories”).
Time constants	0.00 / 0.01 / 0.02 / 0.05 / 0.1 / 0.2 / 0.5 / 1.0 / 2.0 / 5.0 / 10.0 s
Integration time	Automatic
Wavelength verification	Internal holmium filter and deuterium lines
Leak sensor	Yes

Communication

Inputs	Error (IN), Start (IN), Autozero
Outputs	Events 1 – 2 (Relay and TTL compatible, respectively), Error (OUT), + 5 V, Valve + 24 V, Valve (OUT)
Analog outputs	4 × 0 – 5 V, 20 bit, offset adjustable
Control	Mobile Control, software, event control, analog, terminal protocol
Interfaces	LAN (RJ-45), USB (service only), multi-pin connector, analog (RCA cinch connector)

Technical parameters

GLP	Detailed report including lamp recognition, operating hours, lamp operating hours, number of lamp ignitions
Display	Mobile Control (optional)
Ambient conditions	Temperature range: 4 – 40 °C, 39.2 – 104 °F Humidity: below 90 %, non condensing

General

Power supply	100 – 240 V, 50 – 60 Hz, 75 W
Dimensions	361 × 158 × 523 mm (W × H × D)
Weight	12.2 kg

Ordering details

Device	
ADC01	AZURA® Detector DAD 2.1L Diode array detector DAD 2.1L without flow cell 190 – 700 nm, incl. test cell
ADB01	AZURA® Detector MWD 2.1L multi wavelength detector MWD 2.1L, without flow cell 190 – 700 nm, incl. test cell
Accessories	
AMC19XA	10 mm path length, 2 µl, 1/16", 50 bar, LightGuide Flow cell cartridge for AZURA® Detector DAD/MWD
AMD59XA	50 mm path length, 6 µl, 1/16", 50 bar, High Sensitivity LightGuide Flow cell cartridge for AZURA® Detector DAD/MWD
AMC37	10 mm path length, 10 µl, 1/16'', 300 bar, 200 ml/min, PressureProof Flow cell cartridge for AZURA® Detector DAD/MWD
AMC38	10 mm path length, 10 µl, 1/16", 300 bar, PressureProof Flow cell cartridge for AZURA® Detector DAD/MWD
AMB18	3 mm path length, 2 µl, 1/16", 300 bar, PressureProof Flow cell cartridge for AZURA® Detector DAD/MWD
AMKX8KIT	Fiber optics adapter kit for AZURA® Detector DAD/MWD, with fiber optic cables (1 × 400 mm and 1 × 750 mm) and mounting bracket
AMLX8	Test cell for AZURA® Detector DAD/MWD
AZZ00OC	AZURA® Heat Exchanger for analytical PressureProof flow cells, 14 µl

AZURA® Detector UVD 2.1L

The AZURA® UV/VIS Detector UVD 2.1L is a competitively priced HPLC spectrophotometer for routine HPLC applications including fast LC methods. Besides offering excellent technical specifications, this robust detector features a highly flexible and compact design. The UVD 2.1L comes with an installed deuterium lamp which covers a wavelength range from 190 to 750 nm.

Due to a smart design the flow cell is easily accessible and can be changed very quickly. You can choose between a wide range of flow cells for analytical or preparative LC applications with flow rates from 10 µl/min up to 10 l/min.



Key features

- Large choice of flow cells
- Leak management
- 60 years experience
- Made in Germany

KNAUER offers various software control options:
www.knauer.net/software

For detector accessories and flow cells, **see p. 79**

Further information:
www.knauer.net/Detector UVD 2.1L

Specifications

Detection

Detector type	Variable single wavelength UV detector
Detection channels	1
Light source	Deuterium (D_2) lamp with integrated GLP chip
Wavelength range	190 – 750 nm
Spectral bandwidth	11 nm at H_α line (FWHM)
Wavelength accuracy	± 2.5 nm
Wavelength precision	0.3 nm (ASTM E275-93)
Noise	± 15 µAU at 254 nm (ASTM E1657-98)
Drift	300 µAU/h at 254 nm (ASTM E1657-98)
Linearity	> 2.0 AU at 274 nm (ASTM E1657-98)
Maximum data rate	50 Hz (LAN) / 20 Hz (Analog)
Flow cell	Flow cells are not included and need to be ordered separately (see “Accessories”).
Time constants	0.0 / 0.1 / 0.2 / 0.5 / 1.0 / 2.0 / 5.0 / 10.0 s
Integration time	Automatic
Leak sensor	Yes

Communication

Inputs	Error (IN), Start (IN), Autozero, 0 – 10 V Analog (IN)
Outputs	Events 1 – 3, + 5 V, 24 V valve
Analog outputs	1 × 0 – 5 V scalable, 20 bit, offset adjustable
Control	Mobile Control, software, event control, analog, terminal protocol
Programming	Timed: wavelength, events, fraction valve, links, wake up (program, link); 9 programs, 50 program lines

Technical parameters

GLP	Detailed report incl. lamp recognition, operating hours, lamp operating hours, number of lamp ignitions
Display	Mobile Control (optional)
Ambient conditions	Temperature range: 4 – 40 °C, 39.2 – 104 °F Humidity: below 90 %, non condensing

General

Power supply	100 – 240 V, 50 – 60 Hz, 65 W
Dimensions	361 × 158 × 523 mm (W × H × D)
Weight	5.9 kg

Ordering details

Device	
ADA01XA	AZURA® Detector UVD 2.1L with deuterium lamp without flow cell, incl. test cell
ADA04XA	AZURA® Detector UVD 2.1L Fiber Optics Version with deuterium lamp without flow cell
Accessories	
A4061XB	10 mm path length, 10 µl, 1/16”, 300 bar, stainless steel, with heat exchanger one sided inlet and outlet, classical KNAUER flow cell
A4042	3 mm path length, 2 µl, 1/16”, stainless steel, classical KNAUER flow cell
A4126	Test cell Holmium oxide filter
A4146	Test cell, WG 280 filter stray light
A4123	Test cell

AZURA® Detector UVD 2.1S

The AZURA® UVD 2.1S is a highly competitive single variable wavelength UV detector for HPLC. It offers excellent technical specifications for routine laboratory work. With its small footprint, it is one of the smallest detectors for HPLC on the market.

The UVD 2.1S comes in the novel small AZURA® housing. The installed deuterium lamp covers a wavelength range from 190 to 500 nm. The UV detector can be controlled with various CDS software packages via LAN, RS-232 or analog input/output, as well as from the front panel as stand alone device.

Due to a smart design, the flow cell is easily accessible and can be changed very quickly. Choose between a wide range of flow cells for analytical or preparative LC applications with flow rates from 10 µl/min up to 10 l/min. Also available as a module for AZURA® Assistant ASM 2.2L.



Key features

- Compact
- Large choice of flow cells
- Made in Germany

KNAUER offers various software control options:
www.knauer.net/software

For detector accessories and flow cells, **see p. 79**

Further information:
www.knauer.net/Detector UVD 2.1S

Specifications

Detection

Detector type	Variable single wavelength UV detector
Detection channels	1
Light source	Deuterium (D_2) lamp with integrated GLP chip
Wavelength range	190 – 500 nm
Spectral bandwidth	13 nm at H_α line (FWHM)
Wavelength accuracy	± 3 nm
Wavelength precision	0.7 nm (ASTM E275-93)
Noise	± 20 µAU at 254 nm (ASTM E1657-98)
Drift	300 µAU/h at 254 nm (ASTM E1657-98)
Linearity	> 2.0 AU at 274 nm (ASTM E1657-98)
Maximum data rate	50 Hz (LAN) / 20 Hz (Analog) / 10 Hz (RS-232)
Flow cell	Flow cells are not included and need to be ordered separately (see “Accessories”).
Time constants	0.00 / 0.02 / 0.05 / 0.1 / 0.2 / 0.5 / 1.0 / 2.0 s
Integration time	Automatic

Communication

Inputs	Autozero, Start (IN), Error (either IN or OUT), 0 – 10 V Analog IN
Outputs	Error (either OUT or IN)
Analog inputs	Wavelength 0 – 10 V
Analog outputs	1 × ± 2.5 V scalable, 20 bit
Control	Front panel, Mobile Control, software, event control, analog, terminal protocol
Interfaces	LAN (RJ-45), RS-232 (SUB-D 9), multi-pin connector, analog (RCA cinch connector)

Technical parameters

GLP	Lamp operating hours
Display	LED
Ambient conditions	Temperature range: 4 – 40 °C, 39.2 – 104 °F Humidity: below 90 %, non condensing

General

Power supply	External: input 100 – 240 V, output 24 V DC, 60 W
Dimensions	121 × 129 × 187mm (W × H × D)
Weight	1.5 kg

Ordering details

Device	
ADA00	AZURA® Detector UVD 2.1S with deuterium lamp without flow cell, incl. test cell
ADA05	AZURA® Detector UVD 2.1S Fiber Optics Version with deuterium lamp without flow cell
Accessories	
A4061XB	10 mm path length, 10 µl, 1/16”, 300 bar, stainless steel, with heat exchanger one sided inlet and outlet, classical KNAUER flow cell
A4042	3 mm path length, 2 µl, 1/16”, stainless steel, classical KNAUER flow cell
A4045	3 mm path length, 2 µl, 1/16”, 30 bar, biocompatible, classical KNAUER flow cell
A5193	Deuterium lamp, replacement for S2520, 10D, 40D, UVD 2.1S, UVD 2.1L, DAD 2.1L, MWD 2.1L

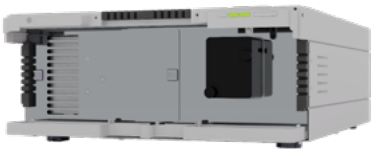
AZURA® Detector VWD 2.1L

The AZURA® VWD 2.1L is a versatile variable wavelength detector designed to adapt seamlessly to your needs.

In combination with a classic KNAUER flow cell, it supports analytical HPLC by detecting up to four compound-specific wavelengths across 190 – 900 nm at data rates of up to 5 Hz (for four UV/VIS channels) or 100 Hz (for one UV/VIS channel).

When equipped with KNAUER's patented CombiLight flow cell, the VWD 2.1L goes even further – simultaneously monitoring conductivity and eluent temperature alongside up to four UV/VIS channels. To complete FPLC systems, a pH glass electrode can be connected for real-time pH monitoring.

Thanks to its flexible design, the VWD 2.1L can be easily converted between flow cell and fiber-optics version. One detector, multiple possibilities – stay flexible and ready for any project that comes your way.



Key features

- Combination of UV/VIS detection with monitoring of conductivity, eluent temperature and pH monitoring
- Patented flow cell technology
- Wide application range
- Large choice of flow cells
- Easily convertible into fiber optics version
- Leak management
- Made in Germany

KNAUER offers various software control options:
www.knauer.net/software

For detector accessories and flow cells, **see p. 79**

Further information:
www.knauer.net/Detector VWD 2.1L

Specifications

Detection

Detector type	Versatile variable wavelength detector
Detection channels	4 UV/VIS, 1 pH Plus 1 conductivity and 1 eluent temperature channel if a CombiLight flow cell is used.
Light source	Deuterium (D2) and halogen lamp with integrated GLP chip
Spectral bandwidth	6 nm at H_α line (FWHM)
Wavelength range	190 – 900 nm
Wavelength accuracy	± 1.5 nm
Wavelength precision	0.4 nm
Conductivity range	0.1 – 1 000 mS/cm
Conductivity accuracy	± 0.5 mS/cm or ± 2 % (the higher value applies) ranging from 0.1 to 200 mS/cm
Temperature range	1 °C to 99 °C
Temperature accuracy	± 1.5 °C, from 4 °C to 40 °C
pH range	0 – 14 pH
pH accuracy	± 0.1 pH, from + 4 °C to + 25 °C
Noise	± 7.5 µAU at 254 nm (ASTM E1657-98) ± 12.5 µAU at 254/280 nm (ASTM E1657-98)
Drift	0.2 µAU/h at 254 nm (ASTM E1657-98), 0.4 µAU/h at 240/280 nm (ASTM E1657-98)
Linearity	> 2.5 AU at 270 nm (ASTM E1657-98)
Maximum data rate	100 Hz (LAN), 1 UV channel 10 Hz (USB), 1 UV channel 5 Hz (LAN), 4 UV channels with DI = 100 nm
Flow cell	Flow cells are not included and need to be ordered separately (see "Accessories").
Time constants	0.01 / 0.02 / 0.05 / 0.1 / 0.2 / 0.5 / 1.0 / 2.0 / 5.0 / 10.0 s
Integration time	Automatic
Leak sensor	Yes

Communication

Inputs	Error (IN), Start (IN), Autozero 0–10 V, Analog IN
Outputs	Events 1 – 3, Error (OUT), Start-Out
Analog outputs	Not available
Control	LAN-DHCP, pin header analog: Control voltage for wavelength selection
Interfaces	2 × LAN (for software control), BNC (for pH electrode), USB (for service), multi-pin connector (for Events)

Technical parameters

GLP	Detailed report with automatic lamp and signal unit detection, operating hours, lamp(s) operating hours, number of lamp(s) ignitions
Display	Mobile Control (optional)
Ambient conditions	Temperature range: 4 – 40 °C, 39.2 – 104 °F Humidity: below 90 %, non condensing

General

Power supply	Power input: 100 – 240 V, output: 50 – 60 Hz , maximum power consumption: 100 watts
Dimensions	361 × 158 × 543 mm (W × H × D)
Weight	11 kg

Ordering details

Device	
ADB71	AZURA® Detector VWD 2.1L, standard version
ADB74	AZURA® Detector VWD 2.1L, fiber optics version
Accessories	
ADZ14	Signal unit for fiber optics
ADZ50	Signal unit for standard and CombiLight flow cells
AMI90	CombiLight test cell
AME91	0.5 mm path length, 168 µl, 1/8" or 3/16", 60 bar, biocompatible, CombiLight flow cell
AMF91	2 mm path length, 168 µl, 1/8" or 3/16", 60 bar, biocompatible, CombiLight flow cell
AMG90	5 mm path length, 51 µl, 1/16" or 1/8", 60 bar, biocompatible, CombiLight flow cell
AMH90	10 mm path length, 23.5 µl, 1/16" or 1/8", 60 bar, biocompatible, CombiLight flow cell
A70091-2	AZURA® pH kit, 100 ml/min, incl. pH electrode, storage vessel, dummy electrode and flow cell
A70091-3	AZURA® pH kit, 1 000 ml/min, incl. pH electrode, storage vessel, dummy electrode and flow cell
A1943	AZURA® pH flow cell, 100 ml/min
A1946	AZURA® pH flow cell, 1 000 ml/min
A1947	AZURA® storage vessel for pH electrode
A1942-1	AZURA® pH dummy electrode
A1933-3	KNAUER pH electrode for AZURA® VWD 2.1L, pH 2.1S and CM 2.1S

AZURA® Detector RID 2.1L

The AZURA® RID 2.1L is a sensitive and competitively priced differential refractometer. It is universally applicable for detecting compounds with little or no UV activity such as alcohols, sugars, lipids or polymers. This instrument is designed for use in analytical HPLC (high performance liquid chromatography) as well as under certain conditions for GPC (gel permeation chromatography) applications.

The intelligently designed optical unit with advanced temperature control ensures high sensitivity, fast baseline stabilization, and excellent reproducibility. Furthermore, the long-life LED, highly pressure resistant flow cell, improved safety features and enhanced diagnostics functions guarantee easy handling and minimal maintenance. The wide linear dynamic range and 10 ml/min maximum flow rate make the AZURA® RID 2.1L the perfect choice for most laboratory tasks.



Key features

- Temperature controlled optical unit
- Long-life LED
- Pressure resistant flow cell
- Made in Germany

KNAUER offers various software control options:
www.knauer.net/software

For detector accessories and flow cells, **see p. 79**

Further information:
www.knauer.net/Detector RID 2.1L

Specifications

Detection	
Detector type	Refractive index detector
Version	Analytical
Light source	Long-life LED
Detection channels	1
Refractive index range	1.00 – 1.75 RIU
Noise	± 2.5 nRIU
Drift	200 nRIU/h
Linearity	> 1 000 µRIU
Flow cell	5 bar back pressure resistance flow cell included
Max. flow rate	10 ml/min (pure water)
Flow cell volume	15 µl (43 µl dispersion volume)
Wetted materials	Stainless steel / quartz / PTFE
Temperature control	OFF, 30 × 55 °C (1 °C increment)
Time constants	0.00 / 0.01 / 0.02 / 0.05 / 0.1 / 0.2 / 0.5 / 1.0 / 2.0 / 5.0 / 10.0 s
Maximum data rate	100 Hz (LAN) / 20 Hz (Analog)
Autozero	Full range
Leak sensor	Yes (internal and external leak management)
Communication	
Inputs	Error (IN), Start (IN), Autozero, Flush (IN)
Outputs	Event 1, Start (OUT), Error (OUT), + 5 V, 24 V valve
Analog outputs	1 × 0 – 2.5 V scalable, 20 bit, offset adjustable
Control	Mobile Control, software, event control, analog, terminal protocol
Interfaces	2 × LAN (RJ-45, dual IP stack), USB (service only), multi-pin connector, analog (cinch connector)
Technical parameters	
GLP	Detailed report including operating hours, light source operating hours
Display	Mobile Control (optional)
Ambient conditions	Temperature range: 4 – 40 °C, 39.2 – 104 °F Humidity: below 90 %, non condensing
General	
Power supply	100 – 240 V, 50 – 60 Hz, 65 W
Dimensions	361 × 158 × 523 mm (W × H × D)
Weight	10.8 kg
Optional accessories	Mobile Control

Ordering details

Device	
ADD31	AZURA® Detector RID 2.1L analytical refractive index detector with flow cell
ADD38	AZURA® Detector RID 2.1L HighFlow preparative refractive index detector with flow cell and external pressure release valve, max. flow rate 100 ml/min

Fluorescence Detector RF-20A/Axs

The fluorescence detector RF-20A/Axs provides world-class sensitivity, excellent maintainability and diverse validation / support functions. It supports a wide range of applications in the wavelength range of 200 to 650 nm (or 200 to 750 nm for RF-Axs) from conventional analysis to high-performance analysis. With a signal-to-noise ratio of 1200 for the water-Raman band, the fluorescence detector is well suited for trace analysis. The xenon lamp and flow cell are directly accessible on the device, thus allowing a quick and easy handling and maintenance of the device by the user, thereby minimizing downtime. The lamp life is 2 000 hours. When replacing the xenon lamp, no positional adjustment is required



Key features

- Pressure resistant flow cell

Further information:
www.knauer.net/Fluorescence detectors

Specifications

Detection	
Detector type	Fluorescence detector
Detection channels	1 (for RF-20A) / 2 (for RF-20Axs)
Number of signals	1 (for RF-20A) / 2 (for RF-20Axs)
Light source	Xenon lamp; RF-20Axs: Low-pressure mercury lamp for wavelength accuracy check
Wavelength range	200 – 650 (RF-20A) / 200 – 750 (RF-20Axs)
Spectral bandwidth	20 nm
Wavelength accuracy	± 2 nm
Wavelength precision	± 0.2 nm, indicates the precision performance when the power is turned ON in the single wavelength mode and the wavelength is changed.
Sensitivity	Can be set at three levels: HIGH (× 1), MED (× 32), LOW (× 1 024)
Wetted materials	SUS316L, PTFE (fluorocarbon polymers), quartz
Flow cell volume	12 µl
Temperature control option	Temperature controlled flow cell for RF-20Axs
Time constants	11 levels can be selected, equivalent to “no filter”, 0.05, 0.1, 0.5, 1.0, 1.5, 2.0, 3.0, 6.0, 8.0 and 10.0 seconds
Autozero	Autozero function, baseline shift function
Communication	
Gain	Can be set at three levels: × 1, × 4, × 16
Technical parameters	
Ambient conditions	Operating temperature: 4 to 35 °C Relative humidity: 20 to 85 % (non condensing)
General	
Power supply	AC 220 – 240 V, 400 VA, 50/60 Hz
Dimensions	260 × 210 × 420 mm (W × H × D)
Weight	16 kg (RF-20A) / 18 kg (RF-20Axs)

Ordering details

Device	
A59200	Fluorescence detector RF-20A, 200 – 650 nm incl. accessories and flow cell
A59201	Fluorescence detector RF-20Axs, 200 – 750 nm, temperature control function, incl. accessories
A59203	Fluorescence detector RF-20A with photomultiplier from 200 – 900 nm incl. accessories and flow cell
A59204	Fluorescence detector RF-20Axs with photomultiplier from 200 – 900 nm, temperature control function, incl. accessories
Accessories	
A59210	Xenon lamp for RF-20A/Axs fluorescence detector
A59211	Flow cell for Semi micro LC cell capacity 3 µl, supports temperature control (RF-20Axs only)
A59212	Inert flow cell for RF-20A/Xs, cell capacity 12 µl, contact materials: PEEK, PTFE

Low-temperature Evaporative Light Scattering Detector Sedex 85LT/90LT/100LT/LC/FP

Evaporative Light Scattering Detection (ELSD) is a universal modern technology with which every analyte that is less volatile than the mobile phase can be detected. Using the unique Low Temperature technology, this detector allows the achievement very high sensitivity. The technology is gradient compatible and is independent of the absorption characteristics of the eluents. Compounds can be universally measured with this detector (carbohydrates, proteins, peptides, polymers, lipids, steroids, etc.), regardless of their fluorescence, absorption or refractive-index characteristics. Comprehensive SOP protocols for GLP conformity and validation procedures are available.



Specifications (for Sedex 85 LT)

Detection	
Detector type	Light scattering detector
Detection channels	1
Light source	Selected high efficiency blue LED (470 nm), elapsed-time count
Sensitivity	< 1 ng caffein (LOD)
Maximum data rate	Digital: 100 Hz/Analog: 30 Hz
Gas requirements	
Gas	Nitrogen or air (Nitrogen preferred)
Gas flow rate	< 3 l/min
Gas inlet pressure	3.5 bar
HPLC flow rate	HPLC nebulizer: 0.2 – 2.5 ml/min (other nebulizers for different flow rates and applications available upon request)
Maintenance	Easily accessible from the front for cleaning
Heated zone	
Temperature range	Ambient to 100 °C
Communication	
Gain	1 to 12 – factor 2^{11} (2 048)
Filter	Moving average (0 – 10 s)
Analog outputs	0 – 1 V
Analog control input	Contact closure, TTL for ready, autozero, power down
Control	RS-232
Power-down methods	Shut-off: gas, LED, heating and/or PMT cleaning mode
Technical parameters	
Display	LCD and keypad
General	
Power supply	230 V/50 Hz, 1.7 A – 115 V/60 Hz, 1.8 A
Dimensions	250 × 480 × 550 mm (W × H × D)
Weight	16 kg

Key features

- Long-life LED
- Attractively priced
- Wide application range
- Large choice of nebulizers

KNAUER offers various software control options:
www.knauer.net/software

Further information:
www.knauer.net/ELSD detectors

Note: This product is only available in Germany.

Ordering details

Device	
A0754-1	Sensitive Light scattering detector ELSD 85LT for univ. detection 0.2 – 2.5 ml/min, 100 Hz including accessories
A0754-3	High sensitive ELSD 90LT for univ. detection for HPLC and UHPLC, low temp. technology, supports high data rates
A0754-5	Light scattering detector ELSD SEDEX LC for universal detection 200 µl/min – 2 ml/min
A0754-6	Ultra high sensitive light scattering detector ELSD SEDEX 100LT for univ. detection 200 µl/min – 2 ml/min 100 Hz including accessories, SAGA
A0754-152	ELSD SEDEX FP, 100 µl/min – 5 ml/min
Accessories	
A1783-4	Sedex Driver for Chromeleon 7.2 and 7.3, for Sedex 85LT / 90LT, Instrument Controller Class 3 necessary
A1783-5	Sedex Driver for Chromeleon 7.2 and 7.3, for Sedex FP / LC / 100LT, Instrument Controller Class 3 necessary
A2618-12	OpenLab® CDS EZChrom Edition driver for Sedex 80LT, 85LT, 90LT from Sedere
A2618-13	OpenLab® CDS EZChrom Edition driver for Sedex FP, LC, 100LT from Sedere
A2628-1	OpenLab® CDS driver for Sedex 80LT, 85LT, 90LT from Sedere
A2628-2	OpenLab® CDS driver for Sedex FP, LC, 100LT from Sedere

High Sensitive Conductivity Detector for Ion Chromatography CDD 10-Avp



The CDD-10-AVP is a highly sensitive conductivity detector applicable to ion chromatography or organic acid analysis. Low noise, low drift and wide dynamic range assure proven performance of the CDD-10-AVP detector. A special features is the VP key for validation.

Flow cell 0.25 µl included.

Specifications

Detection	
Detector type	Conductivity detector
Detection channels	1
Measurement range	0.01 – 52 000 µS/cm
Noise	< 4 nS/cm
Drift	< 25 nS/cm per hour
Flow cell volume	0.25 µl
Time constants	0.05, 0.1, 0.5, 1.0, 1.5, 2.0, 3.0, 6.0, 8.0, 10.0 s
Communication	
Outputs	10 mV recorder terminal, integrator

Further information:
www.knauer.net/Conductivity detector

Ordering details

Device	
A1252-1	Conductivity detector CDD-10 Avp with flow cell 0.25 µl
Accessories	
AZB00XA	AZURA® Interface Box IFU 2.1 LAN, A/D converter, 4 channels

Conductivity Monitor mikron 81

The mikron 81 is a reliable in-line conductivity monitor with a very low footprint and measures with a high linearity in the range from 0.002 to 500 mS/cm. Its cutting edge temperature sensor technology enables highly precise automated temperature correction of the conductivity measurement. The intelligent flow cell design allows for a broad flow rate regime from microliter to lower liter per minute scale. It comes pre-calibrated and ready-to-use with all accessories needed.



Common applications

- Monitoring salt gradients in FPLC systems
- Cleaning-in-place monitoring in process plants and skids
- Value-based triggering within a flow system
- Upscaling (e.g. from 1 ml/min to 1 l/min)
- Ask our support (support@knauer.net) for driver scripts to work without CDS

Further information:
www.knauer.net/Conductivity monitor

Order recommendation

ADG61GD is the ideal choice for low flow applications with ion-sensitive components in the system. It is optimized for monitoring a wide range of conductivity values from 10 µS/cm to 500 mS/cm in a small measurement volume.

ADG61GE is the ideal choice for FPLC applications with high flow and highly viscous solutions. It is designed to monitor conductivity values in the range of 2 µS/cm to 100 mS/cm at low back pressure.

Scope of delivery

ADG61GD and ADG61GE, respectively, comes with:

- Mounting bracked incl. screws for installation to an AZURA® L module (ADG6101)
- 3 m cable for direct connection to a USB-A port of a PC (part of ADG6102)
- User manual in four languages (EN, DE, FR, ES, part of ADG6102)
- Fittings and ferrules to connect either 1/16" or 1/8" OD tubing (FZG10)

Specifications

	ADG61GD	ADG61GE
Flow cell		
Flow cell type	Conductivity flow cell	Conductivity flow cell
Biocompatible	Yes	Yes
Body material	Titanium	Titanium
Capillary connection	UNF 1/4-28, flat bottom	UNF 1/4-28, flat bottom
Wetted materials	Titanium, PEEK	Titanium, PEEK
Flow cell volume	11 µl	53 µl
Max. flow rate	100 ml/min (recommended)	1 000 ml/min (recommended)
Maximum pressure	Max. 171 bar (depending on utilized ferrule and fitting)	Max. 171 bar (depending on utilized ferrule and fitting)
Back pressure	< 0.1 bar at 100 ml/min, ~1 bar at 500 ml/min (water, room temperature)	< 0.1 bar at 1 000 ml/min (water, room temperature)
Note	Color code: blue (50 cm^{-1} nominal cell constant)	Color code: orange (10 cm^{-1} nominal cell constant)
Detection		
Detector type	Conductivity monitor	Conductivity monitor
Sensor	Conductivity and temperature	Conductivity and temperature
Measurement accuracy	Conductivity: ± 2 % or ± 2 mS/cm (whatever is greater) Temperature: ± 0.2 °C	Conductivity: ± 2 % or ± 1 mS/cm (whatever is greater) Temperature: ± 0.2 °C
Measurement precision	Conductivity: ± 0.2 % or ± 0.2 mS/cm (whatever is greater) Temperature: ± 0.1 °C (determined for 1 – 250 mS/cm)	Conductivity: ± 0.2 % or ± 0.1 mS/cm (whatever is greater) Temperature: ± 0.1 °C (determined for 1 – 250 mS/cm)
Measurement range	0.010 – 300 mS/cm (linear), 0.010 – 500 mS/cm (display)	0.002 – 100 mS/cm (linear), 0.002 – 100 mS/cm (display)
Linearity	< 2 % full scale value (0.010 – 500 mS/cm)	< 2 % full scale value (0.002 – 100 mS/cm)
Maximum data rate	Conductivity: 10 Hz (variable in 1 Hz steps) Temperature: 1 Hz (fixed)	Conductivity: 10 Hz (variable in 1 Hz steps) Temperature: 1 Hz (fixed)

Communication

Digital inputs	Via hyperterminal
Digital outputs	Via hyperterminal
Digital control and output	Via PurityChrom or ClarityChrom

Technical parameters

Special features	Free of charge calibration software
GLP	Serial number Firmware version Number of switching cycles Operating hours Date of last maintenance by customer service Date of last validity check
Conformity	CE, UKCA For wetted parts: EN 10204-3.1, USP Class VI, ADI-free
Display	None
Ambient conditions	Operating temperature: 3 – 45 °C, 37.4 – 113 °F Relative humidity: 0 – 90 %, non condensing

General

Power supply	Max. 5 V via USB connection (max. 500 mA, max. 2.5 W power uptake)
Dimensions	32 × 83 mm (diameter × length)
Weight	95 g (monitor + flow cell)

Other

Optional accessories	Adapters to connect 3/16″ or 1/4″ OD tubing

Ordering details

Device

ADG61GD	Conductivity monitor mikron 81 with biocompatible flow cell for up to 100 ml/min
ADG61GE	Conductivity monitor mikron 81 with biocompatible flow cell for up to 1 000 ml/min
ADG61	Conductivity monitor mikron 81 main unit without flow cell

Accessories and spare parts

ADG6103	Gasket for mikron 81 for liquid-tight connection of monitor unit and flow cell
AMN90	Biocompatible flow cell for mikron 81 for up to 100 ml/min
AMO90	Biocompatible flow cell for mikron 81 for up to for up to 1 000 ml/min

AZURA® pH Monitor pH 2.1S

The AZURA® pH 2.1S is a reliable pH monitor which is usually utilized in FPLC to follow buffer gradients.

Please combine an external flow cell according to the systems tubing and flow rate.

The flow cell needs to be ordered separately.



Further information:
www.knauer.net/Other detectors

Specifications

Flow cell

Flow cell type	External pH flow cell
Biocompatible	Yes
Connection of flow cell	1/4-28 UNF or 5/16-24 UNF dependent on flow cell
Capillary connection	1/16", 1/8" and 3/16" dependent on flow cell
Wetted materials	PEEK

Detection

Detector type	pH monitor
Measurement accuracy	± 0.5 pH (within 4 – 25 °C)
Measurement precision	± 0.2 pH (within 4 – 25 °C)
Measurement range	2 – 12
Maximum data rate	5 Hz
Supported electrodes	All pH electrodes with BNC connector and a voltage output of maximal ± 500 mV

Communication

Digital outputs	LAN, RS-232

Technical parameters

GLP	Electronic serial number
Display	LCD, 2 × 8 characters
Ambient conditions	Operating temperature: 4 – 40 °C, 39.2 – 104 °F Relative humidity: below 90 %, non condensing

General

Power supply	100 – 240 V, 50 – 60 Hz, max. 20 W
Dimensions	121 × 129 × 187 mm (W × H × D)
Weight	3.2 kg

Ordering details

Device

ADG31	AZURA® pH Monitor pH 2.1S

Accessories

A70091-2	AZURA® pH kit, 100 ml/min, incl. pH electrode, storage vessel, dummy electrode and flow cell
A70091-3	AZURA® pH kit, 1 000 ml/min, incl. pH electrode, storage vessel, dummy electrode and flow cell
A1943	AZURA® pH flow cell, 100 ml/min
A1946	AZURA® pH flow cell, 1 000 ml/min
A1942-1	AZURA® pH dummy electrode
A1947	AZURA® storage vessel for pH electrode
A1933-3	KNAUER pH electrode for AZURA® VWD 2.1L, pH 2.1S and CM 2.1S
A9854-1	Mounting bracket AZURA® L Bio for manual KNAUER injection valve, pH-flow cell and a prepacked column
A9854-3	Mounting bracket AZURA® L for AZURA® Valve Unifier VU 4.1, AZURA® Conductivity monitor CM 2.1S, AZURA® Degasser DG 2.1S or AZURA® UV detector UVD 2.1S on AZURA® L devices

AZURA® Conductivity Monitor CM 2.1S

The AZURA® CM 2.1S is a reliable conductivity monitor which is usually utilized in FPLC to follow salt gradients. By adding a pH sensor also pH values can be measured.

The contactless measurements of conductivity reduces the risk of carryover to a minimum and makes the device easy to maintain. The fully biocompatible flow cells cover a flow rate of up to 100 ml/min. Choose between ADG30GC and ADG30GD for a CM 2.1S with ready to measure flow cells for either 10 or 100 ml/min maximum flow.



KNAUER offers various software control options:
www.knauer.net/software

Further information:
www.knauer.net/Conductivity monitor

Specifications

Flow cell

Flow cell type	Contactless conductivity flow cell
Biocompatible	Yes
Connection of flow cell	Female 10-32" UNF or M8x1 thread (PEEK), both included in shipment
Capillary connection	1/16" or 1/8" - both included in shipment
Wetted materials	PEEK

Flow cell features by device

Device order number	**ADG30GC**	**ADG30GD**
Flow cell volume	30 µl	300 µl
Max. flow rate	10 ml/min	100 ml/min
Maximum pressure	160 bar	100 bar

Detection

Measurement accuracy	Conductivity: < 5 % full scale end value Temperature: ± 1.0 °C pH: ± 0.5 pH (within 4-25 °C)
Measurement precision	Conductivity: < 2 % of end value or ≤ 5 mS/cm of higher values (measured within 0.1-300 mS/cm; pH: ± 0.2 pH (within 4-25 °C)
Measurement range	0.1-999 mS/cm
pH measurement	2-12
Maximum data rate	5 Hz
Supported electrodes	All pH electrodes with BNC connector and compatible flow cell

Communication

Analog outputs	2 channels (conductivity and pH value - not active if remote controlled by software) DAC 18 bit
Digital outputs	LAN, RS-232

Technical parameters

GLP	Electronic serial number
Display	LCD, 2 × 8 characters
Ambient conditions	Operating temperature: 4-40 °C, 39.2-104 °F Relative humidity: below 90 %, non condensing

General

Power supply	100-240 V, 50-60 Hz, max. 20 W
Dimensions	121 × 129 × 187 mm (W × H × D)
Weight	3.2 kg

Ordering details

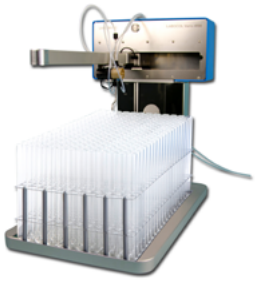
Device	
ADG30GC	AZURA® CM 2.1S with flow cell - up to 10 ml/min - conductivity monitor with optional pH measurement
ADG30GD	AZURA® CM 2.1S with flow cell - up to 100 ml/min - conductivity monitor with optional pH measurement

Accessories

A4156	Flow cell CM 2.1S for flow rates up to 10 ml/min
A4157	Flow cell CM 2.1S for flow rates up to 100 ml/min
A70091-2	AZURA® pH kit, 100 ml/min, incl. pH electrode, storage vessel, dummy electrode and flow cell
A70091-3	AZURA® pH kit, 1 000 ml/min, incl. pH electrode, storage vessel, dummy electrode and flow cell
A5813	Flow splitter for CM 2.1S when used with flow rates over 100 ml/min
A9854-3	Mounting bracket AZURA® L for AZURA® UVD 2.1S and AZURA® CM 2.1S (left-sided on AZURA® L)
A1943	AZURA® pH flow cell, 100 ml/min
A1946	AZURA® pH flow cell, 1 000 ml/min
A1942-1	AZURA® pH dummy electrode
A1947	AZURA® storage vessel for pH electrode
A1933-3	KNAUER pH electrode for AZURA® VWD 2.1L, pH 2.1S and CM 2.1S

Fraction Collector LABOCOL Vario-4000

The LABOCOL Vario-4000 fraction collectors are characterized by their high robustness and optimal ratio of dimensions/benefit. The user is not limited to given rack types. The rack layout can be designed according to individual needs. Freely define the number of fraction vessels and their position. The wide application area make the Vario-4000 series ideal for use in research and development as well as in production. The Vario-4000 models differ in the base area and the flow rate range.



KNAUER offers various software control options:
www.knauer.net/software

For purification accessories,
see p. 89

Further information:
www.knauer.net/FC LABOCOL

Specifications

Fraction collection

Brand	LABOCOL Vario-4000
Max. flow rate	100 ml/min for 1/16"; 500 ml/min for 1/8"
Fraction capacity	Consider list of racks in accessories below
Wetted materials	Stainless steel, PEEK and PTFE
Number of racks	3 (Vario-4000) / 5 (Vario-4000 Plus)
Capillary connection	1/16": 100 ml/min 1/8": 500 ml/min 1/4": 1 000 ml/min

Communication

Control	LAN, RS-232

Technical parameters

Ambient conditions	0 – 40 °C, 32 – 104 °F

General

Power supply	100 – 240 VAC, 50 – 60 Hz, max. 2.5 A			
Dimensions	Vario-4000 Vario-4000 Plus min. H*: 52 cm max. H*: 67 cm	30 × 50 cm (W × D) 46 × 50 cm (W × D)	Max. floor space Vario-4000 Vario-4000 Plus	 24 × 41 cm (W × D) 40 × 41 cm (W × D)
Weight	8 kg (Vario-4000) / 10 kg (Vario-4000 Plus)			

* with touch panel

Ordering details

Device

A591022	Fraction collector LABOCOL Vario-4000, for 1/16" or 1/8" tubing
A591024	Fraction collector LABOCOL Vario-4000, for 1/4" tubing
A591023	Fraction collector LABOCOL Vario-4000 Plus, for 1/16" or 1/8" tubing
A591026	Fraction collector LABOCOL Vario-4000 Plus, for 1/4" tubing

Accessories

A591029	Touchpanel for LABOCOL Vario-4000/Vario-4000 Plus
A5910221	Enclosure for LABOCOL Vario-4000/Vario-4000 Plus, customized dimensions, made of acrylic glass, 2x front doors, hole cut-outs on rear side for point suction and cable feed-through
A59130	Rack standard for 80 tubes 18 mm/max. 36 ml/15 ml Falcons for LABOCOL Vario-4000/Vario-4000 Plus
A59131	Rack micro for 125 tubes 10.5 mm/max. 9 ml for LABOCOL Vario-4000/Vario-4000 Plus
A59132	Rack prep for 20 tubes 36 mm/max. 140 or 240 ml for LABOCOL Vario-4000/Vario-4000 Plus
A59133	Rack semiprep for 39 tubes 26 mm/max 80 ml for LABOCOL Vario-4000/Vario-4000 Plus
A59134	Rack for 24 Falcon® tubes of 50 ml for LABOCOL Vario-4000/Vario-4000 Plus
A20521	Micro test tubes, 9 ml, 100 pcs, L 150 mm, OD 10.5 mm for rack A59131
A20522	Preparative tubes, 25 pcs, L 284 mm × OD 36 mm, V 240 ml for rack A59132

Fraction Collector Foxy® R1/R2

The Foxy® R1 fraction collector can be adapted to a broad spectrum of applications. Fractions can be collected into 96 well microplates, standard tube sizes, and bottles. For essentially unlimited volumes, funnel racks can direct fluids to any collection vessel or downstream process.



KNAUER offers various software control options:
www.knauer.net/software

For purification accessories, see p. 89

Further information:
www.knauer.net/FC Foxy

Specifications

Fraction collection

Brand	Foxy R1
Fractionation modes	Drop counting, time intervals, volume intervals, level
Max. flow rate	25 ml/min or 125 ml/min
Fraction capacity	Consider list of racks in accessories below
Diverter valve	Drop former (NC): 110 µl waste (NO): 130 µl
Wetted materials	Valve: PEEK and perfluoroelastomer (FFKM), Supplied ferrules: ETFE, Supplied valve tubing: PTFE Supplied drain tubing: vinyl
Fractionation control	Operator: front panel control via touch screen LCD integrated systems: direct communication via Ethernet (TCP/IP) and RS-232 serial communications
Maximum test tube height	160 mm
RFID rack recognition	No
Number of racks	1
Capillary connection	1/16": 25 ml/min 1/8": 125 ml/min 1/4": 1 000 ml/min (Foxy R2 only)

Communication

Control	LAN, RS-232

Technical parameters

Conformity	CE, CSA
Display	Touch screen LCD displays
Ambient conditions	0 – 40 °C, 32 – 104 °F

General

Power supply	100 – 240 V AC, 50 – 60 Hz, max. 1 A
Dimensions	R1: 311 × 330 × 355 mm (W × D × H) R2 1/8": 311 × 533 × 378 mm (W × D × H) R2 1/4": 311 × 533 × 394 mm (W × D × H)
Weight	R1: 7.1 kg R2 1/8": 10.3 kg R2 1/4": 10.4 kg

Ordering details

Device

A59100	Fraction collector Foxy® R1 for 1/16" or 1/8" tubing
A59102	Fraction collector Foxy® R2 for 1/16" or 1/8" tubing
A591021	Fraction collector Foxy® R2 for 1/4" tubing

Accessories

A59122	Cooling option for Foxy® R1 with cooling hood, cooling plate and accessories
A59117	Cooling rack for 144 tubes 1.5 ml for Foxy® R1*
A59118	Cooling rack for 72 Falcons 15 ml for Foxy® R1*
A59119	Cooling rack for 96-Well Microplates for Foxy® R1*
A59105	Rack for 100 vials 16 mm/max. 20 ml for Foxy® R1/R2
A59104	Rack for 144 vials 13 mm/max. 9 ml for Foxy® R1/R2
A59111	Rack for 2 microwell plates 96 well for Foxy® R1/R2
A59114	Rack for 2 × 9 bottles 480 ml for Foxy® R2 (not suitable for Foxy® R1, bottles too tall)
A59110	Rack for 36 Falcon 50 ml for Foxy® R1/R2
A59108	Rack for 36 vials 25 mm/max. 70 ml for Foxy® R1/R2
A59107	Rack for 60 tubes 1.5 ml for Foxy® R1/R2
A59106	Rack for 72 Falcons 15 ml for Foxy® R1/R2
A59109	Rack with 36 funnels with vinyl tubing for Foxy® R1/R2
A591092	Scintillation rack for 36 vials 28 mm for Foxy® R1/R2
A70055	Thermostatting unit -20 to 40 °C
A70050	Thermostatting unit -40 to 200 °C

* for Foxy R1 with cooling option

AZURA® Fraction Collector FC 6.1

NEW

The new FC 6.1 is a small, versatile fraction collector designed for FPLC and HPLC applications. It can be used for flow rates from 0.1 - 250 ml/min. With its easy-to-change drop formers, the FC 6.1 can be quickly equipped for biocompatibility. It offers space for a main rack and a small side rack, which are accessible from three sides and the top. Thanks to height adjustability, fractionation in bottles up to 1l is also possible. The FC 6.1 can be placed on the KNAUER AZURA® devices and is supported in PurityChrom® 6, Mobile Control and ClarityChrom.



KNAUER offers various software control options:
www.knauer.net/software

For purification accessories, **see p. 89**

Further information:
www.knauer.net/FC 6.1

Specifications

Fraction collection

Brand	AZURA® FC 6.1
Fractionation modes	Drop counting, time intervals, volume intervals, fill level
Max. flow rate	Best working up to 100 ml/min, max. 250 ml/min
Diverter valve	Valve: 50 µl Drop former (inner diameter 0.75 mm, length 22 mm): 10 µl
Wetted materials	Valve: ▪ AFA00: PPS, EPDM ▪ AFA01: PEEK, FFKM Supplied ferules: ETFE Supplied valve and drain tubing: ETFE or FEP
Number of racks	1 main rack, 1 side rack
Rack type	Main rack, side rack (included in FC 6.1): 4 round bottles 100 ml each 3 round/square bottles 250 ml each
RFID rack recognition	No
Capillary connection	1/16": 100 ml/min 1/8": 250 ml/min UNF 1/4" - 28, flat botton
Leak sensor	No

Communication

Control	LAN, 3-way pin header plug (Analog IN, OUT, GROUND)

Technical parameters

Ambient conditions	4 – 40 °C (39.2 – 104 °F), below 90 %, non-condensing

General

Power supply	100 – 240 V AC, 50 – 60 Hz, max. 1.5 A
Max. power consumption	100 W
Dimensions	Ground plate: approx. 36 × 42 cm Height without tubing: 35 cm Height with tubing: approx. 45 cm Swivel range: 42 cm of the ground plate plus 15 cm on both sides
Weight	5.06 kg
Software	PurityChrom, Mobile Control, ClarityChrom

Ordering details

Device	
AFA00	AZURA® FC 6.1 BIO Fraction Collector for FPLC Applications, aqueous eluents, not resistant to acetonitrile
AFA01	AZURA® FC 6.1 Fraction Collector for HPLC applications/solvents
Accessories	
AFR01	Rack for 165 tubes (11 mm), 1 / 1.5 / 2 ml each
AFR02	Rack for 99 tubes (17 mm), 15 ml each
AFR03	Rack for 30 tubes (31 mm), 50 ml each
AFR04*	Rack for 15 round bottles (56 mm), 100 ml each
AFR05*	Rack for 8 round (72 mm) or square (65 mm) bottles, 250 ml each

* no use of side racks

AZURA® Degasser DG 2.1S

Dissolved gases in the solvent can cause bubbles in the flow path of pumps and detectors. Reliable chromatographic separation therefore requires degassing of the solvent. The small analytical 2-channel degasser DG 2.1S is equipped with two degassing chambers and can thus degas two solvents simultaneously.



KNAUER offers various software control options:
www.knauer.net/software

Specifications

Degasser module

Degasser channels	2
Max. flow rate/channel	10 ml/min
Recommended flow rate/channel	2.8 ml/min
Degassing method	Gas permeation through a fluoropolymere membrane
Degassing chamber volume	285 µl
Solvent applicability	Universal, except hydrochloric acid, halogenated hydrocarbons, hexafluoro isopropanol (HFIP)
Wetted materials	PTFE, PPS, PEEK, Systec AF™
Pressure decline	1.37 mm (Hg/ml/min)
Maximum pressure stability	70 psi

Technical parameters

Display	1 LED
Ambient conditions	Temperature range: 4 – 40 °C, 39.2 – 104 °F Relative humidity: below 90 %, non-condensing

General

Power supply	85 – 265 V, 50 – 60 Hz, 20 W
Dimensions	121 × 138 × 190 mm (W × H × D)
Weight	2.3 kg
Connector	1/4”-28 UNF female port

Feature overview

Order no.	Degasser type	Channels	Max. flow rate	Chamber volume
AZE02	analytical	2	10 ml/min (recommended 2 ml/min)	285 µl per channel
AZE03-1	analytical	4	10 ml/min (recommended 2 ml/min)	285 µl per channel
A5335	analytical, for GPC	2	10 ml/min (recommended 3 ml/min)	480 µl per channel
A5328	semi-preparative	2	30 ml/min (recommended 15 ml/min)	5.3 ml per channel
AZE02-1	preparative	2	200 ml/min (recommended 75 ml/min)	23 ml per channel
AZE03	preparative	4	200 ml/min (recommended 75 ml/min)	23 ml per channel

Ordering details

Device

AZE02	Biocompatible 2 channel degasser
AZE03-1	Analytical 4 channel degasser, biocompatible
A5335	Analytical 2 channel GPC degasser
A5328	Semi-preparative 2 channel degasser
AZE02-1	Preparative 2 channel degasser, biocompatible
AZE03	Preparative 4 channel degasser, biocompatible

AZURA® Valve Unifier VU 4.1

The valve drive AZURA® Valve Unifier VU 4.1 enables automatic valve switching. Due to its low switching time, the flow path is blocked only for a very short time, and pressure peaks are reduced to a minimum. Valves are identified via RFID technology, which guarantees an easy valve exchange of KNAUER valves. An additional feature is the easy monitoring of GLP data, which simplifies maintenance such as the exchange of a rotor seal. The display enables user-friendly standalone operation. In addition, the valve drive can be operated with software as well with an optional touch display (Mobile Control), via LAN or analog input/output, by which it can be integrated into nearly every LC system.



Specifications

Communication

Interfaces	LAN, display, terminal strip
Control	Display, software, event control
Inputs	Binary control; Home, Backward/Inject, Forward/Load, Start (IN)
Outputs	Trigger out, Event

General

Power supply	External DC 24 V, 65 W
Dimensions	80 × 123 × 192 mm (W × H × D)
Weight	1.9 kg
Ambient conditions	Temperature range: 4 – 40 °C; 39.2 – 104 °F, below 90 % humidity (non condensing)

Key features

- One valve drive for all valves
- Ultra fast switching cycle
- Easy maintenance
- Compact
- Multiple interfaces and drivers available

KNAUER offers various software control options:
www.knauer.net/software

For valve accessories, **see p. 61**

Further information:
www.knauer.net/VU 4.1

Further information:
www.knauer.net/valves



Valve drive VU 4.1 (AWA01XA) with 6 port 2-position valve (AVC28AC)



AVS34CE AVN94CE AVR38AC AVC38AC AVS62CE

Ordering details

Device

AWA01XA	VU 4.1 valve drive for V 4.1 valves

Accessories

A9854-3	Mounting bracket AZURA® L for AZURA® Valve Unifier VU 4.1 (both-sided) or AZURA® UVD 2.1S and AZURA® CM 2.1S (left-sided on AZURA® L)
A9854-4	Double Mounting bracket AZURA® L for two AZURA® Valve Unifier VU 4.1 (both-sided)

Valves for Valve Unifier VU 4.1



AVJ26AE



AVG24CE



AVK25AE



AVJ36AE

Manual valves*

Ports	Stator material	Rotor material	Max. pressure [bar]	Bore size [mm]	Capillary connection	Order number
6	SST DLC[1]	POM	100	0.75	1/16″ (UNF 10-32)	AVJ23AF
6	SST DLC[1]	PEEK	500	0.75	1/16″ (UNF 10-32)	AVJ26AE
6	SST DLC[1]	Vespel®	1 200	0.3	1/16″ (UNF 10-32)	AVI28AC
6	PEEK	PEEK	240	0.75	1/16″ (UNF 10-32)	AVG24CE
6	SST DLC[1]	PEEK	300	1.5	1/8″ (UNF 1/4-28, coned)	AVK25AE
6	PEEK	PEEK	100	2.0	1/8″ (UNF 1/4-28, coned)	AVL23CE
8	SST DLC	PEEK	500	0.75	1/16″ (UNF 10-32)	AVJ36AE
8	SST DLC[1]	Vespel®	1 200	0.3	1/16″ (UNF 10-32)	AVI38AC

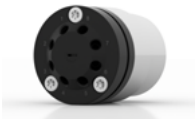
* The mounting bracket A9853 is required to mount the manual valves to an AZURA® L device.
[1] Stainless steel coated with diamond-like carbon



AVD26AE



AVD24CE



AVD36AE

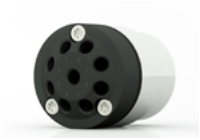


AVF32CE

2-position valves

Ports	Stator material	Rotor material	Max. pressure [bar]	Bore size [mm]	Capillary connection	Order number
6	SST DLC[1]	POM	100	0.75	1/16″ (UNF 10-32)	AVD23AF
6	SST DLC[1]	PEEK	500	0.75	1/16″ (UNF 10-32)	AVD26AE
6	SST DLC[1]	PEEK	500	0.75	1/16″ (UNF 10-32)	AVD26AH*
6	SST DLC[1]	Vespel®	1 200	0.3	1/16″ (UNF 10-32)	AVC28AC
6	PEEK	PEEK	240	0.75	1/16″ (UNF 10-32)	AVD24CE
6	SST DLC[1]	PEEK	300	1.5	1/8″ (UNF 1/4-28, coned)	AVE25AE
6	SST DLC[1]	PEEK	300	1.5	1/8″ (UNF 1/4-28, coned)	AVE25AI**
6	PEEK	PEEK	100	2.0	1/8″ (UNF 1/4-28, coned)	AVF23CE
8	SST DLC[1]	PEEK	500	0.75	1/16″ (UNF 10-32)	AVD36AE
8	SST DLC[1]	Vespel®	1 200	0.3	1/16″ (UNF 10-32)	AVC38AC
8	PEEK	PEEK	50	2.0	1/8″ (UNF 1/4-28, coned)	AVF32CE
10	SST DLC[1]	Vespel®	1 200	0.3	1/16″ (UNF 10-32)	AVC48AC
10	PEEK	PEEK	100	0.75	1/16″ (UNF 10-32)	AVD43CE

* Break-free valve design.
**Special version of AVE25AE with 2-channel rotor seal instead of 3 channels.
[1] Stainless steel coated with diamond-like carbon



AVT84AH



AVS26AE



AVS62CE



AVU32CE

Multiposition valves

Ports	Stator material	Rotor material	Max. pressure [bar]	Bore size [mm]	Capillary connection	Order number
2	SST DLC[1]	PEEK	300	0.75	1/16″ (UNF 10-32)	AVS85AH*
2	SST DLC[1]	PEEK	200	1.5	1/8″ (UNF 1/4-28 coned)	AVT84AH*
2	SST DLC[1]	PEEK	100	2.3	1/4-28″ (UNF 1/8 coned)	AVU83AH
4	PEEK	PEEK	20	3.0	3/16″ (UNF 5/16-24 flat bottom)	AVW01GE
4	PEEK	Vespel	50	3.0	3/16″ (UNF 5/16-24 flat bottom)	AVW01GC
6	SST DLC[1]	POM	100	0.75	1/16″ (UNF 10-32)	AVS23AF
6	SST DLC[1]	PEEK	500	0.75	1/16″ (UNF 10-32)	AVS26AE
6	SST DLC[1]	Vespel	1 200	0.3	1/16″ (UNF 10-32)	AVR28AC
6	SST DLC[1]	PEEK	300	1.5	1/8″ (UNF 1/4-28 coned)	AVT25AE
6	SST DLC[1]	Vespel	300	0.75	1/16″ (UNF 10-32)	AVS25AC
8	SST DLC[1]	PEEK	300	0.75	1/16″ (UNF 10-32)	AVS35AE
8	SST DLC[1]	PEEK	500	0.75	1/16″ (UNF 10-32)	AVS36AE
8	SST DLC[1]	Vespel	1 200	0.3	1/16″ (UNF 10-32)	AVR38AC
8	PEEK	PEEK	240	0.75	1/16″ (UNF 10-32)	AVS34CH* NEW
8	PEEK	PEEK	240	0.75	1/16″ (UNF 10-32)	AVS34CE
8	SST DLC[1]	PEEK	300	0.75	1/16″ (UNF 10-32)	AVS35AH*
8	SST DLC[1]	PEEK	200	1.5	1/8″ (UNF 1/4-28 coned)	AVT34AE
8	SST DLC[1]	PEEK	200	1.5	1/8″ (UNF 1/4-28 coned)	AVT34AH*
8	PEEK	PEEK	50	2.0	1/8″ (UNF 1/4-28, flat-bottom)	AVU32GE
8	PEEK	PEEK	50	2.0	1/8″ (UNF 1/4-28 coned)	AVU32CE
8	SST DLC[1]	Vespel	200	1.5	1/4-28″ (UNF 1/8 coned)	AVT34AC
12	SST DLC[1]	PEEK	100	1.5	1/8″ (UNF 1/4-28 coned)	AVT53AE
12	PEEK	PEEK	100	1.5	1/8″ (UNF 1/4-28 coned)	AVT53CE
16	SST DLC[1]	POM	100	0.75	1/16″ (UNF 10-32)	AVQ63AF
16	SST DLC[1]	PEEK	500	0.6	1/16″ (UNF 10-32)	AVQ66AE
16	PEEK	PEEK	50	0.75	1/16″ (UNF 10-32)	AVS62CE
16	PEEK	PEEK	150	0.75	1/16″ (UNF 10-32)	AVS63CE

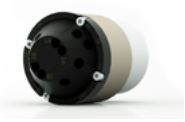
[1] Stainless steel coated with diamond-like carbon
* Break-free valve design



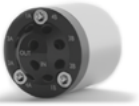
AVN94CE



AVN96AE



AVZ52CE



AVM48AC

Special purpose valves*

Valves	Capillary connection	Max. pressure [bar]	Bore size [mm]	Order number
Multi-injection valve, biocompatible. Allows manual and automated sample loading as well as direct injection.	1/16" (UNF 10-32)	240	0.75	AVN94CE
Multi-injection valve, stainless steel. Allows manual and automated sample loading as well as direct injection.	1/16" (UNF 10-32)	500	0.75	AVN96AE
Column selection valve for high pressure applications. Can be used for up to 4 columns and bypass.	1/16" (UNF 10-32)	1200	0.2	AVM48AC
Biocompatible column selection valve. Allows for switching between 7 columns and 1 bypass with reverse flow option.	1/16" (UNF 10-32)	200	0.75	AVN64CE
Symmetrical 2-Pos. Valve for column switching and backflushing or 2D LC. DLC coated SSt stator and Vespel rotor seal.	1/16" (UNF 10-32)	1200	0.3	AVM38AJ

* For detailed information please check our website: **www.knauer.net/valves**

K-7400S Semi-Micro Osmometer

KNAUER is the pioneer in the field of osmometry and known for its reliable and user friendly instruments for many decades.

Our freezing point osmometer K-7400S allows the easy and fast determination of the osmolality of various aqueous solutions. Also, the freezing point depression of the samples can be read. The proven technology of freezing point determination in combination with the robust and intelligent design of the device allows reproducible measurements.

The instrument is equipped with a peltier cooler and an integrated microprocessor controlling the automated measurement. The freezing point osmometer is a standalone device that optionally can be equipped with a printer. Furthermore, the device can be controlled via the EuroOsmo 7400 software. The software automatically plots the temperature curve for each measurement and calibration and allows saving of the measured values. In addition, the data can optionally be exported into various file formats for archival storage.

The measurement specifications of the KNAUER Semi-Micro Osmometer K-7400S complies with the European Pharmacopoeia for osmolality (Ph. Eur. 2.2.35, 10/2021).



Key features

- Made in Germany
- 60 years experience
- Fast measurements

For more osmometry accessories and standards, **see p. 117**

Further information:
www.knauer.net/osmometry

Specifications

Technical parameters

Sample volume	50 or 150 µl
Osmolality range	0 – 2 000 mOsmol/kg
Resolution	Osmolality: integer value without decimal part, e.g. 850 mOsmol Temperature: value with three digits, e.g. – 1 576 °C
Test time	~ 2 min
Precision	SD ≤ 4 mOsmol/kg [0 – 400 mOsmol/kg] RSD ≤ 1 % [400 – 2 000 mOsmol/kg]
Linearity	± 1 % [0 – 1 500 mOsmol/kg] ± 1.5 % [0 – 2 000 mOsmol/kg]
Calibration	Two-point calibration (0 mOsmol/kg and one freely selectable osmolality) Optional: Three-point calibration (0 Osmol/kg and two freely selectable osmolalities)

General

Power supply	100 – 240 V, 50 – 60 Hz, 70 W
Dimensions	160 × 182 × 340 mm (W × H × D)
Weight	5.3 kg
Ambient conditions	Temperature range: 10 – 35 °C Relative humidity: 20 – 80 % (non-condensing)

Communication

Interfaces	RS-232 port
Control	Keypad (LED display, 2 rows with 24 characters) optional: EuroOsmo7400 Software

Ordering details

Device	
A0006AC	Osmometer for the determination of osmolality or freezing point of aqueous solutions, including calibration standards (400 & 850 mOsmol/kg) and sample tubes (500 pcs.)
Accessories	
A0840-2	Measuring head 150 µl for plastic sample tubes; compatible with the K-7400 and the K-7400S Semi-Micro Osmometer
A0840-4	Measuring head 50 µl for plastic sample tubes; compatible with the K-7400 and the K-7400S Semi-Micro Osmometer
A3705	EuroOsmo 7400 – software for K-7400 and K-7400S Osmometer
A3711	Plain paper printer for freezing point osmometer K-7400 and K-7400S
A13270	Barcode scanner with USB cable, for EuroOsmo 7400
A0272	500 pack of plastic sample tubes for Semi-Micro Osmometer K-7400S

ionBench LC and ionBench MS for AZURA® HPLC and MS systems

NEW

The ionBench LC and ionBench MS are mobile benches for liquid chromatography systems or MS instruments. The benches are ready to use and a compact solution to increase the flexibility of your laboratory. They simplify moving the systems and ensure easy access to the back of the instruments.

The ionBench LC is height-adjustable and thus allows for easy access to the solvent bottles and facilitates the change of these. A slot at the front of the bench keeps waste lines organized. When placed close to a mass spectrometer the dead volume between the LC and the MS can be minimized.

The ionBench MS includes an integrated noise reduction enclosure for the mass spectrometer's vacuum pump, featuring a vibration-dampening system and an overheating temperature alarm. The noise reduction enclosure can also accommodate a nitrogen generator, e.g. for AZURA® HPLC systems with an ELSD detector.



Key features ionBench LC (A70001)

- Hight adjustable
- Chemical resistant
- Minimize dead volume

Key features ionBench MS (A70002)

- Makes your lab quieter
- Vibration free
- Chemically resistant work surface

Further information:
www.knauer.net/A70001

Further information:
www.knauer.net/A70002

Specifications

Technical parameters

	ionBench LC	ionBench MS
Work surface	67 × 65 cm	90 × 88 cm
Hight adjustment	56–86 cm, controlled by a control module with 3 positions memorization	-
Weight capacity	max. 250 kg	max. 500 kg
Noise reduction	-	- 15 db(A) (80 % reduction in noise perception)
Dampening system	-	Vibration dampening platform prevents vibration from being transmitted from the vacuum pump to the mass spec. Reduces vibrations by 99 %.
Overheating temperature alarm	-	Audible (continuous alarm at ~ 90 dB) and visual (red LED)
Accessories (incl. in scope of delivery)	IEC power strip, fixed underneath bench work surface	Power Strip Germany, 8 outlets, fixed behind the bench
Power supply	-	110 / 220 V, 50/60 Hz
Dimensions	670 × 560–860 × 650 mm (W × H × D)	900 × 860 × 880 mm (W x H x D) Noise enclosure: 615 × 545 × 680 mm (W × H × D)
Weight	30 kg	102 kg
Other	Tubing path – Front slot for the solvent waste line(s)	-

Ordering details

Device

A70001	ionBench LC for AZURA Analytical HPLC Systems
A70002	ionBench MS, with integrated noise enclosure for vacuum pump

AZURA® SmartClick



The smart, space-saving way to organize and integrate your AZURA® L system.

The **AZURA® SmartClick** is the innovative solution for optimizing HPLC/FPLC system integration. With its modular, click-on design, SmartClick lets you attach chromatography columns, air sensors, external pressure sensors, and a wide range of accessories directly to your system – turning separate parts into a single, streamlined solution.

SmartClick supports a variety of modules, including holders for laboratory vessels, small reagent bottles, pipette holders, and monitoring devices. Everything you need is securely and conveniently positioned exactly where you need it, optimizing workflows, minimising potential error sources, and enhancing safety.

SmartClick can be used exclusively with AZURA® L devices. A maximum of two SmartClick Rails can be attached to SmartClick Edges. The number of modules on a rail depends on the module size. Find even more modules at Better Basics Laborbedarf GmbH.

AZURA® SmartClick Adapter adapts the following devices to SmartClick:

- Interface box
- Pressure control
- AZURA® pH sensor
- AZURA® Organizer
- Distribution box

Key features

- Modular, click-on design: Accessories right at your system
- LC system solution: Turning separate parts into a single, streamlined solution
- Optimizing workflows: Everything you need is exactly where you need it
- Enhancing safety: Secure and convenient attachments

Further information:
www.knauer.net/SmartClick

Ordering details

A700881	AZURA® SmartClick Start-up kit – includes one pair of SmartClick Edge and one SmartClick Rail
A700882	AZURA® SmartClick Rail
A700883	AZURA® SmartClick Edge
A700884	AZURA® SmartClick Adapter
A7008841	AZURA® SmartClick Holder for VU 4.1 Valve Drive
A7008842	AZURA® SmartClick Tubing/Capillary Guide Clamp
A7008843	AZURA® SmartClick Hook
A700885	AZURA® SmartClick Holder for laboratory threaded bottles – GL45
A700886	AZURA® SmartClick Syringe Filter Box
A700887	AZURA® SmartClick Holder for pipettes with hook
A700888	AZURA® SmartClick Storage Box
A7008881	AZURA® SmartClick Storage Box Insert – for centrifuge tubes 6 × 15 ml, 4 × 50 ml
A7008882	AZURA® SmartClick Storage Box Insert – for microreaction vessels 20 × 2 ml
A7008891	AZURA® SmartClick Holder for chromatography column Ø 5 – 10 mm
A7008892	AZURA® SmartClick Holder for chromatography column Ø 10 – 15 mm
A7008893	AZURA® SmartClick Holder for chromatography column Ø 15 – 25 mm
A7008894	AZURA® SmartClick Holder for chromatography column Ø 25 – 35 mm
A7008895	AZURA® SmartClick Holder for chromatography column Ø 35 – 46 mm
A7008896	AZURA® SmartClick Holder for chromatography column Ø 46 – 56 mm
A70088961	AZURA® SmartClick Holder for chromatography column Ø 56 – 66 mm

Maintenance kits

Each maintenance kit contains all parts that are to be replaced according to the maintenance plan. Included parts are wear parts.

Pump and pump head maintenance

Maintenance kits for AZURA® Pumps

Maintenance kit for pump P 2.1L, 80P – incl. all wear parts: tooth belts, silicon tube	ARP00
Maintenance kit for pump P 2.1S, P 4.1S, P 4.2S, 40P – incl. all wear parts: tooth belt, silicon tube	ARP10
Maintenance kit for pump P 6.1L – isocratic LPG; 40P – including all wear parts: tooth belt, silicon tube, filter cartridge	ARP20
Maintenance kit for pump P 6.1L – HPG – including all wear parts: tooth belt, silicon tube, filter cartridge	ARP21
Maintenance kit for pump P 6.1L – isocratic, LPG; 40P – metal free – including all wear parts: tooth belt, silicon tube, filter cartridge	ARP22
Maintenance kit for pump P 6.1L – HPG – metal free – including all wear parts: tooth belt, silicon tube, filter cartridge	ARP23
Maintenance kit for pump P 8.1L – incl. all wear parts: tooth belt, filter cartridge, rotor seal	ARP31

Maintenance kits for mixing chambers

Maintenance kit for Dynamic Mixer 1/16" SST – incl. all wear parts: sieves, gaskets	ARM01
Maintenance kit for Dynamic Mixer 1/16" Titanium – incl. all wear parts: sieves, gaskets	ARM02
Maintenance kit for Dynamic Mixer 1/8" SST, Titanium – incl. all wear parts: gaskets	ARM03


A9670


A9671


A9672


A9673

Tool kits for pump head maintenance

Maintenance tool kit for 10 ml pump heads	A9670
Maintenance tool kit for 50 ml pump heads	A9671
Maintenance tool kit for 100 ml pump heads	A9672
Maintenance tool kit for 250 ml pump heads	A9673
Maintenance tool kit for 500 ml pump heads	A9674
Maintenance tool kit for 1 000 ml pump heads	A9675



ARH30



ARH38



ARH40



ARH48

Maintenance kits for AZURA® Pump heads

Maintenance kit for pump head 5 ml (P 8.1L) – AHA70 – incl. all wear parts: check valves, piston rods, gaskets, O-rings, backing rings	ARH20
Maintenance kit for pump head 5 ml/10 ml – AHA60, AHB32, AHB32DA, AHB40, AHB40CA, AHB40CB, AHB40CB – incl. all wear parts: check valves, piston rods, gaskets, O-rings, backing rings, compression springs	ARH30
Maintenance kit for pump head 10 ml – AHB40BA – incl. all wear parts: check valves, piston rods, gaskets, O-rings, backing rings, compression springs	ARH31
Maintenance kit for pump head 10 ml – AHB43 – incl. all wear parts: check valves, piston rods, gaskets, O-rings, backing rings, compression springs	ARH32
Maintenance kit for pump head 10 ml – AHB40FA, AHB32GA, AHB32FA – incl. all wear parts: check valves, piston rods, gaskets, O-rings, backing rings, compression springs	ARH33
Maintenance kit for pump head 50 ml – AHC20, AHC20CA, AHC22, AHC20CB – incl. all wear parts: check valves, piston rods, gaskets, O-rings, backing rings, compression springs	ARH34
Maintenance kit for pump head 50 ml – AHC23 – incl. all wear parts: check valves, piston rods, gaskets, O-rings, backing rings, compression springs	ARH35
Maintenance kit for pump head 50 ml – AHC20FA – incl. all wear parts: check valves, piston rods, gaskets, O-rings, backing rings, compression springs	ARH36
Maintenance kit for pump head 50 ml – AHC20BA – incl. all wear parts: check valves, piston rods, gaskets, O-rings, backing rings, compression springs	ARH37
Maintenance kit for pump head 50 ml – AHC22FA – incl. all wear parts: check valves, piston rods, gaskets, O-rings, backing rings, compression springs	ARH38
Maintenance kit for pump head 100 ml – A4029-1 – incl. all wear parts: check valves, piston rods, gaskets, O-rings, compression springs	ARH40
Maintenance kit for pump head 100 ml – A4023V5 – incl. all wear parts: check valves, piston rods, gaskets, O-rings, backing rings	ARH41
Maintenance kit for pump head 100 ml – A4029V2 – incl. all wear parts: check valves, piston rods, gaskets, O-rings, compression springs	ARH42
Maintenance kit for pump head 250 ml – A4021-1 – incl. all wear parts: check valves, piston rods, gaskets, O-rings, compression springs	ARH43
Maintenance kit for pump head 250 ml – A4021V2 – incl. all wear parts: check valves, piston rods, gaskets, O-rings, compression springs	ARH44
Maintenance kit for pump head 500 ml – A4038-1 – incl. all wear parts: check valves, piston rods, gaskets, O-rings, compression springs	ARH45
Maintenance kit for pump head 500 ml – A4038V2 – incl. all wear parts: check valves, piston rods, gaskets, O-rings, compression springs	ARH46
Maintenance kit for pump head 1 000 ml – A4022-1 – incl. all wear parts: check valves, piston rods, gaskets, O-rings, compression springs	ARH47
Maintenance kit for pump head 1 000 ml – A4022V2 – incl. all wear parts: check valves, piston rods, gaskets, O-rings, compression springs	ARH48

Detector maintenance

Maintenance kits for detectors

Maintenance kit for detector UVD 2.1S, UVD 2,1L, MWD 2.1L, DAD 2.1L, 10D, 40D – incl. all wear parts: deuterium lamp	ARD10
Maintenance kit for detector DAD 6.1L – incl. all wear parts: only deuterium lamp	ARD11
Maintenance kit for detector DAD 6.1L – incl. all wear parts: deuterium lamp, halogen lamp	ARD12
Maintenance kit for detector RID 2.1L – incl. all wear parts: LED	ARD20
Maintenance kit for detector 50D, VWD 2.1L – incl. all wear parts: only deuterium lamp	ARD30
Maintenance kit for detector 50D, VWD 2.1L – incl. all wear parts: deuterium lamp, halogen lamp	ARD31

Autosampler maintenance

Maintenance kits for autosamplers

Maintenance kit for Autosampler AS 6.1L, S3950 (700 bar) – incl. all wear parts: sample needle, air needle, rotor seal, syringe, syringe valve, buffer tubing, wash position	ARA10
Maintenance kit for Autosampler AS 6.1L (1240 bar), S3950 (1000 bar) – incl. all wear parts: sample needle, air needle, rotor seal, syringe, syringe valve, buffer tubing, wash position	ARA11
Maintenance kit for Autosampler S3950 – biocompatible – incl. all wear parts: sample needle, air needle, rotor seal, syringe, syringe valve, buffer tubing, wash position	ARA12
Maintenance kit for Autosampler AS 6.1L, S3950 – biocompatible, preparative – incl. all wear parts: sample needle, air needle, rotor seal, syringe, syringe valve, buffer tubing, wash position	ARA13
Maintenance kit for Autosampler AS 6.1L, S3950 – preparative – incl. all wear parts: sample needle, air needle, rotor seal, syringe, syringe valve, buffer tubing, wash position	ARA14
Maintenance kit for Autosampler AS 6.1L (862 bar) – incl. all wear parts: sample needle, air needle, rotor seal, syringe, syringe valve, buffer tubing, wash position	ARA15
Maintenance kit for Autosampler AS-1 – incl. all wear parts: sample needle, air needle, rotor seal, syringe, syringe valve, buffer tubing, wash position	ARA16
Maintenance kit for Autosampler S3950 – incl. all wear parts: sample needle, air needle, rotor seal, syringe, syringe valve, buffer tubing, wash position	ARA17
Maintenance kit for AZURA® Autosampler AS 6.1L – biocompatible – incl. all wear parts: sample needle, air needle, rotor seal, syringe, syringe valve, buffer tubing, wash position	ARA18
Maintenance kit for AZURA® Autosampler AS 8.1L up to 1 379 bar – incl. needle seat, Injection valve/rotor seal, syringe valve, ZDV Syringe 100 µl	ARA20

Valve maintenance

Maintenance kits for VICI Valves

Maintenance kit for valve A5850 – incl. all wear parts: rotor seal	ARV20
Maintenance kit for valve A5854 – incl. all wear parts: rotor seal	ARV21
Maintenance kit for valve A5858 – incl. all wear parts: rotor seal	ARV22
Maintenance kit for valve A5859 – incl. all wear parts: rotor seal	ARV23
Maintenance kit for valve A5860 – incl. all wear parts: rotor seal	ARV24
Maintenance kit for valve AVZ52CE, M6032-1 – incl. all wear parts: rotor seal	ARV25
Maintenance kit for valve EVZ34CE, M6035-1 – incl. all wear parts: rotor seal	ARV26
Maintenance kit for valve M6036 – incl. all wear parts: rotor seal	ARV27

KNAUER

Maintenance kits for KNAUER Valves

Valve article no.	Ports	Positions	Pressure [bar]	Version	Maintenance kit (rotor seal) article number
AVC28AC	6	2	1 200	for VU 4.1	ARV36
AVC38AC	8	2	1 200	for VU 4.1	ARV31
AVC48AC	10	2	1 200	for VU 4.1	ARV59
AVD23AF	6	2	100	for VU 4.1	ARV48
AVD24CE	6	2	240	for VU 4.1	ARV35
AVD26AE	6	2	500	for VU 4.1	ARV33
AVD26AH	6	2	500	for VU 4.1	ARV58
AVD36AE	8	2	500	for VU 4.1	ARV34
AVD43CE	10	2	100	for VU 4.1	ARV69
AVE25AE	6	2	300	for VU 4.1	ARV49
AVE25AI	6	2 / 2 channel	300	for VU 4.1	ARV51
AVF23CE	6	2	100	for VU 4.1	ARV50
AVF32CE	8	2	50	for VU 4.1	ARV52
AVG24CE	6	2	240	manual	ARV35
AVI28AC	6	2	1 200	manual	ARV36
AVI38AC	8	2	1 200	manual	ARV31
AVJ23AF	6	2	100	manual	ARV48
AVJ26AE	6	2	500	manual	ARV33
AVJ36AE	8	2	500	manual	ARV34
AVK25AE	6	2	300	manual	ARV49
AVL23CE	6	2	100	manual	ARV50
AVM38AJ	8	Special	1 200	for VU 4.1	ARV72
AVM48AC	10	Special	1 200	for VU 4.1	ARV60
AVN64CE	16	Special	200	for VU 4.1	ARV63
AVN94CE	8	Special	240	for VU 4.1	ARV40
AVN96AE	8	Special	500	for VU 4.1	ARV55
AVQ63AF	16	Multi	100	for VU 4.1	ARV56
AVQ66AE	16	Multi	500	for VU 4.1	ARV43
AVR28AC	6	Multi	1 200	for VU 4.1	ARV38
AVR38AC	8	Multi	1 200	for VU 4.1	ARV39
AVS23AF	6	Multi	100	for VU 4.1	ARV53
AVS25AC	6	Multi	300	for VU 4.1	ARV71
AVS26AE	6	Multi	500	for VU 4.1	ARV46
AVS34CE	8	Multi	240	for VU 4.1	ARV32
AVS34CH	8	Multi / break-free	240	for VU 4.1	ARV57
AVS35AE	8	Multi	300	for VU 4.1	ARV32
AVS35AH	8	Multi / break-free	300	for VU 4.1	ARV57
AVS36AE	8	Multi	500	for VU 4.1	ARV32
AVS62CE	16	Multi	50	for VU 4.1	ARV44
AVS63CE	16	Multi	150	for VU 4.1	ARV44
AVS85AH	2	Multi	300	for VU 4.1	ARV57
AVT25AE	6	Multi	300	for VU 4.1	ARV54
AVT34AC	8	Multi	200	for VU 4.4	ARV66
AVT34AE	8	Multi	200	for VU 4.1	ARV45
AVT34AH	8	Multi / break-free	200	for VU 4.1	ARV42

Valve article no.	Ports	Positions	Pressure [bar]	Version	Maintenance kit (rotor seal) article number
AVT53AE	12	Multi	100	for VU 4.1	ARV47
AVT53CE	12	Multi	100	for VU 4.1	ARV47
AVT84AH	2	Multi / break-free	200	for VU 4.1	ARV42
AVU32CE	8	Multi	50	for VU 4.1	ARV41
AVU32GE	8	Multi	50	for VU 4.1	ARV41
AVU83AH	2	Multi	100	for VU 4.1	ARV65
AVW01GC	4	Multi	50	for VU 4.1	ARV70
AVW01GE	4	Multi	20	for VU 4.1	ARV68

KNAUER

Spare parts and kits

Spare part kits contain parts that are replaced during repairs.

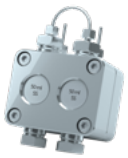
Pump spare parts



AHB40XA



AHB32



AHC20



AHB40CA

Replacement pump heads for analytical AZURA® pumps

Pump head 10 ml, stainless steel, 862 bar	AHB40XA
Pump head 10 ml, stainless steel, 400–700 bar	AHB40
Pump head 10 ml, ceramic with PEEK bushings, 400 bar	AHB32
Pump head 10 ml, ceramic with Ti-bushings, 400 bar	AHB32DA
Pump head 10 ml, Hastelloy-C, 400 bar, for corrosive chemicals	AHB43
Pump head 50 ml, stainless steel, 300 bar	AHC20
Pump head 50 ml, ceramic, 200 bar	AHC22
Pump head 50 ml/min, Hastelloy® C, 300 bar, for corrosive chemicals	AHC23
Pump head 10 ml, stainless steel, 700 bar, for aqueous solutions	AHB40FA
Pump head 10 ml, ceramic with Titanium bushings, 400 bar, for aqueous solutions	AHB32GA
Pump head 50 ml, stainless steel, 300 bar, for aqueous solutions	AHC20FA
Pump head 50 ml, ceramic, 200 bar, for aqueous solutions	AHC22FA
Pump head 5 ml, stainless steel, 1 000 bar	AHA60
Pump head 10 ml, stainless steel, for normal phase applications	AHB40BA
Pump head 10 ml, stainless steel, 700 bar, for high-temperature applications	AHB40CA
Pump head 50 ml, stainless steel, for normal phase applications	AHC20BA
Pump head 50 ml, stainless steel, 300 bar, for high-temperature applications	AHC20CA



A4029-1



A4029V2



A4021-1



A4021V2

Replacement pump heads for preparative AZURA® pumps

Pump head 100 ml, stainless steel, 400 bar	A4029-1
Pump head 100 ml, titanium, 400 bar	A4029V2
Pump head 250 ml, stainless steel, 200 bar	A4021-1
Pump head 250 ml, titanium, 200 bar	A4021V2
Pump head 500 ml, stainless steel, 100 bar	A4038-1
Pump head 500 ml, titanium, 100 bar	A4038V2
Pump head 1 000 ml, stainless steel, 50 bar	A4022-1
Pump head 1 000 ml, titanium, 50 bar	A4022V2

Filters and filter cartridges for pumps

Filter cartridge for pump P 6.1L/40P, titanium frit, 2 µm pore size, 50 ml/min maximum flow, high capacity filter, 60 µl volume, 3 pcs.	A9661
Filter cartridge for pump P 6.1L/40P, stainless steel frit, 2 µm pore size, 10 ml/min maximum flow, volume optimized filter, 20 µl volume, 3 pcs.	A96601
Empty filter cartridge for pump P 6.1L/40P, PEEK	A9652


A06840


A06841


A068411


A1122

Check valves for pumps

Check valve unit for 10/50 ml pump heads, for dosing applications, bore: Ø 1.4 mm, ball: Ø 1.75 mm	A06840
Check valve unit for 10 ml pump heads, for HPLC applications, bore: Ø 0.7 mm, ball: Ø 1.75 mm	A06841
Spring-loaded check valve unit for 10 ml / 50 ml pump heads, for normal phase applications, bore: Ø 1.4 mm, ball: Ø1.75 mm	A068411
Check valve unit (KEL-F) for 10 ml pump head, for aggressive substances, bore: Ø 0.7 mm, ball: Ø 1.75 mm	A068412
Check valve unit for 50 ml pump heads, for HPLC applications, bore: Ø 1.2 mm, ball: Ø 1.75 mm	A06842
Check valve unit (KEL-F) for 50 ml pump head, for aggressive substances, bore: Ø 1.2 mm, ball: Ø 1.75 mm	A068422
Check valve unit stainless steel/PEEK for 500 ml and 1 000 ml pump heads, bore: Ø 2.9 mm, ball: Ø 4.76 mm	A1080
Check valve unit titanium/PEEK for 500 ml and 1 000 ml pump heads, bore: Ø 2.9 mm, ball: Ø 4.76 mm	A1080V1
Check valve unit titanium/KEL-F for 500 ml and 1 000 ml pump heads, bore: Ø 3.0 mm, ball: Ø 4.17 mm	A1080V2
Check valve unit stainless steel/KEL-F for 500 ml and 1 000 ml pump heads, bore: Ø 3.0 mm, ball: Ø 4.17 mm	A1080V3
Check valve unit stainless steel/PEEK for 100 ml and 250 ml pump heads, bore: Ø 2.0 mm, ball: Ø 3.17 mm	A1122
Check valve unit titanium/PEEK for 100 ml and 250 ml pump heads, bore: Ø 2.0 mm, ball: Ø 3.17 mm	A1122-1
Check valve unit titanium/KEL-F for 100 ml and 250 ml pump heads, bore: Ø 2.0 mm, ball: Ø 3.17 mm	A1122-2
Check valve unit stainless steel/KEL-F for 100 ml and 250 ml pump heads, bore: Ø 2.0 mm, all: Ø 3.17 mm	A1122-3

Rebuild kits for pumps

Rebuild-Kit for one pump head AZURA® Pump P 2.1L and BlueShadow 80P (100 ml/250 ml), Venting screw KEL-F, Check valve unit KEL-F, O-ring	A58211
Rebuild-Kit for one pump head AZURA® Pump P 2.1L and BlueShadow 80P (500ml/1 000 ml), Venting screw KEL-F, Check valve unit KEL-F, O-ring	A58212
Rebuild-Kit Kel-F for AZURA® Pump P 2.1S/P 4.1S/P 4.2S/P 6.1L/40P, 10 ml/min pump head	A5821-1
Rebuild-Kit Kel-F for AZURA® Pump P 2.1S/P 4.1S/P 4.2S/P 6.1L/40P, 50 ml/min pump head	A5821-2
Rebuild-Kit for aqueous eluents (for P 2.1S/P 4.1S/P 6.1L/40P with 10 ml pump head)	A5823
Rebuilt-Kit for aqueous eluents (for AZURA® P 2.1S, P 4.1S, P 4.2S, P 6.1L and BlueShadow 40P with 50 ml pump head)	A5823-1

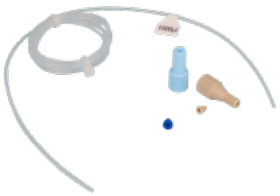
Spare part kits pressure sensor / purge valve, incl. capillary for AZURA® P 4.2S

Pressure sensor 800 bar, stainless steel, for APK20EA, APK20EG, APK20FA, APK20FG	A8880131
Purge valve, stainless steel, for APK90EA, APK90EG, APK90FA, APK90FG	A8880132
Pressure sensor 800 bar, titanium, for APK20EF, APK20EH	A8880133
Pressure sensor 800 bar, PEEK, for APK20EB, APK20FB, APK20FI	A8880134
Purge valve, PEEK, for APK90EB, APK90FB	A8880135
Pressure sensor 800 bar, Hastelloy-C, for APK20EC, APK20FC	A8880136
Purge valve, Hastelloy-C, for APK90EC, APK90FC	A8880137

Capillary (pump head - pressure sensor / purge valve) for AZURA® P 4.2S

Capillary, stainless steel, ID 0.5 mm, for APK20EA, APK20EG, APK20FA, APK20FG, APK90EA, APK90EG, APK90FA, APK90FG	A80404
Capillary, titanium, ID 0.7 mm, for APK20EF, APK20EH	A80405
Capillary, PEEK, ID 0.25 mm, for APK20EB, APK90EB	A80406
Capillary, Hastelloy-C, ID 0.5 mm, for APK20EC, APK20FC, APK90EC, APK90FC	A80407
Capillary, PEEK, ID 0.5 mm, for APK20FB, APK20FI, APK90FB	A80408

Autosampler spare parts



A500526

Buffer tubing kits

Buffer tubing for AZURA® Autosampler AS 6.1L, 500 µl incl. fittings; Spare part for AAA00AA, AAA01AA, AAA10AA, AAA11AA, AAA20AA, AAA21AA, AAA50AA & AAA51AA	A500525
Buffer tubing for AZURA® Autosampler AS 6.1L, 1 000 µl, incl. fittings; alternative to standard configuration	A500526
Buffer tubing for AZURA® Autosampler AS 6.1L, 2 000 µl, incl. fittings; Spare part for AAA31AA, AAA40AA, AAA41AA	A500527
Buffer tubing for AZURA® Autosampler AS 8.1L	A80511



A500846

Syringes

100 µl Syringe for AZURA® Autosampler AS 6.1L autosamplers and other KNAUER Autosamplers; alternative to standard configuration	A500846
250 µl Syringe for AZURA® Autosampler AS 6.1L; Spare part for AAA00AA, AAA01AA, AAA10AA, AAA11AA, AAA20AA, AAA21AA, AAA50AA & AAA51AA	A500847
500 µl Syringe for AZURA® Autosampler AS 6.1L; alternative to standard configuration	A500864
1 000 µl Syringe for AZURA® Autosampler AS 6.1L; alternative to standard configuration	A500865
2 500 µl Syringe for AZURA® Autosampler AS 6.1L; Spare part for AAA31AA, AAA40AA, AAA41AA	A500866
ZDV Syringe 100 µl for AZURA® Autosampler AS 8.1L	A80497
ZDV Syringe 250 µl for AZURA® Autosampler AS 8.1L	A80498
ZDV Syringe 500 µl for AZURA® Autosampler AS 8.1L	A80502



A500519

Rotor seals for AZURA® Autosampler

Rotor seal for AZURA® Autosampler AS 6.1L, 700 bar, Vespel; spare part for AAA00AA, AAA01AA	A500519
Rotor seal for AZURA® Autosampler AS 6.1L, 1000 bar, Vespel; spare part for AAA50AA & AAA51AA	A500520
Rotor seal for AZURA® Autosampler AS 6.1L, 1240 bar, Vespel; spare part for AAA10AA, AAA11AA	A500521
Rotor seal for AZURA® Autosampler AS 6.1L, 345 bar, PEEK; spare part for AAA20AA, AAA21AA	A500522
Rotor seal for AZURA® Autosampler AS 6.1L, 350 bar, ValconH; spare part for AAA40AA, AAA41AA	A500523
Rotor seal for AZURA® Autosampler AS 6.1L, 350 bar, ValconE; spare part for AAA31AA	A500524
Rotor seal for UHPLC injection valve for AZURA® Autosampler AS 8.1L	A80494



A50078



A50077

Sample loops

Sample loop incl. fittings, 10 µl, stainless steel; spare part for AAA10AA, AAA11AA	A50078
Sample loop incl. fittings, 10 ml, stainless steel; spare part for AAA40AA, AAA41AA	A500509
Sample loop incl. fittings, 10 ml, PEEK; spare part for AAA31AA	A500511
Sample loop incl. fittings, 100 µl, stainless steel; spare part for AAA00AA, AAA01AA, AAA50AA, AAA51AA	A50077
Sample loop incl. fittings, 100 µl, PEEK; spare part for AAA20AA, AAA21AA	A500510
Sample loop 250 µl, stainless steel for AZURA® Autosampler AS 6.1L & 3950, incl. fittings	A500528

Sample needles

Sample needle for preparative AZURA® Autosampler AS 6.1L; spare part AAA40AA, AAA41AA	A500516
Sample needle for biocompatible autosampler AZURA® Autosampler AS 6.1L; spare part AAA20AA, AAA21AA	A500517
Sample needle for bio-preparative AZURA® Autosampler AS 6.1L; spare part AAA31AA	A500518
Sample needle for analytical AZURA® Autosampler AS 6.1L; spare part for AAA00AA, AAA01AA, AAA10AA, AAA11AA, AAA50AA, AAA51AA	A64700
AZURA® AS 8.1L UHPLC Extension Loop Assy 100 µl	A80505
AZURA® AS 8.1L UHPLC Needle 30 µl incl. needle seat	A80507



A50058

Air needles

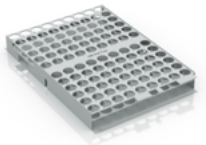
Air needle for AZURA® Autosampler AS 6.1L; 50 mm protrusion length; Spare part for AAA40AA, AAA41AA and for biocompatible AAA31AA	A500529
Air needle for AZURA® Autosampler AS 6.1L; 56 mm protrusion length	A500530
Air needle for AZURA® Autosampler AS 6.1L; 62 mm protrusion length - standard for all autosampler versions (except AAA40AA, AAA41AA, AAA31AA)	A50058
Air needle for AZURA® Autosampler AS 6.1L; 68 mm protrusion length	A500531
Air needle for AZURA® Autosampler AS 6.1L; 74 mm protrusion length	A500532
Air needle for AZURA® Autosampler AS 6.1L; 80 mm protrusion length	A500533
Set of air needles for AZURA® Autosampler AS 6.1L, 1 pc. of each length	A50059



A50050



A500502



A500505



A500507

Vial plates

Vial plate for 48 × 1.5 ml vials for Autosampler 3950 and AZURA® Autosampler AS 6.1L, 1 pc.	A50050
Vial plate for 84 × 1.5 ml and 3 x 10 ml vials for Autosampler 3950 and AZURA® Autosampler AS 6.1L, 1 pc.	A500501
Prep vial plate for 12 × 10 ml for Autosampler 3950 and AZURA® Autosampler AS 6.1L, 1 pc.	A500502
Vial plate for 108 × 1.5 ml vials for Autosampler 3950 and AZURA® Autosampler AS 6.1L, 1 pc.	A500505
Prep vial plate for 30 × 10 ml for Autosampler 3950 and AZURA® Autosampler AS 6.1L, 1 pc.	A500507
Vial plate 12 × 2 ml (2 pcs.) for AZURA® AS 8.1L	A80499
Vial plate 54 × 2 ml (2 pcs.) for AZURA® AS 8.1L	A80500
Vial plate 3 × 10 ml (2 pcs.) for AZURA® AS 8.1L	A80525
Vial plate 15 × 10 ml (2 pcs.) for AZURA® AS 8.1L	A80524
Set vial plate 10 ml for AZURA® AS 8.1L	A80523

Other autosampler spare parts

Fuse (2.5 A) for AZURA® Autosampler AS 6.1L, 2 pcs.	A500534
Rectangular bottle (250 ml, PE) for wash or transport solution	A500535
Waste tube for AZURA® Autosampler AS 6.1L, silicone, 1 m	A500536
Waste tube for AZURA® Autosampler AS 6.1L, PTFE, 1 m	A500537
Fuses 5AT (2 pcs.) for AZURA® Autosampler AS 8.1L	A80506



A50810



A50809



A50806

A50807

Liquid Handler spare parts

Tip/Injection needle – Liquid Handler LH 2.1	A50810
Dispenser syringe 12.5 ml – Liquid Handler LH 2.1	A50809
Dispenser syringe 5 ml – Liquid Handler LH 2.1	A50811
Dispenser syringe 2.5 ml – Liquid Handler LH 2.1	A50812
Dispenser syringe 1 ml – Liquid Handler LH 2.1	A50813
Buffer tubing, ID 2.1 mm, AD 1/8", 600 cm, 21 ml, FEP - Liquid Handler LH 2.1	A50814
Liquid Handler LH 2.1 Rack fixation for docking a LH 2.1 rack to the drainage tray	A50806
Tubing for tip, ID 1.5 mm, 180" 457 cm, FEP - Liquid Handler LH 2.1	A50807
Tubing for washing solution, ID 2 mm, 70" 177 cm, FEP – Liquid Handler LH 2.1	A50808
Wash Station for Liquid Handler LH 2.1	A50815



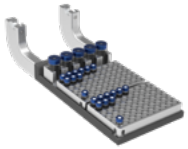
A5104



A5101



A5105 with A51051



A5103 with A51042

Accessories for Liquid Handler LH 8.1

Fast wash station for 2 solvents	A5101
Manual wash station with 2 reservoirs and waste line	A5102
Manual wash station with 6 × 10 ml reservoirs	A51021
Manual sample rack holder	A5103
Robotic Cooler with three drawers	A5104
Valve drive for Injection valve	A5105
Sample Rack for 60 Vials	A51041
Sample Rack for 130 Vials	A51042
Sample Rack for microtiter plate 384	A51043
Sample Rack for microtiter plate 96 or deep well plate 96	A51044
Injection valve (without valve drive)	A51051
Injection port zero dead volume HPE	A510513

Syringes

25 µl syringe, 22s gauge, polished and coated for chemical inertness	A510542
50 µl syringe, 22s gauge, polished and coated for chemical inertness	A510543
100 µl syringe, 22 gauge, polished and coated for chemical inertness	A510544-1
100 µl syringe, gauge 22, starter model	A510548

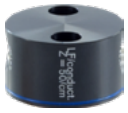
Detector spare parts

Spare parts kits for flow cells

Spare part kit for analytical flow cells, 10 mm (A4061, A4061V1, A4061XB)	A1131
Spare part kit for PEEK TRI-Clamp flow cells (A4152-1, A4154-1)	A1132-1
Spare part kit for preparative flow cells (A4066, A4067, A4068)	A1132
Spare part kit for semi-preparative flow cell, 3 mm (A4042, A4045)	A1132V3
Spare part kit for analytical flow cell, 10 mm (A4130)	A1540
Spare part kit for analytical flow cell, 3 mm (A4131, A4132)	A1540V1



ADG61


AMN90


AMO90

Conductivity monitor mikron 81 spare parts

Conductivity monitor mikron 81 main unit without flow cell	ADG61
Gasket for mikron 81 for liquid-tight connection of monitor unit and flow cell	ADG6103
Biocompatible flow cell for mikron 81 for up to 100 ml/min	AMN90
Biocompatible flow cell for mikron 81 for up to for up to 1 000 ml/min	AMO90


AZL01


AZL02


A5193


A5194

Lamps

HBST deuterium lamp for AZURA® Detector DAD 6.1L	AZL01
Deuterium lamp, replacement, for S2520, 10D, 40D, UVD 2.1S, UVD 2.1L, DAD 2.1L, MWD 2.1L	A5193
HBST deuterium lamp for Smartline UV and UV/VIS detectors 2550, BlueShadow 50D and VWD 2.1L	A5194
Halogen lamp for AZURA® Detector DAD 6.1L	AZL02
Halogen lamp for Smartline UV/VIS detector 2550, BlueShadow 50D and VWD 2.1L	A5195
LED for Sedex 80LT and Sedex 85LT light scattering detectors	A07541
Xenon lamp for RF-20A/Axs fluorescence detector	A59210

KNAUER

Valve spare parts

Spare parts for valves

Valve article no.	Ports	Positions	Pressure [bar]	Version	Spare part (stator) article no.
AVC28AC	6	2	1 200	for VU 4.1	A8880063
AVC38AC	8	2	1 200	for VU 4.1	A8880062
AVC48AC	10	2	1 200	for VU 4.1	A8880082
AVD23AF	6	2	100	for VU 4.1	A8880077
AVD24CE	6	2	240	for VU 4.1	A8880070
AVD26AE	6	2	500	for VU 4.1	A8880077
AVD26AH	6	2	500	for VU 4.1	A8880077
AVD36AE	8	2	500	for VU 4.1	A8880069
AVD43CE	10	2	100	for VU 4.1	A8880155
AVE25AE	6	2	300	for VU 4.1	A8880075
AVE25AI	6	2 / 2 channel	300	for VU 4.1	A8880075
AVF23CE	6	2	100	for VU 4.1	A8880081
AVF32CE	8	2	50	for VU 4.1	A8880067
AVG24CE	6	2	240	manual	A8880070
AVI28AC	6	2	1 200	manual	A8880063
AVI38AC	8	2	1 200	manual	A8880062
AVJ23AF	6	2	100	manual	A8880077
AVJ26AE	6	2	500	manual	A8880077
AVJ36AE	8	2	500	manual	A8880069
AVK25AE	6	2	300	manual	A8880075
AVL23CE	6	2	100	manual	A8880081
AVM38AJ	8	Special	1 200	for VU 4.1	A8880175
AVM48AC	10	Special	1 200	for VU 4.1	A8880099
AVN64CE	16	Special	200	for VU 4.1	A8880147
AVN94CE	8	Special	240	for VU 4.1	A8880066
AVN96AE	8	Special	500	for VU 4.1	A8880080
AVQ63AF	16	Multi	100	for VU 4.1	A8880072
AVQ66AE	16	Multi	500	for VU 4.1	A8880072
AVR28AC	6	Multi	1 200	for VU 4.1	A8880063
AVR38AC	8	Multi	1 200	for VU 4.1	A8880062
AVS23AF	6	Multi	100	for VU 4.1	A8880077
AVS25AC	6	Multi	300	for VU 4.1	A8880077
AVS26AE	6	Multi	500	for VU 4.1	A8880077
AVS34CE	8	Multi	240	for VU 4.1	A8880071
AVS34CH	8	Multi / break-free	240	for VU 4.1	A8880071
AVS35AE	8	Multi	300	for VU 4.1	A8880069
AVS35AH	8	Multi / break-free	300	for VU 4.1	A8880069
AVS36AE	8	Multi	500	for VU 4.1	A8880069
AVS62CE	16	Multi	50	for VU 4.1	A8880073
AVS63CE	16	Multi	150	for VU 4.1	A8880073
AVS85AH	2	Multi	300	for VU 4.1	A8880069
AVT25AE	6	Multi	300	for VU 4.1	A8880075
AVT34AC	8	Multi	200	for VU 4.4	A8880074
AVT34AE	8	Multi	200	for VU 4.1	A8880074
AVT34AH	8	Multi / break-free	200	for VU 4.1	A8880074
AVU32CE	8	Multi	50	for VU 4.1	A8880067

Valve article no.	Ports	Positions	Pressure [bar]	Version	Spare part (stator) article no.
AVU32GE	8	Multi	50	for VU 4.1	A8880068
AVU83AH	2	Multi	100	for VU 4.1	A8880148
AVW01GC	4	Multi	50	for VU 4.1	A8880154
AVW01GE	4	Multi	20	for VU 4.1	A8880154

KNAUER

Accessories

Pump accessories



AZC00



A5325



A2056



A5324

Eluent trays & bottles

AZURA® Eluent tray E 2.1L for AZURA® devices with a capacity of 6 × 1 l bottles or 4 × 2.5 l bottles or 2 × 5 l bottles (delivery without bottles)	AZC00
Eluent bottle 1 000 ml, clear glass, incl. cap for eluent tubing, GL45	A5325
250 ml bottle for piston back flushing	A2056
Set of 4 eluent bottles 1 000 ml, incl. caps for eluent tubing, GL45	A5324
Set of 2 eluent bottles 1 000 ml, incl. caps for eluent tubing, GL45	A5324-1
Set of 4 eluent bottles 1 000 ml, 1 bottle 250 ml for piston backflushing, incl. caps for eluent tubing, GL45	A5324-2
Recommended for AZURA® systems: Set of 4 eluent bottles 1 000 ml, 1 bottle 250 ml for piston backflushing, incl. closed caps, GL45 use with AZURA® tubing kit	A5324-3
Set of 2 eluent bottles 1 000 ml, 1 bottle 250 ml for piston backflushing, incl. closed caps, GL45 use with AZURA® tubing kit	A5324-4
Set of eluent supply bottles, 3 × 2.5 l brown glass bottles (borosilicate glass) with special round bottom for minimal eluent remainder, for preparative HPLC/FPLC, includes screw-type cap	A70037
Eluent supply bottle plastic 2 l incl. cap and tubing for IC and ECD systems	A70038
Eluent supply bottle 2 000 ml, GL45 thread, round, clear glass, without screw cap	A59158-1
Waste can 2.5 l with GL45 screw top, UN-approved, 153 × 115 × 202 mm	A59173
Waste can, 10 l with GL45 screw top, UN-approved, 192 × 317 × 231 mm	A59256



A5390



A5398



A5396

Mass flow controllers*

Mini CORI-Flow (M13) Mass flow controller incl. mounting block, flow: 1–50 ml/min, stainless steel 316	A5390
Mini CORI-Flow (M14) Mass flow controller incl. mounting block, flow: 2–833 ml/min, stainless steel 316	A5391
Mini CORI-Flow (M14) Mass flow controller incl. mounting block, flow: 2–833 ml/min, stainless steel 316, Profibus	A5391P
Mini CORI-Flow (M13) Mass flow controller incl. mounting block, flow: 1–50 ml/min, stainless steel 316, Profibus	A5393
Mini CORI-Flow (M12) Mass flow controller incl. mounting block, flow: 0.03–3.3 ml/min, stainless steel 316	A5394
Mini CORI-Flow (M13) Mass flow controller incl. mounting block, flow: 1–50 ml/min, Hastelloy-C22	A5395
Mini CORI-Flow (MI140) Mass flow controller, flow: 2–833 ml/min, stainless steel 316, Profinet	A5398
HI-TEC Bright display for Mini CORI-Flow mass flow controller (display, setpoint and counter)	A5396

* analog and bus versions on request



AZZ00MB



AZZ00MC



AZZ10ME



A5830

Static mixers

AZURA® HPLC mixer up to 1 240 bar, 50 µl mixing volume, stainless steel	AZZ00MB
AZURA® HPLC mixer up to 1 240 bar, 100 µl mixing volume, stainless steel	AZZ00MC
AZURA® HPLC mixer up to 1 240 bar, 200 µl mixing volume, stainless steel	AZZ00MD
AZURA® HPLC mixer up to 1 240 bar, 400 µl mixing volume, stainless steel	AZZ00MF
AZURA® HPLC mixer up to 1 240 bar, 600 µl mixing volume, stainless steel	AZZ00MG
AZURA® HPLC mixer up to 40 MPa, 250 µl mixing volume, PEEK (biocompatible)	AZZ10ME
HyperShear Static Mixer, 1.5 ml, 1 – 40 ml/min, max. 414 bar, stainless steel and PEEK, incl. mounting brackets for AZURA® L devices (A9853-8)	A5830



A0285

Dynamic mixers

Dynamic mixing chamber (250 V), titanium, analytical, 1/16“, up to 420 bar, 1 740 µl mixing volume	A0275
Dynamic mixing chamber (115 V), titanium, analytical, 1/16“, up to 420 bar, 1 740 µl mixing volume	A02751
Dynamic mixing chamber (250 V), stainless steel, analytical, 1/16“, up to 420 bar, 1 740 µl mixing volume	A0285
Dynamic mixing chamber (115 V), stainless steel, analytical, 1/16“, up to 420 bar, 1 740 µl mixing volume	A02851
Dynamic mixing chamber (250 V), titanium, preparative, 1/8“, up to 250 bar, 5.9 ml mixing volume	A70581
Dynamic mixing chamber (115 V), titanium, preparative, 1/8“, up to 250 bar, 5.9 ml mixing volume	A705811
Dynamic mixing chamber (250 V), stainless steel, preparative, 1/8“, up to 250 bar, 5.9 ml mixing volume	A0581
Dynamic mixing chamber (115 V), stainless steel, preparative, 1/8“, up to 250 bar, 5.9 ml mixing volume	A05811
Extension unit for dynamic mixer A70581/A705811	A2515



A3373



A3374



A3375



A3364

Solvent filters & inlet tubing

Mobile Phase Filter, stainless steel, 2 µm, 1/8" pipe OD, suitable for all analytical HPLC systems, max. flow rate 50 ml/min	A3373
Mobile Phase Filter, stainless steel, 20 µm, for 1/8" OD, compatible with the AZURA® Tubing Kit (A9650), suitable for all analytical and semi preparative HPLC systems, max. flow rate 100 ml/min	A3374
Mobile Phase Filter, stainless steel, 10 µm, for 1/8" OD, compatible with the AZURA® Tubing Kit (A9650), suitable for all analytical HPLC systems, max. flow rate 50 ml/min	A3375
Mobile Phase Filter, biocompatible PE, 20 µm, 1/8" pipe OD, suitable for all FPLC systems, max. flow rate 500 ml/min	A3364
AZURA® Tubing kit with cap and solvent filter (A3375, stainless steel, 10 µm), suitable for all analytical HPLC systems	A9650
AZURA® Tubing kit bio with cap and insert, solvent filter inlet and fittings, 1 set	A96507
Inlet-bushing kit with 1/4"-PTFE Tubing and 20 µm stainless steel solvent filter (up to 250 ml/min)	A58207



AZZ00NA



AZZ00NB



AZZ10NB



FZZ2

Pulse dampers

KNAUER Pulse Damper, low volume, 275 µl, stainless steel, 1/16", 1 000 bar	AZZ00NA
KNAUER Pulse Damper, high volume, 290 µl, stainless steel, 1/16", 1 000 bar	AZZ00NB
This pulse damper combines high damping performance with reliable, membrane-free assembly. Fully biocompatible, it can be easily integrated into all AZURA® FPLC systems.	AZZ10NB
KNAUER Pulse Damper mounting bracket	FZZ2



A9861



A9868



A98611



A58267

Pump head inlet fittings

Pump head inlet for AZURA® Pump P 2.1L, BlueShadow 80P, 1/4" (NPT), stainless steel	A9861
Pump head inlet for AZURA® Pump P 2.1L, Set, 1/2"-20 UNF, PEEK with CTFE (Kel-F) adapter, including tubing 1/4" PTFE	A9868
Inlet bushing for prep pump heads, adapter to 3/8" tube stub	A98611
Inlet bushing for binary LPG prep pump heads, LPG inlet to 3/8" tube stub	A98612
Inlet bushing for LPG prep pump heads, LPG ternary inlet to 3/8" tube stub	A98613
Male connector to connect a 1/4" OD capillary to a 1/4'' male NPT adapter (material: stainless steel) for A9861	A58267
Male connector to connect a 4 mm OD capillary to a 1/4" male NPT adapter (material: stainless steel) for A9861	A58268
Male connector to connect a 1/8" OD capillary to a 1/4" male NPT adapter (material: stainless steel) for A9861	A58269
Inlet-bushing kit for P 2.1S, P 4.1S, P 6.1L, 40P and S1050 pumps for pump heads 10 ml (1/8" capillaries)	A58202
Inlet-bushing kit for P 2.1S, P 4.1S, P 6.1L, 40P and S1050 pumps for pump heads 10 ml (1/16" capillaries)	A58203
Inlet-bushing kit for P 2.1S, P 4.1S, P 6.1L, 40P and S1050 pumps for pump heads 50 ml (1/8" capillaries)	A58204
Inlet-bushing kit for P 2.1S, P 4.1S, P 6.1L, 40P and S1050 pumps for pump heads 50 ml (1/16" capillaries)	A58205
Inlet-bushing kit with 1/4"-PTFE tubing and 20 µm stainless steel solvent filter (up to 250 ml/min)	A58207



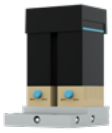
A5822



A7200

Pump head outlet fittings

Outlet-bushing kit 1/8" tube stub for S1800, 80P and P 2.1L pumps	A5822
Adapter to connect a capillary with 1/16" OD (thread: 10-32 UNF) to AZURA® Pump P 2.1L or BlueShadow Pump 80P outlet (1/8", M8x1 thread), material: stainless steel, 2 pcs.	A7200



AZZ00AA



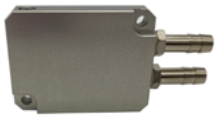
AZZ00AB



AZZ10AB

LPG modules

LPG module for Pump P 2.1L binary up to 800 ml/min, stainless steel	AZZ00AA
LPG module for Pump P 2.1L ternary up to 220 ml/min, stainless steel	AZZ00AB
LPG module for Pump P 2.1L ternary up to 220 ml/min, PEEK	AZZ10AB



A2034-1



A2035-1



A57024



A57036-1

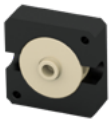
Temperature control

Pump head cooling and heating device for 100/250/500/1 000 ml/min pump heads	A2034-1
Pump head cooling and heating device for 10 and 50 ml/min pump heads	A2035-1
Temperature controller for column heating sleeve	A57024
St~100 Transformer for small diameter column heating sleeves	A57024-3
Heating solution for 10 and 50 ml/min pump heads, includes temperature controller, heating plate and insulation sleeve	A57036-1
Heating solution for 10 and 50 ml/min pump heads, includes heating plate and insulation sleeve (without temperature controller)	A57037-1

Detector accessories



AMC19XA



A4045



A4061V2



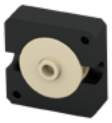
A4061XB

Flow cells 1/16"

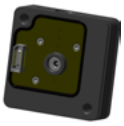
0.5 mm path length, 3 µl, 1/16", 200 bar, stainless steel, classical KNAUER flow cell	A4069
0.5 mm path length, 3 µl, 1/16", 100 bar, biocompatible, classical KNAUER flow cell	A4095
3 mm path length, 2 µl, 1/16", 300 bar, PressureProof Flow cell cartridge for AZURA® Detector DAD/MWD	AMB18
3 mm path length, 2 µl, 1/16", stainless steel, classical KNAUER flow cell	A4042
3 mm path length, 2 µl, 1/16", 30 bar, biocompatible, classical KNAUER flow cell	A4045
5 mm path length, 51 µl, 1/16" or 1/8", 60 bar, biocompatible, CombiLight flow cell	AMG90
10 mm path length, 23.5 µl, 1/16" or 1/8", 60 bar, biocompatible, CombiLight flow cell	AMH90
10 mm path length, 10 µl, 1/16", 300 bar, stainless steel, for BlueShadow Detector 50D, S2550 and MW-1, classical KNAUER flow cell	A4061V2
10 mm path length, 10 µl, 1/16", 300 bar, stainless steel, with heat exchanger one sided inlet and outlet, classical KNAUER flow cell	A4061XB
10 mm path length, 2 µl, 1/16", 50 bar, LightGuide Flow cell cartridge for AZURA® Detector DAD/MWD	AMC19XA
10 mm path length, 10 µl, 1/16", 300 bar, 200 ml/min, PressureProof Flow cell cartridge for AZURA® Detector DAD/MWD	AMC37
10 mm path length, 10 µl, 1/16", 300 bar, PressureProof Flow cell cartridge for AZURA® Detector DAD/MWD	AMC38
50 mm path length, 6 µl, 1/16", 50 bar, High Sensitivity LightGuide Flow cell cartridge for AZURA® Detector DAD/MWD	AMD59XA



A4066



A4067



AME91



AMF91

Flow cells 1/8"

0.5 mm path length, 168 µl, 1/8" or 3/16", 60 bar, biocompatible, CombiLight flow cell	AME91
2 mm path length, 1/8", 200 bar, stainless steel, changeable to 0.5/1.25 mm, classical KNAUER flow cell	A4066
2 mm path length, 1/8", 100 bar, biocompatible, changeable to 0.5/1.25 mm, classical KNAUER flow cell	A4067
2 mm path length, 168 µl, 1/8" or 3/16", 60 bar, biocompatible, CombiLight flow cell	AMF91
5 mm path length, 51 µl, 1/16" or 1/8", 60 bar, biocompatible, CombiLight flow cell	AMG90
10 mm path length, 23.5 µl, 1/16" or 1/8", 60 bar, biocompatible, CombiLight flow cell	AMH90



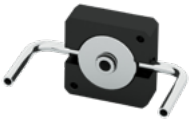
AME91



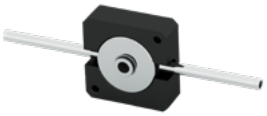
AMF91

Flow cells 3/16”

0.5 mm path length, 168 µl, 1/8” or 3/16”, 60 bar, biocompatible, CombiLight flow cell	AME91
2 mm path length, 168 µl, 1/8” or 3/16”, 60 bar, biocompatible, CombiLight flow cell	AMF91



A4068



A4068-2

Flow cells 1/4”

2 mm path length, 1/4” angular connections, 200 bar, stainless steel, changeable to 0.5/1.25 mm, without fittings, classical KNAUER flow cell	A4068
2 mm path length, 1/4” straight connections, 200 bar, stainless steel, changeable to 0.5/1.25 mm, without fittings, classical KNAUER flow cell	A4068-2



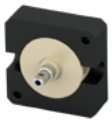
A4044



A4044HT



AMKX8KIT



A4047

Flow cells 1/16” fiber optics

0.5 mm path length, 3 µl, 1/16”, 200 bar, stainless steel, fiber optic connectors, classical KNAUER flow cell	A4089
0.5 mm path length, 3 µl, 1/16”, 100 bar, biocompatible, fiber optic connectors, classical KNAUER flow cell	A4096
3 mm path length, 2 µl 1/16”, 30 bar, biocompatible, fiber optic connectors, classical KNAUER flow cell	A4047
3 mm path length, 2 µl, 1/16”, 300 bar, 85 °C, stainless steel, fiber optic connectors, classical KNAUER flow cell	A4044HT
3 mm path length, 2 µl, 1/16”, 300 bar, stainless steel, fiber optic connectors, classical KNAUER flow cell	A4044
10 mm path length, 10 µl, 1/16”, 300 bar, stainless steel, fiber optic connectors, classical KNAUER flow cell	A4074
Fiber optics adapter kit for AZURA® Detector DAD/MWD, with fiber optic cables (1 × 400 mm and 1 × 750 mm) and mounting bracket	AMKX8KIT



A4078



A4079

Flow cells 1/8" fiber optics

2 mm path length, 1/8", 200 bar, stainless steel, fiber optic connectors, changeable to 0.5/1.25 mm, classical KNAUER flow cell	A4078
2 mm path length, 1/8", 100 bar, biocompatible, fiber optic connectors, changeable to 0.5/1.25 mm, classical KNAUER flow cell	A4079



A4081



A4153



A4152



A4152-1

Flow cells larger than 1/8" fiber optics

2 mm path length, 1/4" angular connections, 200 bar, stainless steel, fiber optic connectors, changeable to 0.5/1.25 mm	A4081
2 mm path length, 1/4" straight connections, 200 bar, stainless steel, fiber optic connectors, changeable to 0.5/1.25 mm	A4081V2
2 mm path length, 1/4" TRI-Clamp connections, 80 bar, stainless steel, fiber optic connectors, changeable to 0.5/1.25 mm	A4153
2 mm path length, 3/8" TRI-Clamp connections, 80 bar, stainless steel, fiber optic connectors, changeable to 0.5/1.25 mm	A4152
7 mm path length, 3/8" TRI-Clamp connections, 10 bar, biocompatible, fiber optic connectors	A4152-1
2 mm path length, 1/2" TRI-Clamp connections, 80 bar, stainless steel, fiber optic connectors, changeable to 0.5/1.25 mm	A4154
10 mm path length, 3/8" TRI-Clamp connections, 10 bar, biocompatible, fiber optic connectors	A4154-1
2 mm path length, 3/4" TRI-Clamp connections, 80 bar, stainless steel, fiber optic connectors, changeable to 0.5/1.25 mm	A4155



A4104

Nano flow cell

3 mm path length, 6 nl, 375 µm OD, 50 µm ID, 300 bar, fused silica, fiber optic connectors	A4104

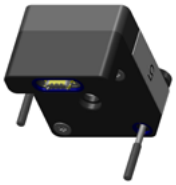


A0740

Fiber optic cables

Fiber optic cables (2 pcs.), 750 mm, 2 × SMA 905 600/660, polymicro	A454000750
Fiber optic cables (2 pcs.), custom-made sizes, 2 × SMA 905 600/660, polymicro	A4540XXXXX*
Fiber optic cables (2 pcs.), 750 mm, 2 × SMA 905 600/660, polymicro, up to 85 °C	A0740HT

* XXXXX refers to the length of the fiber optic cables in mm, e.g. for 500 mm order A454000500, for 3000 mm A454003000



ADZ50



ADZ14

Signal units

Signal unit for standard KNAUER or CombLight flow cells, for AZURA® VWD 2.1L	ADZ50
Signal unit for fiber optic cables and flow cell, for AZURA® VWD 2.1L	ADZ14



A4123



A4125



A4126



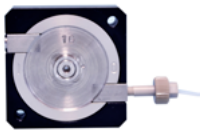
A4128

Test cells

Standard test cell for AZURA® UVD 2.1S, UVD 2.1L and Smartline 200, 2520, 2500, 2550 and 2600	A4123
Test cell with fiber optic connectors for AZURA® Detector UVD 2.1S, UVD 2.1L and Smartline 2520, 2500 and 2600	A4125
Standard test cell with holmium filter for AZURA® UVD 2.1S, UVD 2.1L and Smartline 200, 2520, 2500, 2550 and 2600	A4126
Test cell with holmium filter and fiber optic connectors for AZURA® Detector UVD 2.1S, UVD 2.1L and Smartline 200, 2520, 2500, 2550 and 2600 detectors	A4128
Standard test cell with stray light filter, WG280, for AZURA® UVD 2.1S, UVD 2.1L and Smartline 200, 2520, 2500, 2550 and 2600	A4146
Test cell with stray light filter, WG280, and fiber optic connectors for AZURA® Detector UVD 2.1S, UVD 2.1L and Smartline 2520, 2500 and 2600	A4148
Standard CombiLight test cell for AZURA® VWD 2.1L	AMI90



A9842



A9843



A9844

Waste tubing kits

Waste tubing kit for LightGuide flow cells, 1/16", 0.5 mm ID, 1 500 mm length	A9842
Waste tubing kit for UV flow cells, 1/16", 0.5 mm ID, 1 500 mm length	A9843
Waste tubing kit for UV flow cells, 1/8", 2 mm ID, 1 500 mm length	A9844
Waste tubing kit for AZURA® Detector RID 2.1L, 1/16", 0.9 mm ID, 1 500 mm length	A9841
Waste tubing kit CombiLight flow cells with UNF 1/4-28, 1/16" or 1/8", 0.5 or 2 mm ID, 2 000 mm length	A98431
Waste tubing kit CombiLight flow cells with UNF 5/16-24, 1/8" or 3/16", 2 or 3.2 mm ID, 2 000 mm length	A98441



AZZ00OC

External heat exchangers

AZURA® Heat Exchanger for analytical PressureProof flow cells, 14 µl internal volume	AZZ00OC



A1947



A1942-1

pH accessories

AZURA® pH kit, 100 ml/min, incl. pH electrode, storage vessel, dummy electrode and flow cell	A70091-2
AZURA® pH kit, 1000 ml/min, incl. pH electrode, storage vessel, dummy electrode and flow cell	A70091-3
AZURA® pH flow cell, 100 ml/min	A1943
AZURA® pH flow cell, 1000 ml/min	A1946
AZURA® storage vessel for pH electrode	A1947
AZURA® pH dummy electrode	A1942-1
KNAUER pH electrode for AZURA® VWD 2.1L, pH 2.1S and CM 2.1S	A1933-3

Adjustable flow splitters

Analytical post-column flow splitters for flow rates of 0.25 – 5.0 ml/min

The default inlet flow for calibration is 1.0 ml/min. However, please always specify your inlet flow (0.25 – 5 ml/min) before order. This way we can assure optimum pressure drop across the splitter, even if the inlet flow differs from the default calibration inlet flow rate.
For orders of spare parts, please provide the part number and the serial number of the splitter.

Flow splitter	Split ratio [min.]	Split ratio [max.]
A1770-1	50:1	1 000:1
A1770-2	15:1	300:1
A1770-3	5:1	100:1
A1770-4	1:1	20:1

Port size:	1/16" OD; UNF 10-32 thread
Max. operating pressure:	350 bar / 5 000 psi
Wetted materials:	Stainless steel, PEEK, Teflon
Dimensions (H × W × D):	12.2 cm × 10.2 cm × 13 cm (4.8" × 4" × 5.1")

Semi-preparative post-column flow splitters for flow rates of 5.0 – 40 ml/min

The default inlet flow for calibration is 20.0 ml/min. However, please always specify your inlet flow (5.0 - 40 ml/min) before order. This way we can assure optimum pressure drop across the splitter, even if the inlet flow differs from the default calibration inlet flow rate.
For orders of spare parts please provide the part number and the serial number of the splitter.

Flow splitter	Split ratio [min.]	Split ratio [max.]
A5816-2	1 000:1	20.000:1
A5816-3	100:1	2 000:1
A5816-4	15:1	300:1
A5816-5	1:1	20:1

Port size:	1/16" OD; UNF 10-32 thread
Max. operating pressure:	350 bar / 5 000 psi
Wetted materials:	Stainless steel, PEEK, Teflon
Dimensions (H × W × D):	12.2 cm × 10.2 cm × 13 cm (4.8" × 4" × 5.1")

Preparative post-column flow splitters

The preparative flow splitters are set to a custom split ratio. Therefore, please always specify your inlet flow and the desired split ration before order.
Please note, that the port of the low flow rate outlet has a UNF 10-32 thread and is for 1/16" OD capillaries. For the ports of the Inlet and the high flow rate outlet we offer also versions with UNF 5/16-24 thread for 1/8" OD capillaries as indicated in the table below.
For orders of spare parts please provide the part number and the serial number of the splitter.

Flow splitter	Flow rate	Port size (inlet & outlet of high flow)
A5815-1	40 – 125 ml/min	1/16" OD UNF 10-32 thread
A5815-2	75 – 200 ml/min	1/16" OD UNF 10-32 thread
A5815-3	75 – 200 ml/min	1/8" OD UNF 5/16-24 thread
A5815-4	100 – 1 000 ml/min	1/16" OD UNF 10-32 thread
A5815-5	100 – 1 000 ml/min	1/8" OD UNF 5/16-24 thread

Max. operating pressure:	350 bar / 5 000 psi
Wetted materials:	Stainless steel, PEEK, Teflon
Dimensions (H × W × D):	12.2 cm × 10.2 cm × 13 cm (4.8" × 4" × 5.1")

Further information:
www.knauer.net/en/Systems-Solutions/Accessories-and-supplies/FlowSplitter



A5815-1

Valve accessories

Fittings for 1/8" valves of V 4.1 valve generation



A7205



A7206



A7207



A7212

Bushings for 1/8" UNF 1/4-28 coned

1/8" Bushing, short, for UNF 1/4-28 thread, SSt	A7205
1/8" Bushing, long, for UNF 1/4-28 thread, SSt	A7206
1/8" Bushing, long, UNF 1/4-28 thread, SSt, for biconical sealing	A7207
1/8'' Blind fitting, for UNF 1/4-28 thread, SSt	A7208
1/8'' Bushing with integrated sealing ring, for UNF 1/4-28 thread, PEEK	A7209
1/8" Bushing for bionical sealing, UNF 1/4-28 thread, PEEK	A7210
1/8'' Bushing with integrated seal ring, for UNF 1/4-28 thread, PCTFE	A7211
1/8" Blind plug, for UNF 1/4-28 thread, PEEK	A7212



A7213



A7214



A7215



A7217

Ferrules, seal rings and clamp rings for 1/8" UNF 1/4-28 coned

1/8" Ferrule for wrench-tight fittings, for ports with UNF 1/4-28 thread, SSt	A7213
Split-grooved clamp ring for 1/8" capillary, for ports with UNF 1/4-28 thread, SSt	A7214
Split-grooved clamp ring for 1/8'' capillary, for ports with UNF 1/4-28 thread, Titanium	A7215
Biconical seal ring for 1/8" capillary, for ports with UNF 1/4-28 thread, PTFE	A7216
Seal ring for 1/8" capillary, for ports with UNF 1/4-28 thread, PEEK	A7217



A7218



A7219



A7220



A7221

Adapters and couplings for 1/8" UNF 1/4-28 coned

Coupling to connect two 1/8" capillaries, 1/8" (UNF 1/4-28) to 1/8" (UNF 1/4-28), SSt	A7218
Coupling to connect two 1/8" capillaries, 1/8" (UNF 1/4-28) to 1/8" (UNF 1/4-28), Titanium	A7219
Coupling to connect two 1/8" capillaries, 1/8" (UNF 1/4-28) to 1/8" (UNF 1/4-28), PEEK	A7220
Coupling to connect 1/16" with 1/8" capillary 1/8" (UNF 1/4-28) to 1/16" (UNF 10-32), SSt	A7221
Coupling to connect 1/16" with 1/8" capillary, 1/8" (UNF 1/4-28) to 1/16" (UNF 10-32), Titanium	A7222
Coupling to connect 1/16" with 1/8" capillary, 1/8" (UNF 1/4-28) to 1/16" (UNF 10-32), PEEK	A7223
T-connector to connect 3 capillaries with 1/8'' OD (material: stainless steel, Swagelok®)	A58260
Adapter to connect a capillary with 1/16" OD (thread: 10-32 UNF) to 1/8" (1/4-28 UNF coned), PEEK	A7224

KNAUER

Sample loops

Sample loops 1/16" SST incl. fittings

These stainless steel sample loops are designed to be used for 1/16" injection valves with a UNF 10-32 thread.

Sample loop,1 µl, stainless steel, 0.1 mm ID	A05642
Sample loop, 2 µl, stainless steel, 0.1 mm ID	A05643
Sample loop, 5 µl, stainless steel, 0.25 mm ID	A05644
Sample loop, 10 µl, stainless steel, 0.25 mm ID	A05645
Sample loop, 20 µl, stainless steel, 0.25 mm ID	A05646
Sample loop, 50 µl, stainless steel, 0.45 mm ID	A05647
Sample loop, 100 µl, stainless steel, 0.45 mm ID	A05648
Sample loop, 200 µl, stainless steel, 1 mm ID	A0565
Sample loop, 350 µl, stainless steel, 1 mm ID	A142615
Sample loop, 500 µl, stainless steel, 1 mm ID	A0566
Sample loop, 700 µl, stainless steel, 1 mm ID	A142616
Sample loop, 1000 µl, stainless steel, 1 mm ID	A0567
Sample loop, 2000 µl, stainless steel, 1 mm ID	A0568
Sample loop, 5000 µl, stainless steel, 1.6 mm ID	A0586-2

Sample loops 1/8" SST incl. fittings

These stainless steel sample loops are designed to be used for 1/8" injection valves. Based on the port geometry of the valve we offer two variants. For our older valve generations (e.g. V 2.1 valves) please use the one with M8x1 fittings.
For our current V 4.1 valve generation please choose the variant with UNF 1/4-28 fittings. If you are not sure which sample loop to select, you can check the thread specification for your individual valve on our website.

1 ml sample loop, stainless steel, 2.2 mm ID, incl. M8x1 fittings	A1043
1 ml sample loop, stainless steel, 2.2 mm ID, incl. UNF 1/4-28 fittings	A142609
2 ml sample loop, stainless steel, 1.6 mm ID, incl. M8x1 fittings	A1044
2 ml sample loop, stainless steel, 1.6 mm ID, incl. UNF 1/4-28 fittings	A142610
10 ml sample loop, stainless steel, 1.76 mm ID, incl. M8x1 fittings	A0843
10 ml sample loop, stainless steel, 1.76 mm ID, incl. UNF 1/4-28 fittings	A142611

Sample loops 1/16" PEEK incl. fittings

Sample loop, 10 µl, PEEK, 345 bar, 0.25 mm ID	A1058
20 µl, PEEK, 345 bar, 0.25 mm ID	A1059-1
20 µl, PEEK, 345 bar, 0.5 mm ID	A1059
Sample loop, 50 µl, PEEK, 240 bar, 0.75 mm ID	A1060
Sample loop, 100 µl, PEEK , 240 bar, 0.75 mm ID	A0508
Sample loop, 200 µl, PEEK, 240 bar, 0.75 mm ID	A1061
Sample loop, 500 µl, PEEK, 240 bar, 0.75 mm ID	A1057
Sample loop, 1000 µl, PEEK, 240 bar, 0.75 mm ID	A0423
Sample loop, 2000 µl, PEEK, 240 bar , 0.75 mm ID	A0785

Sample loops 1/8" PEEK incl. fittings

5 ml sample loop, PEEK, 50 bar, 1/16" ID, incl. M8x1 fittings	A78980
5 ml sample loop, PEEK, 50 bar, 1/16" ID, incl. UNF 1/4-28 fittings	A142612
10 ml sample loop, PEEK, 50 bar, 1/16" ID, incl. M8x1 fittings	A78985
10 ml sample loop, PEEK, 50 bar, 1/16" ID, incl. UNF 1/4-28 fittings	A142613

Syringes & injection ports



A0723

TECH TIP

For full-loop injections an overfilling with sample of two to five loop volumes is recommended to ensure precise and reproducible results. Therefore, choose a syringe that exceeds the loop volume by the mentioned factor.

Injection syringes for 1/16" injection port

Injection syringe 10 µl	A0723
Injection syringe 25 µl	A0724
Injection syringe 50 µl	A0725
Injection syringe 100 µl	A0726
Injection syringe 250 µl	A0727
Injection syringe 500 µl	A0728
Injection syringe 1 000 µl	A0729
Injection syringe 2 500 µl	A0730

VariLoops NEW

Multiple injection loop, 20 ml, stainless steel, 1/16"	A1055AA
Multiple injection loop, 40 ml, stainless steel, 1/16"	A1055XB
Multiple injection loop, 20 ml, stainless steel, 1/8"	A1160AA
Multiple injection loop, 40 ml, stainless steel, 1/8"	A1160XB
Replacement frit for VariLoop, 2 µm, 1/16"	A10551



A0653

Luer-Lock glass syringes for 1/8" injection port

Luer-Lock glass syringe, 10 ml	A0573
Luer-Lock glass syringe, 20 ml	A0653



A0555



A0328



A03281



A0505

Loop filling ports

Guide for the injection port of manual injection valves to make the insertion of different sized injection needles safer.	A0555
Injection Port, stainless steel, 1/16"	A0328
Injection Port, PEEK, 1/16"	A03281
Injection Port, stainless steel, 1/8"	A0505
Injection Port, PEEK, 1/8"	A05051
Injection Port, UNF 1/4-28, PEEK, 1/8"	A05053

Column and eluent tempering



A70060-1



A70054V4



A57024



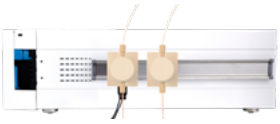
A57026

Eluent & column heating

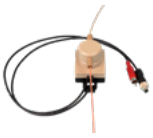
AZURA® ELH 2.1L one-heating section device with plugs for one column heating sleeve and two external temperature sensors	A70060-1
AZURA® ELH 2.1L two-heating section device with plugs for two column heating sleeves and two external temperature sensors	A70060-2
Eluent heating device (1 channel), 1/16", ambient to 100 °C, clean room compatible, including two external temperature sensors, 5.7" display	A70054V3
Eluent heating device (2 channels), 1/16", ambient to 100 °C, clean room compatible, including two external temperature sensors, with plugs to connect one column heating sleeve, 5.7" display	A70054V4
Eluent heating device (2 channels), 1/16", ambient to 60 °C, clean room compatible, including two external temperature sensors, with plugs to connect one column heating sleeve, 5.7" display	A70054V6
Temperature controller for column heating sleeve	A57024
Heating sleeve for HPLC column 150 × 20 mm HM D = 25..57* L = 193 mm 100 °C, 230 V, 200 W, Pt100	A57026
Heating sleeve for HPLC column 250 × 20 mm HM D = 25..57* L = 293 mm 100 °C, 230 V, 200 W, Pt100	A57027
Heating sleeve for HPLC column 150 × 30 mm HM D = 38..70* L = 203 mm 100 °C, 230 V, 400 W, Pt100	A57028
Heating sleeve for HPLC column 250 × 30 mm HM D = 38..70* L = 303 mm 100 °C, 230 V, 500 W, Pt100	A57029
Heating sleeve for HPLC column 150 × 50 mm HM D = 60..100* L = 211 mm 100 °C, 230 V, 500 W, Pt100	A57030
Heating sleeve for HPLC column 250 × 50 mm HM D = 60..100* L = 311 mm 100 °C, 230 V, 800 W, Pt100	A57031
Heating sleeve for HPLC costum-made up to 350 × 50 mm	A57032
Heating sleeve for HPLC costum-made up to 350 × 50 mm (moisture-proof, for clean room use)	A57034

NEW

Purification accessories



AZG10



AZG10-1



AZG10-2



A70083

Pressure control for delta pressure measurement up to 250 ml/min for 1/16" and 1/8", incl. interface box	AZG10
External pressure sensor up to 250 ml/min for 1/16" and 1/8"	AZG10-1
External pressure sensor for up to 1 000 ml/min for 1/4", 0 – 10 bar, analog output	AZG10-2
External pressure sensor for up to 1 000 ml/min for 1/4", 0 – 10 bar, LAN	AZG10-3
External pressure sensor for up to 1 000 ml/min for 1/4", 0 – 10 bar, LAN, biocompatible	AZG10-4
External pressure sensor for up to 1 000 ml/min for 1/8", 0 – 50 bar, LAN	AZG10-5
Air sensor (1/16", 1/8") for AZURA® Bio LC with one air sensor and wiring for up to 4 air sensors	A700921
Additional air sensor for AZURA® Bio LC for 1/16" or 1/8" tubing	A700922
Air sensor (1/4") for AZURA® Bio LC with one air sensor and wiring for up to 4 air sensors	A70083
Additional Air Sensor for AZURA® BIO LC for 1/4" tubing	A70083-1
AZURA® Organizer for attachment of columns and FPLC accessories to an AZURA® device or system	A70085
Clamp for AZURA® Organizer 12 mm	A70085-1
Clamp for AZURA® Organizer 16 mm	A70085-2
Clamp for AZURA® Organizer 25 mm	A70085-3
AZURA® Click rail to attach IFU 2.1 LAN, air sensors, pressure sensors, pH flow cells or the AZURA® Organizer to AZURA® L devices	A70089
Don't forget to order! Power Supply: Distribution Box 24 V for 6 devices like air sensor, external pressure sensor, IFU 2.1 LAN	AZS80SA

Lab equipment



A0569



A0851



A0809



A9865

Capillary and tube cutter

Tube cutter, suitable for all tubes	A0569
Capillary cutter for PEEK capillaries and tubings with OD up to 4 mm	A0851
Metal capillary cutting pliers for 1/16" capillaries	A0809
Metal capillary cutter for 1/8" capillaries	A9865

Tools



X0219



X0003



X0030

Wrenches & tightening tools

Torque wrench basic tool, 1 – 25 Nm, without plug-in head	X0219
Open-jaw plug-in head for Torque wrench X0219, 1 – 17 mm (for 100 – 1 000 ml pump head in-/outlet and LPG block)	X0220
Open-jaw plug-in head for Torque wrench X0219, 1 – 10 mm (for Smartline I pump heads)	X0221
Open-jaw plug-in head for Torque wrench X0219, 1 – 13 mm (for 10 – 50 ml Smartline II/ AZURA® pump heads in-/outlet)	X0222
Double open-end wrench, 1/4" and 5/16"	X0003
Double open-end wrench, 8/10 mm	X0030
Double open-end wrenches, 2 pcs., 1/4" and 5/16"	A0138
Tightening tools for PEEK fittings, blue, 1/16" fittings 1/4" hex head nut (10 – 32 threads)	A25030
Tightening tools for PEEK fittings, green, 1/32" fittings 3/16 hex head nut (6 – 40 threads)	A25031

A0137



A9864



A9870

Capillary graters and benders

Capillary grater for degrating of 1/16" stainless steel capillaries, can also be used to remove column filters	A0137
Capillary grater for degrating of 1/8" stainless steel capillaries	A9864
Tube bender for 1/8" and 3/16" tubings with an bend radius of 90°	A9870



A1033



A1033-2

Tool kits for AZURA® systems

Tool kit AZURA® for systems with PEEK or pre-cut capillary kits	A1033
Tool kit AZURA® for 1/16" systems, stainless steel	A1033-1
Tool kit AZURA® for 1/8" systems, stainless steel	A1033-2

Racks



A70010



A70011



Application example (devices not included in the scope of delivery)



A9860

LC racks - space saving solution for AZURA system setup

The Benchtop Racks area solution to install AZURA® L systems at space-limited sites, especially in cold rooms.

Benchtop rack: AZURA® S 300 × 160 × 210 mm (WHD), designed to place an AZURA® S device with a height of 129 mm beneath it	A70016
Benchtop rack: AZURA® L low 480 × 190 × 420 mm (WHD), designed to place AZURA® S or AZURA® L devices with a low height of 150 mm beneath it	A70010
Benchtop rack: AZURA® L high 480 × 430 × 420 mm (WHD), designed to place the Foxy fraction collector or AZURA® L devices beneath it	A70011
Benchtop rack: custom-made with individual dimensions	A70015
Product Riser AZURA®: Set of 4 feet that lift the device to a height of 28 mm for easy handling of the waste tube of the drainage system – for L devices before 2018	A9860

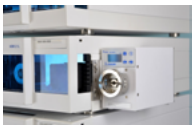
Mounting brackets



A9853



A9854-2



A9854-3



A9853-5

AZURA® mounting brackets

Mounting bracket AZURA® L for KNAUER manual injection valves	A9853
Mounting bracket AZURA® L for VICI valve drives	A9853-2
Mounting bracket AZURA® L for columns with 20 mm OD	A9853-3
Mounting bracket AZURA® L for KNAUER flow cells	A9853-5
Mounting bracket AZURA® L for prep sample loop	A9853-6
Mounting bracket AZURA® L for Hypershear mixing chambers	A9853-8
Mounting bracket AZURA® L Bio for manual KNAUER injection valve, pH flow cell and a prepacked column	A9854-1
Mounting bracket AZURA® S for manual KNAUER injection valve	A9854-2
Mounting bracket AZURA® L for AZURA® Valve Unifier VU 4.1, AZURA® Conductivity monitor CM 2.1S, AZURA® Degasser DG 2.1S or AZURA® UV detector UVD 2.1S on AZURA® L devices	A9854-3
Double mounting bracket AZURA® L for two AZURA® Valve Unifier VU 4.1 (both-sided)	A9854-4

Column holders



A9847



A1319



A70190



A0070A

LC column holder/multi column base

Column holder: Magnetic clip, for all KNAUER columns with 3, 4 and 4.6 mm ID, compatible with all AZURA® devices	A9847
Prism column holder for horizontal storage of HPLC columns on the lab bench, the most price attractive alternative to store your HPLC columns	A3983
Glass column holder, Stand, plate and 2 clamps, can hold one glass column with dimensions of 10 – 40 mm ID	A1319
Multi Column Base Bio 60 × 40 × 130 cm (W × D × H) for up to 3 MPLC columns with connection for cooling device	A70190
Multi Column Base including bosshead and clamps, serves as a holder for up to 3 columns with inner diameter up to 50 mm, especially made for preparative column solutions	A0070A
External pre-column holder for all KNAUER 3, 4 and 4.6 mm ID pre-column cartidges	A0037-3



A4364



A4368



A2820



A2820A

Accessories for LC column holder

3-finger clamps, long shaft, finger with silicone coating, clamp width 12–100 one piece	A4364
3-finger clamps, short shaft, finger with silicone coating, clamp width 12–100, one piece	A4364-1
Clamp for Multi Column Base, short shaft, to fix an HPLC column or other accessories to the Multi Column Base, for up to 20 mm ID columns	A4368
Clamp for Multi Column Base, long shaft, to fix an HPLC column or other accessories to the Multi Column Base, for up to 20 mm ID columns	A2820
Bosshead squared for Multi Column Base, used in combination with clamps with a long shaft on the Multi Column Base	A2820A

Installation accessories



A1071



A9862

Installation accessories

HPLC standard accessory kit	A1071
Installation Box Kit, box for small parts, KNAUER file folder and support sticker	A9862

KNAUER

Consumables

Fittings and bushings

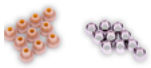
KNAUER K-connect fittings

The K-Connect system consists of a bushing, a split-grooved clamping ring, and a polymer sealing. The split-grooved clamping ring and polymer sealing are slipped over the capillary "back to back", while the bushing tightens all parts. K-Connect fingertight fittings can optionally be tightened further using wrenches if a higher backpressure resistance is needed.



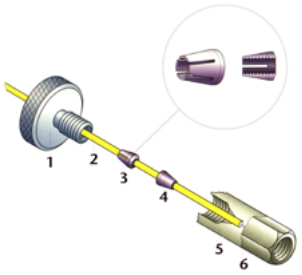
Article number	Description	Material	For capillary OD [inch]	Thread	Included ferrules	Max. back-pressure [bar]	Amount in set	Picture
A9646	Fingertight Fitting, long	PEEK	1/16"	UNF 10/32	Biconical sealing rings A1022	n/a	2	
A9646-1	Fingertight Fitting, long	PEEK	1/16"	UNF 10/32	Biconical sealing rings A1070	n/a	10	
A9645	Fingertight Fitting, long	SST	1/16"	UNF 10/32	Split-grooved clamping rings A0484 and polymer sealing rings A0139	1200	2	
A9645-1	Fingertight Fitting, long	SST	1/16"	UNF 10/32	Split-grooved clamping rings A0484 and polymer sealing rings A0139	1200	10	
A9647	Standard Fitting	SST	1/16"	UNF 10/32	Stainless steel ferrules A0110	1200	2	
A9647-1	Standard Fitting	SST	1/16"	UNF 10/32	Stainless steel ferrules A0110	1200	10	

Flat bottom fittings

Article number	Description	Material	For capillary OD [inch]	Thread	Included ferrules	Amount in set	Picture
A5829	Bushings flat bottom, super flangeless	PEEK	1/8"	1/4-28	without ferrules	10	
A58291	Bushings flat bottom, super flangeless	PEEK	1/16"	1/4-28	without ferrules	10	
A58292	Ferrules for super flangeless fittings, with lock ring	PEEK/Stainless steel	1/16"			10	
A58293	Ferrules for super flangeless fittings, with lock ring	PEEK/Stainless steel	1/8"			10	
A58294	Ferrules for super flangeless fittings, with lock ring	ETFE/Stainless steel	1/8"			10	

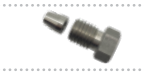
Dynaseal fittings

The DYNASEAL connecting system connects capillaries made out of stainless steel and PEEK, as well as PTFE and Tefzel tubings with low dead volumes. It allows maintenance-free operation and provides a long life. Suitable for UNF-threads of type 10/32.
The system consists of a bushing ① , a split-grooved clamping ring ③ and a polymer ferrule ④. The split-grooved clamping ring and polymerferrule are slipped over the capillary ② "back to back", while the bushing tightens all parts. Thus, leak-free operation is made possible.
DYNASEAL connections are pressure stable up to 450 bar. DYNASEAL can be optionally used with double-cone sealings made out of PEEK. In this case, pressure stability is accordingly reduced to 150 bar.



Article number	Description	Material	For capillary OD [inch]	Thread	Included ferrules	Max. back-pressure [bar]	Amount in set	Picture
A0108	Dynaseal bushings, short	SST	1/16"	UNF 10/32	Split-grooved clamping rings A0484 & polymer sealing rings A0139	450	4	
A1021	Dynaseal bushings, short	SST	1/16"	UNF 10/32	without ferrules	depends on ferrule	10	
A0181	Dynaseal bushings, long	SST	1/16"	UNF 10/32	Split-grooved clamping rings A0484 & polymer sealing rings A0139	450	3	
A1064	Dynaseal bushings, long	SST	1/16"	UNF 10/32	without ferrules	depends on ferrule	5	
A1020	Dynaseal bushings, short	SST	1/16"	UNF 10/32	Biconical sealing rings A1022	150	10	
A1069	Dynaseal bushings, long	SST	1/16"	UNF 10/32	Biconical sealing rings A1022	150	5	
A0736	Dynaseal bushings, long	SST	1/8"	M8x1	Split-grooved clamping rings A1239 & polymer Sealing rings A0232	n/a	4	
A0735	Dynaseal bushings, long		1/8"	M8x1	without ferrules	depends on ferrule	4	
A0644	Dynaseal bushings, short	SST	1/8"	M8x1	Split-grooved clamping rings A1239 & polymer sealing rings A0232	n/a	4	
A1201	Dynaseal bushings, long, hex head	SST	1/8"	M8x1	without ferrules	depends on ferrule	4	
A1201-1	Dynaseal bushings, long, hex head	PEEK	1/8"	M8x1	without ferrules	depends on ferrule	4	

Standard fittings, stainless steel

Article number	Description	Material	For capillary OD [inch]	Thread	Included ferrules	Max. back-pressure [bar]	Amount in set	Picture
A0112	Bushings short, wrench caliber 1/4"	SST	1/16"	UNF 10/32	without ferrules	1200	10	
A0113	Bushings short, wrench caliber 1/4"	SST	1/16"	UNF 10/32	without ferrules	1200	25	
A0115	Bushings long, wrench caliber 1/4"	SST	1/16"	UNF 10/32	without ferrules	1200	3	
A0116	Bushings long, wrench caliber 1/4"	SST	1/16"	UNF 10/32	without ferrules	1200	10	
A0830	Bushings, wrench caliber 10	SST	1/8"	M8x1	without ferrules	n/a	10	
A7227	Fitting Set for 1/8"	SST	1/8"	UNF 5/16-24	with ferrules	400	2	

Standard fittings, PEEK & polymer

Article number	Description	Material	For capillary OD [inch]	Thread	Included ferrules	Max. back-pressure [bar]	Amount in set	Picture
A0141	Bushings knurled, short	Polymer	1/16"	UNF 10/32	without ferrules	depends on ferrule	10	
A0142	Bushings knurled, short	Polymer	1/16"	UNF 10/32	without ferrules	depends on ferrule	30	
A0144	Bushings knurled, long	Polymer	1/16"	UNF 10/32	without ferrules	depends on ferrule	10	
A0145	Bushings knurled, short	Polymer	1/16"	UNF 10/32	integrated sealing cone	n/a	10	
A0584	Bushings short	PEEK	1/16"	UNF 10/32	integrated sealing cone	n/a	10	
A0733	Bushings short	Polymer	1/8"	M8x1	integrated sealing cone	n/a	10	
A2501	Bushings short, hex-head	PEEK	1/16"	UNF 10/32	integrated sealing cone	350	1	
A25011	Bushings short, hex-head	PEEK	1/16"	UNF 10/32	integrated sealing cone	350	5	

Article number	Description	Material	For capillary OD [inch]	Thread	Included ferrules	Max. back-pressure [bar]	Amount in set	Picture
A2502	Bushings long,hex-head	PEEK	1/16″	UNF 10/32	integrated sealing cone	350	1	
A25021	Bushings long, hex-head	PEEK	1/16″	UNF 10/32	integrated sealing cone	350	5	
A7226	Fitting Set for 1/8″	PEEK	1/8″	UNF 5/16-24	incl. ferrules and clamp rings	100	2	
A142607	Ferrule for VICI valve	CTFE	1/4″		suited for nut A142608	34	1	
A142608	Nut for 1/4″ valves	CTFE	1/4″	UNF 1/2-20	suited for ferrule A142607	34	1	

Fittings



A9646



A9646-1



A9645



A9645-1

K-Connect system

K-Connect fingertight fitting, PEEK, long, set of 2, incl. ferrule, UNF 10/32 thread for 1/16 inch K-Connect and PEEK capillaries	A9646
K-Connect fingertight fitting, PEEK, long, set of 10, incl. ferrule, UNF 10/32 thread for 1/16 inch K-Connect and PEEK capillaries	A9646-1
K-Connect fingertight fitting, stainless steel, long, set of 2, incl. ferrules, UNF 10/32 thread for 1/16 inch K-Connect capillaries	A9645
K-Connect fingertight fitting stainless steel, long, set of 10, incl. ferrules, UNF 10/32 thread for 1/16 inch K-Connect capillaries	A9645-1
K-Connect standard fitting, stainless steel, set of 2, incl. ferrule, UNF 10/32 thread for 1/16 inch K-Connect capillaries	A9647
K-Connect standard fitting, stainless steel, set of 10, incl. ferrule, UNF 10/32 thread for 1/16 inch K-Connect capillaries	A9647-1



A0108



A0181



A1020



A1069

DYNASEAL system

DYNASEAL connection system, 1/16", 4 short bushings, 4 clamping rings and 8 sealing rings	A0108
DYNASEAL connection system, 1/16", 3 long bushings, 3 clamping rings and 4 sealing rings	A0181
DYNASEAL connection system, 1/16", 10 short bushings, 10 biconical sealing rings	A1020
DYNASEAL connection system, 1/16", 5 long bushings, 5 biconical sealing rings	A1069
DYNASEAL connection system, 1/8", M8x1, 4 long bushings, 4 clamping rings and 8 sealing rings	A0736
DYNASEAL connection system, 1/8", M8x1, 4 short bushings, 4 clamping rings and 8 sealing rings	A0644

Ferrules and clamping rings



A0484

Split-grooved clamping rings

4 Split-grooved clamping rings for capillaries with 1/16" OD	A0484
4 Split-grooved clamping rings for capillaries with 1/8" OD	A1239
100 Split-grooved clamping rings for capillaries with 1/16" OD	A0482



A0139



A1062



A0232

Sealing rings

30 Sealing rings for capillaries with 1/16" OD, PETP	A0139
100 Sealing rings for capillaries with 1/16" OD, PETP	A0140
10 Sealing rings for capillaries with 1/16" OD, PEEK	A1062
10 Sealing rings for capillaries with 1/8" OD, PETP	A0232
10 Sealing rings for capillaries with 1/8" OD, PEEK	A1063



A1070



A1022



A0738

Biconical sealing rings

10 Biconical sealing rings for 1/16", PEEK	A1070
10 Biconical sealing rings for 1/16", PETP	A1022
10 Biconical sealing rings for 1/8", PETP	A0738



A0112



A0115

Bushings for capillaries, SST

10 Bushings for capillaries with 1/16" OD, stainless steel, wrench caliber 1/4", UNF 10-32, short	A0112
25 Bushings for capillaries with 1/16" OD, stainless steel, wrench caliber 1/4", UNF 10-32, short	A0113
3 Bushings for capillaries with 1/16" OD, stainless steel, wrench caliber 1/4", UNF 10-32, long	A0115
10 Bushings for capillaries with 1/16" OD, stainless steel, wrench caliber 1/4", UNF 10-32, long	A0116
10 Bushings for capillaries with 1/8" OD, stainless steel, M8x1, wrench caliber 10	A0830



A0110



A0874



A01101

Ferrules for capillaries

30 Ferrules for capillaries with 1/16" OD, stainless steel	A0110
100 Ferrules for capillaries with 1/16" OD, stainless steel	A0111
10 Ferrules for capillaries with 1/8" OD, stainless steel	A0874
10 Ferrules for capillaries with 1/16" OD, Hastelloy	A01101
10 Ferrules for capillaries with 1/16" OD, titanium	A01102



A0141



A0142



A0144



A0145

Bushings for capillaries, PEEK & polymer

Bushings for 1/16" capillaries, PETP, fingertight, UNF 10-32, short, 10 pcs.	A0141
Bushings for 1/16" capillaries, PETP, knurled, UNF 10-32, short, 30 pcs.	A0142
Bushings for 1/16" capillaries, PETP, fingertight, UNF 10-32, long, 10 pcs.	A0144
Bushings for 1/16" capillaries, PETP, with integrated sealing cone, fingertight, UNF 10-32, short, 10 pcs.	A0145
Bushings for 1/16" capillaries, PEEK, with integrated sealing cone, fingertight, UNF 10-32, 10 pcs.	A0584
Bushings for 1/8" capillaries, PETP, with integrated sealing cone, fingertight, M8x1, short,10 pcs.	A0733
Bushing for 1/16" capillaries, PEEK, with integrated sealing cone, wrench tight (Hex), UNF 10-32, short, 5 pcs.	A25011
Bushing for 1/16" capillaries, PEEK, long, wrench tight (Hex), with integrated sealing cone, 5 pcs.	A25021



A5829



A58291



A58292



A58293

Flat bottom fittings and adapters

Bushings flat bottom for 1/8" capillaries, PEEK, Super flangeless, 1/4 - 28, 10 pcs.	A5829
Bushings flat bottom for 1/16" capillaries, PEEK, Super flangeless, 1/4 - 28, 10 pcs.	A58291
Ferrules for 1/16" capillaries and flat bottom bushings, PEEK, with lock ring (stainless steel), for Super flangeless bushings, 10 pcs.	A58292
Ferrules for 1/8" capillaries and flat bottom bushings, PEEK, with lock ring (stainless steel), for Super flangeless bushings, 10 pcs.	A58293
Ferrules for 1/8" capillaries and flat bottom bushings, ETFE, with lock ring (stainless steel), for Super flangeless bushings, 10 pcs.	A58294
Adapter PEEK 1/8" flat bottom internal on 1/16" external 10/32 thread	A1982
Adapter to connect an 1/16"-OD capillary onto an female, coned M8x1 port	A05841
Adapter female 1/4-28 flat bottom to 1/2-20 UNF, for 1/4" VICI valves & pump head inlet A9868, PEEK	A142605
Adapter flat bottom for 1/4" UNF 1/2-20 female to 3/16" UNF 5/16-24 male, PCTFE	A142705

Blind fittings & connectors



A0146



A0582



A0734

Blind fittings / Plugs

10 Blind plugs, 1/16", knurled, UNF 10-32, short, PETP	A0146
30 Blind plugs, 1/16", knurled, UNF 10-32, short, PETP	A0147
10 Blind plugs, 1/16", knurled, UNF 10-32, short, PEEK	A0582
10 Blind plugs, 1/8", knurled, M8x1, short, PETP	A0734

Couplings & adapters



A0148



A0233



A0233-1



A1407

Couplings and adapters

Coupling to connect 2 capillaries with 1/16" OD (material: PEEK/PETP, thread: UNF10-32), including 2 bushings and sealing rings, 0.5 mm bore, suitable for classical HPLC, 1 pc.	A0148
Coupling to connect 2 capillaries with 1/16" OD (material: PEEK/PETP, thread: UNF10-32), including 2 bushings and sealing rings, 0.5 mm bore, suitable for classical HPLC, 5 pcs.	A0149
Coupling to connect 2 capillaries with 1/16" OD (material: PEEK, thread: 10-32 UNF), including 2 one-piece PEEK fittings, 0.5 mm bore, suitable for classical HPLC, 1 pc.	A0233
Coupling to connect 2 capillaries with 1/16" OD (material: PEEK, thread: 10-32 UNF), without fittings, 0.5 mm bore, suitable for classical HPLC, 1 pc.	A0233-1
Coupling to connect 2 capillaries with 1/16" and 1/8" OD (material: PEEK, thread: 10-32 UNF, M8x1), including 2 one piece fittings (1x 1/16", 1x 1/8"), 1 mm bore, 1 pc.	A1407
Coupling to connect 2 capillaries with 1/8" OD (material: PEEK, thread: M8x1), including 2 one piece fittings 1/8", 2 mm bore, suitable for preparative HPLC, 1 pc.	A14071
Adapter, female, for 1/4" VICI valves and pump head inlet, 1/4-28 flat bottom to 1/2-20 UNF, (material: PEEK)	A142605



A0117V1



A2512



A0845



A0480

Couplings, SST/Titanium

Coupling to connect 2 capillaries with 1/16" OD (material: titanium, thread: 10-32 UNF), including 2 bushings and ferrules, 0.5 mm bore, suitable for classical HPLC, 1 set	A0117V1
Coupling to connect 2 capillaries with 1/16" OD (material: stainless steel, thread: 10-32 UNF), including 2 bushings and ferrules, 0.5 mm bore, suitable for classical HPLC, 1 set	A0117
Coupling to connect 2 capillaries with 1/16" OD (material: stainless steel, thread: 10-32 UNF), including 2 bushings and ferrules, 0.5 mm bore, suitable for classical HPLC, 5 sets	A0118
Coupling to connect 2 capillaries with 1/16" OD (material: stainless steel, thread: 10-32 UNF), including 2 bushings and ferrules, 0.5 mm bore, suitable for classical HPLC, 25 sets	A0119
Coupling to connect 2 capillaries with 1/8" OD (material: stainless steel, thread: M8x1), including 2 bushings and ferrules, 2 mm bore, suitable for preparative HPLC, 1 set	A2512
Coupling to connect a capillary with 1/16" OD to a capillary with 1/8" OD (material: stainless steel, thread: M8x1, 10-32 UNF), 1 mm bore, 1 set	A2513
Coupling Dynaseal to connect a capillary with 1/16" OD to a capillary with 1/8" OD (material: stainless steel, thread: M8x1, 10-32 UNF), including Dynaseal bushings and ferrules (1x 1/16", 1x 1/8"), 1 mm bore, 1 set	A0485
Coupling Dynaseal to connect 2 capillaries with 1/8" OD (material: stainless steel, thread: M8x1), including 2 Dynaseal bushings and ferrules, 2 mm bore, suitable for preparative HPLC, 1 set	A0480



A58263



A58264



A582886



A582891

SST Swagelok® unions & reducing unions

Union to connect 2 capillaries with 1/4" OD, material: stainless steel, Swagelok®	A58263
Reducer to connect a capillary with 3/8" OD to a capillary with 1/4" OD, material: stainless steel, Swagelok®	A58264
Reducer to connect a capillary with 8 mm OD to a capillary with 1/4" OD, material: stainless steel, Swagelok®	A58265
Reducer to connect a capillary with 1/8" OD to a capillary with 1/4" OD, material: stainless steel, Swagelok®	A58266
Reducer to connect a capillary with 1/16" OD to a 1/8" OD pipe, material: stainless steel, Swagelok®	A58270
Reducer to connect a capillary with 1/8" OD to a 1/4" pipe union, material: stainless steel, Swagelok®	A58271
Reducer for 1/4" OD capillary to 1/8" OD pipe socket, material: stainless steel, Swagelok®	A582713
Reducer to connect a 1/16" tube socket to 1/4" pipe union, material: stainless steel, Swagelok®	A58273
Bulkhead Union for 1/8", stainless steel, Swagelok®	A58281
Bulkhead Union for 1/4", stainless steel, Swagelok®	A582811
Reducer to connect a capillary with 4 mm OD to a 1/8" pipe union, material: stainless steel, Swagelok®	A58282
Reducer to connect a capillary with 10 mm OD to a capillary with 1/4" OD, material: stainless steel, Swagelok®	A58257
Reducer to connect a capillary with 12 mm OD to a capillary with 1/4" OD, material: stainless steel, Swagelok®	A58258
Reducer to connect a capillary with 1/4" OD to a capillary with 1/8" OD, material: stainless steel, Swagelok®	A582881
Reducer to connect a capillary with 1/4" OD to a 3/16" OD pipe, material: stainless steel, Swagelok®	A582895
Reducer to connect a capillary with 1/4" OD to a capillary with 3/16" OD, material: stainless steel, Swagelok®	A582894
Bulkhead Union to connect two capillaries with 1/16" OD, material: stainless steel, Swagelok®	A582882
Male connector to connect a 1/4" OD capillary to a 1/4" male NPT adapter (material: stainless steel) for A9861	A58267
Male connector to connect a 4 mm OD capillary to a 1/4" male NPT adapter (material: stainless steel) for A9861	A58268
Male connector to connect a 1/8" OD capillary to a 1/4" male NPT adapter (material: stainless steel) for A9861	A58269
Blind plug for 1/4" OD capillary, material: stainless steel, Swagelok®	A582883
Blind plug for 1/8" OD capillary, material: stainless steel, Swagelok®	A582884
Blind plug for 1/16" OD capillary, material: stainless steel, Swagelok®	A582885
Blind plug for 6 mm OD capillary, material: stainless steel, Swagelok®	A582892
Blind plug for 12 mm OD capillary, material: stainless steel, Swagelok®	A582893
Ferrule set for a capillary with 1/4" OD, 1 front Ferrule/ 1 back Ferrule, material: stainless steel, Swagelok®	A582886
Ferrule set for a capillary with 1/8" OD, 1 front Ferrule/ 1 back Ferrule, material: stainless steel, Swagelok®	A582887
Ferrule set for a capillary with 1/16" OD, 1 front Ferrule/ 1 back Ferrule, material: stainless steel, Swagelok®	A582888
Gap Inspection Gauge for 1/8" OD, 2 mm and 3 mm female nuts, material: stainless steel, Swagelok®	A582890
Gap Inspection Gauge for 1/4", 3/8", 1/2" OD, 6 mm and 12 mm female nuts, material: stainless steel, Swagelok®	A582891
Tube Fitting Union to connect two 1/8" OD capillaries, material: stainless steel, Swagelok®	A582671

Connectors


A2511


A0120


A58260


A58261

Metal T-connectors

T-connector to connect 3 capillaries with 1/8" OD (material: stainless steel, thread: M8x1), including 3 bushings and ferrules	A2511
T-connector to connect 3 capillaries with 1/16" OD (material: stainless steel, thread: 10-32 UNF), including 3 bushings and ferrules	A0120
Reducer to connect a capillary with 12 mm OD to a capillary with 1/4" OD, material: stainless steel, Swagelok®	A58258
T-connector to connect 3 capillaries with 1/4" OD (material: stainless steel, Swagelok®)	A58261
T-connector to connect 3 capillaries with 1/4" OD (material: titanium, Swagelok®)	A58262


A150-1


A2511-1


A0150

Polymer T-connectors

T-connector to connect 3 capillaries with 1/16" OD (material: PETP/POM, thread: 10-32 UNF, coned), inclusive 3 bushings and sealing rings	A0150
T-connector to connect 3 capillaries with 1/16" OD (material: PEEK, thread: 10-32 UNF, coned), without bushings	A150-1
T-connector to connect 3 capillaries with 1/8" OD (material: PEEK, thread: M8x1, coned), including 2 one piece 1/8"-PEEK fittings	A2511-1

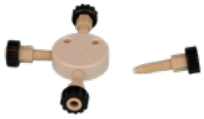

A0121


A1096


A58272

SST X-connectors

X-connector to connect 4 capillaries with 1/16" OD (material: stainless steel, thread: 10-32 UNF), including 4 bushings and ferrules	A0121
X-connector to connect 4 capillaries with 1/8" OD (material: stainless steel, thread: M8x1), including 4 bushings and ferrules	A1096
X-connector to connect 4 tubings with 1/4" OD (material: stainless steel, Swagelok®) for 1 000 ml/min systems	A58272



A0151

Polymer X-connectors

X-connector to connect 4 capillaries with 1/16" OD (material: PEEK, thread: 10-32 UNF), including 4 one-piece fittings	A0151



A5800



A5805



A5804



A5805

Pressure release valves

Pressure Release Valve Kit for AZURA® pump P 2.1L and 80P (25 to 50 bar), 1/8", stainless steel, cross piece titanium	A5800
Pressure Release Valve Kit for AZURA® pump P 2.1L and 80P (3.4 to 24 bar), 1/8", stainless steel, cross piece titanium	A5801
Pressure Release Valve for AZURA® pump P 2.1L and 80P (without spring), 1/4", stainless steel	A5802
Backpressure Regulator/pressure release valve kit for 1/16" OD tubing, stainless steel, provides a constant backpressure of 3 bar, contains pressure release valve tee and fittings for 1/16"	A58051
Backpressure Regulator/pressure release valve for 1/16" OD tubing, stainless steel, provides a constant backpressure of 52 bar, contains pressure release valve tee and fittings for 1/16"	A58051-1
Spring for pressure release valve, 25 - 50 bar	M1070
Spring for pressure release valve, 3.4 - 24 bar	M1080
Backpressure Regulator/pressure relief valve for 1/8" and 1/16" OD tubing, 134 µl volume, PEEK, provides a constant backpressure of 1.4 bar (20 psi), contains pressure release valve tee and fittings for 1/8" and 1/16"	A58041
Backpressure Regulator for 1/16" OD tubing, 134 µl volume, PEEK, provides a constant backpressure of 0.3 bar (5 psi), contains Y assembly and fittings	A5804-1



A70087



A70088



A70084

Backpressure regulators

Backpressure Regulator for 1/16" OD tubing, 10 - 32 threads, PEEK, Range 1 – 20 bar (15 – 300 psi)	A70087
Backpressure Regulator for 1/16" OD tubing, 10 - 32 threads, PEEK, Range 20 – 103 bar (300 – 1 500 psi)	A70088
Backpressure Regulator for 1/16" OD tubing, 10 - 32 threads, stainless steel, Range 90 – 300 bar (1 300 – 4 200 psi)	A70084
Spare membranes for Backpressure Regulators A70084, A70087, A70088	A70082
Backpressure Regulator/pressure relief valve for 1/8" and 1/16" OD tubing, 134 µl volume, PEEK, provides a constant backpressure of 1.4 bar (20 psi), contains pressure release valve tee and fittings for 1/8" and 1/16"	A5804
Backpressure Regulator for 1/16" OD tubing, 134 µl volume, PEEK, provides a constant backpressure of 0.3 bar (5 psi), contains Y assembly and fittings	A5804-1

Capillaries and start-up kits



A0130



KNAUER Capillaries, straight

Capillaries 1/16", SST

Stainless steel, 1/16" OD, 0.1 mm ID, 300 cm length, 1 pc.	A0130
Stainless steel, 1/16" OD, 0.25 mm ID, 300 cm length, 1 pc.	A0131
Stainless steel, 1/16" OD, 0.5 mm ID, 300 cm length, 1 pc.	A0132
Stainless steel, 1/16" OD, 0.7 mm ID, 300 cm length, 1 pc.	A0133
Stainless steel, 1/16" OD, 1 mm ID, 300 cm length, 1 pc.	A0134
Stainless steel, 1/16" OD, 0.1 mm ID, 10 cm length, 10 pcs.	A0123
Stainless steel, 1/16" OD, 0.1 mm ID, 20 cm length, 10 pcs.	A0124
Stainless steel, 1/16" OD, 0.1 mm ID, 30 cm length, 10 pcs.	A0125
Stainless steel, 1/16" OD, 0.25 mm ID, 10 cm length, 10 pcs.	A0126
Stainless steel, 1/16" OD, 0.25 mm ID, 20 cm length, 10 pcs.	A0127
Stainless steel, 1/16" OD, 0.25 mm ID, 30 cm length, 10 pcs.	A0128

Capillaries 1/16", titanium

Titanium, 1/16" OD, 0.7 mm ID, 50 cm length, 1 pc.	A0506

Capillaries 1/4", SST

Stainless steel, 1/4" OD, 4.6 mm ID, 100 cm length, straight, 1 pc.	A01322-4
Stainless steel, 1/4" OD, 4.6 mm ID, 150 cm length, straight, 1 pc.	A01322-5
Stainless steel, 1/4" OD, 4.6 mm ID, 200 cm length, straight, 1 pc.	A01322-6

Capillaries 1/8", SST

Stainless steel, 1/8" OD, 1.6 mm ID, 150 cm length, oval bent, 1 pc.	A0639
Stainless steel, 1/8" OD, 2.2 mm ID, 150 cm length, oval bent, 1 pc.	A0640
Stainless steel, 1/8" OD, 2.2 mm ID, 100 cm length, straight, 1 pc.	A0640-4
Stainless steel, 1/8" OD, 2.2 mm ID, 150 cm length, straight, 1 pc.	A0640-5
Stainless steel, 1/8" OD, 2.2 mm ID, 200 cm length, straight, 1 pc.	A0640-6
Stainless steel, 1/8" OD, 1.6 mm ID, 100 cm length, straight, 1 pc.	A0639-4
Stainless steel, 1/8" OD, 1.6 mm ID, 150 cm length, straight, 1 pc.	A0639-5
Stainless steel, 1/8" OD, 1.6 mm ID, 200 cm length, straight, 1 pc.	A0639-6



AZF110 / AZF120

Trouble-free (U)HPLC start-up kits for AZURA® Analytical systems

Start-up Kit for AZURA® Analytical ULDC and UHPLC systems	AZF110
Start-up Kit for AZURA® Analytical HPLC 862 bar systems	AZF120



Trouble-free connection



Trouble-free connection

Trouble-free (U)HPLC connections with reusable zero dead volume fitting for 1/16”

	Recommended for AZURA® Analytical systems	
MarvelXACT™, stainless steel, 0.254 mm ID, 150 mm length		AZF121
MarvelXACT™, stainless steel, 0.254 mm ID, 250 mm length		AZF122
MarvelXACT™, stainless steel, 0.254 mm ID, 350 mm length		AZF123
MarvelXACT™, stainless steel, 0.254 mm ID, 500 mm length		AZF124
MarvelXACT™, stainless steel, 0.254 mm ID, 600 mm length	Pump - Autosampler	AZF125
MarvelXACT™, stainless steel, 0.125 mm ID, 150 mm length		AZF111
MarvelXACT™, stainless steel, 0.125 mm ID, 250 mm length		AZF112
MarvelXACT™, stainless steel, 0.125 mm ID, 350 mm length	Column - Detector / Detector 1 - Detector 2	AZF113
MarvelXACT™, stainless steel, 0.125 mm ID, 500 mm length	Autosampler - Column ≤ 150 mm length / Column - Detector	AZF114
MarvelXACT™, stainless steel, 0.125 mm ID, 600 mm length	Autosampler - Column ≥ 250 mm length	AZF115
MarvelXACT™, stainless steel, 0.1 mm ID, 150 mm length		AZF101
MarvelXACT™, stainless steel, 0.1 mm ID, 250 mm length		AZF102
MarvelXACT™, stainless steel, 0.1 mm ID, 350 mm length	Column - Detector / Detector 1 - Detector 2	AZF103
MarvelXACT™, stainless steel, 0.1 mm ID, 500 mm length	Autosampler - Column ≤ 150 mm length / Column - Detector	AZF104
MarvelXACT™, stainless steel, 0.1 mm ID, 600 mm length	Autosampler - Column > 250 mm length	AZF105
MarvelXACT™, PEEK-lined stainless steel, 0.075 mm ID, 500 mm length	Autosampler - Column ≤ 150 mm length / Column - Detector (UHPLC and ULDC)	AZF05
MarvelXACT™ PEEK-lined stainless steel, 0.075 mm ID, 350 mm length	Column - Detector (UHPLC and ULDC) / Detector 1 - Detector 2	AZF05-1
MarvelXACT™, PEEK-lined stainless steel, 0.075 mm ID, 600 mm length		AZF05-2
Flexible stainless steel capillary, 0.18 mm ID, 900 mm length, without fittings	Pump - Autosampler for systems with two AZURA® L detectors or benchtop installation	AZF55-1

AZURA® Capillary start-up kit, SST

AZURA® Start-up kit 1/16", stainless steel, capillary kit	A9849
AZURA® Start-up kit 1/16", stainless steel, semi-prep, capillary kit	A9849-1
AZURA® Start-up kit 1/8", stainless steel, capillary kit	A9850
AZURA® Start-up kit 1/16", stainless steel, 0.25 mm ID precut capillaries	AZF70
AZURA® Accessory kit for ScaleUp system, 1/16" (0.25, 0.5, and 0.7 mm ID), stainless steel	A9850-1
AZURA® Start-up kit 1/4" HPG, stainless steel, set of capillaries and fittings	A9850-2
AZURA® Start-up kit 1/4" LPG, stainless steel, Set of capillaries and fittings	A9850-3



A50041



A9849-2



A70501

AZURA® Capillary start-up kits for special HPLC systems

AZURA® GPC Cleanup Start-up kit, Tefzel-(ETFE) tubing, OD 1/16", ID 0.7 mm	A50041
AZURA® Accessory kit for ScaleUp system, 1/16" (0.25, 0.5, and 0.7 mm ID), stainless steel	A9850-1
AZURA® Capillary Start-up kit for educational system 1/16", stainless steel	A9849-2
AZURA® Start-up kit, PEEK, for Analytical HPLC System, up to 5 ml/min or 300 bar	A70501

Tubing

Articles grouped under the expression "by meter" can be shipped in the desired length, by simply ordering it multiple times.
E.g. ordering 3.4 x A2528 will result in capillary with a length of minimum 3.4 meters.

Note: If you need tubings with an exact length, please contact KNAUER directly.



A70500



A70500A



A70600



A70300

Tubing start-up kits for FPLC

AZURA® FPLC Start-up kit, PEEK, 1/16" for 10 ml/min FPLC systems	A70500
AZURA® FPLC Start-up kit, transparent FEP, 1/16" for FPLC systems up to 10 ml/min and 20 bar	A70500A
AZURA® FPLC Start-up kit, PEEK/Tefzel, 1/16" for 50 ml/min FPLC systems	A70600
AZURA® FPLC Start-up kit, FEP/PEEK, 1/8" for 100 ml/min - 500 ml/min FPLC systems	A70300
AZURA® FPLC Start-up kit, PEEK, 1/8" for 100 ml/min - 500 ml/min FPLC systems, up to 100 bar.	A70300A
AZURA® FPLC Start-up kit, PEEK/Tefzel, 1/16" for FPLC systems up to 100 ml/min	A70300B
AZURA® FPLC Start-up kit, 1/4" for 1000 ml/min FPLC systems	A70400
AZURA® Start-up kit 1/4" HPG, PFA, set of capillaries and fittings	A9850-4

A9869



A9869-1

Tubing, various OD, FEP

2.1 mm ID, 300 cm length, FEP tubing, 1/8" OD	A9869
0.81 mm ID, 300 cm length, FEP tubing, 1/16" OD	A9869-1



A2522

Tubing 1/16" OD, PEEK, by meter

0.13 mm ID, variable length, max. pressure 420 bar, red striped	A2522
0.18 mm ID, variable length, max. pressure 400 bar, yellow striped	A2523
0.25 mm ID, variable length, max. pressure 385 bar, blue striped	A2524
0.50 mm ID, variable length, max. pressure 350 bar, orange striped	A2525
0.75 mm ID, variable length, max. pressure 240 bar, green striped	A2526
1.00 mm ID, variable length, max. pressure 165 bar, gray striped	A2527
1.40 mm ID, variable length, max. pressure 50 bar, black striped	A2528

Tubing 1/8" OD, PEEK, by meter

0.75 mm ID, variable length, max. pressure 345 bar, natural	A2541
1.59 mm ID, variable length, max. pressure 220 bar, natural	A2540
2.00 mm ID, variable length, max. pressure 165 bar, natural	A2542



A0182-1

Tubing 1/16" OD, Tefzel™, by meter

0.25 mm ID, variable length, max. pressure 185 bar	A0182-1
0.75 mm ID, variable length, max. pressure 115 bar	A0183-1
1.0 mm ID, variable length, max. pressure 85 bar	A04781-1

Tubing 1/8" OD, ETFE, by meter

1.6 mm ID, variable length, max. pressure 70 bar	A0478-1

Tubing, various OD, PTFE, by meter

0.45 mm ID, variable length, max. pressure 150 bar, 1.6 mm (1/16") OD	A0152-1
0.9 mm ID, variable length, 1.6 mm OD	A04782-1
1.45 mm ID, variable length, max. pressure < 10 bar, 2 mm OD	A0153-1
1.5 mm ID, variable length, max. pressure 35 bar, 3.2 mm (1/8") OD	A0732-1
2 mm ID, variable length, 1/8" OD	A0873-1
3 mm ID, variable length, max. pressure 20 bar, 4 mm OD	A0154-1
7 mm ID, variable length, 9 mm OD	A1099-1
1.6 mm ID, variable length, 1/8" OD, black, anti-static	A3306
4.4 mm ID, variable length, 1/4" OD, black, anti-static	A3307

Tubing, various OD, PFA, by meter

PFA tubing, 1/4" OD, 4 mm ID, translucent, max. pressure 15.4 bar, variable length	A31891
PFA tubing, 1/8" OD, 1.6 mm ID, translucent, variable length	A31892
PFA tubing, 1/4" OD, 4.8 mm ID, translucent, max. pressure 15.4 bar, variable length	A31891-1
PFA tubing, 1/8" OD, 2.4 mm ID, translucent, variable length	A31892-1

Inline and pre-column filter, shut-off valves, and adapters



A3381



A00161



B2



A00164-1

Inline filters, SST, for HPLC

Inline Filter (prep.) 5-10 µm, stainless steel, max. flow rate 1000 ml/min (for 1/8" tubing)	A3381
Replacement frit for A3381 5-10 µm, stainless steel, max. flow rate 1000 ml/min	A33811
Inline Filter, PEEK body, stainless steel frit, 1/16", to protect your column, with 2 µm pore size, 3 pcs., easily connected directly to any column	A00161
UHPLC/HPLC pre-column filter, universal, 0.5 µm titanium frit, set of 5, stainless steel body, up to 1 034 bar	B2
Inline Filter, stainless steel, frit 0.5 µm, 0.2 µl, for 1/16" capillaries, 0.25 mm bore, up to 1375 bar	A00164
Frit 0.5 µm, 0.2 µl for Inline Filter, stainless steel with 0.25 mm bore up to 1375 bar, 5 pcs.	A00164-1



A3378



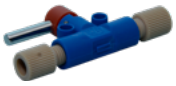
A3378-1



A00162

Inline and pre-column filter, biocompatible, for FPLC

Inline Filter, PEEK/Titanium, 1/16", biocompatible, to protect your column, with 2 µm pore size, titanium frit	A3378
Inline Filter, PEEK/Titanium, 1/16", biocompatible, to protect your column, with 10 µm pore size, titanium frit	A3379
Replacement Frits 2 µm for Inline Filter, PEEK/Titanium, biocompatible	A3378-1
Replacement Frits 10 µm for Inline Filter, PEEK/Titanium, biocompatible	A3379-1
Inline Filter, PEEK body, titanium frit, 1/16", to protect your column, with 0.5 µm pore size, 3 pcs., easily connected directly to any column	A00162
Inline Filter, PEEK body, titanium frit, 1/16", to protect your column, with 2 µm pore size, 3 pcs., easily connected directly to any column	A00163



A5811

Shut-off valves

Shut-off valve, PEEK, 1/16", including connectors (1/4–28 flat bottom)	A5811
Shut-off valve, PEEK, 1/8", including connectors (1/4"–28 flat bottom)	A5812



A1980



A7237



A7238

Adapters

Luer Adapter to 10-32, ETFE, female Luer to male 10/32 threads for injection, simply screw the adapter in the port of your injection valve	A1980
Adapter to connect a capillary with 1/8" OD (thread: 1/4-28 UNF coned) to 1/16" V4.1 valve (thread: 10-32 UNF coned), material: stainless steel	A7237
Adapter to connect a capillary with 1/8" OD (thread: 1/4-28 UNF) to 1/16" V4.1 valve (thread: 10-32 UNF coned), material: PEEK	A7238

Safety-caps



A59257



A59257-1



A59259



A59258

Safety caps sets for AZURA analytical systems

For isocratic systems, incl. filters, bottles and fittings	A59257
Safety Caps Set for HPG/LPG systems, incl. filters, bottles and fittings (4 pcs.)	A59257-1
For HPG systems, incl. filters, bottles and fittings (2 pcs.)	A59257-2
Eluent waste kit for all AZURA® Analytical systems, incl. filter, waste can and cap	A59258
Safety Cap set for AZURA® Preparative systems, for one eluent line, incl. filter, bottle and fittings	A59259
Waste Cap set for AZURA® Preparative systems, incl. filter, canister and fittings	A59259-1



A59260



A59261



A59231



A59234

Safety caps

Eluent Safety Cap, GL45 Thread, 2 ports, 1/4”-28 connection, including air valve and fittings	A59260
Eluent Safety Cap Prep, GL45 Thread, 2 ports for 1/4” tubing, including air filter and fittings	A59261
Eluent Safety Cap Prep, GL45 Thread, 2 ports for 1/4” tubing, including air filter and fittings	A59262
Eluent Safety Cap Filter, spare part, 6 months usable	A59263
VICI Cap, GL45 Thread, 3 ports, 1/4”-28 connection, including O-ring EPDM, nuts and ferrules	A59231
VICI Safety Cap with stopcocks, GL45 Thread, 3 ports, 1/4”-28 connection, including O-ring EPDM, nuts and ferrules	A59234
VICI Safety Cap with stopcocks, GL45 Thread, 4 ports, 1/4”-28 connection, including O-ring EPDM, nuts and ferrules	A59235
VICI Waste Cap, GL45 Thread, 3 ports 1/4”-28 connection, 1 × 10M × 1 for barbed hose adapter, including O-ring EPDM, nuts and ferrules	A59236
VICI Cap, GL45 Thread, 2 ports, 1/4”-28 connection, including O-ring EPDM, nuts and ferrules	A59232
VICI Cap, GL45 Thread, 2 ports, 1/4”-28 connection, including O-ring EPDM, nuts and ferrules	A59230
VICI Safety Cap with stopcocks, GL45 Thread, 2 ports, 1/4”-28 connection, including O-ring EPDM, nuts and ferrules	A59233



A59240



A59241



A59242



A59243

Safety caps accessories

VICI Safety Air Inlet Valve with 4 mm filter, fit any VICI cap or VICI safety cap	A59240
VICI Safety Air Inlet Valve with 15 mm filter, fit any VICI cap or VICI safety cap	A59241
VICI Safety Exhaust Filter filled with absorbent, fit any VICI cap or VICI safety cap	A59242
VICI Safety Exhaust Filter with detector, filled with absorbent, fit any VICI cap or VICI safety cap	A59243
O-ring FEP coated for sealing all VICI caps or VICI safety caps, improved chemical resistance	A59244
VICI 1/4-28 flangeless nuts, PPS, for 1/16" tubing, for VICI caps, 10 pcs.	A59245
VICI 1/4-28 flangeless nuts, PPS, for 1/8" tubing, for VICI caps, 10 pcs.	A59246
VICI inverted ferrules, ETFE, for 1/16" tubing, suitable for A59245, for VICI caps, 10 pcs.	A59247
VICI inverted ferrules, ETFE, for 1/8" tubing, suitable for A59246, for VICI caps, 10 pcs.	A59248
VICI plugs, PEEK, 1/4"-28, 1 pc., to closing unused ports for VICI caps	A59249
VICI barbed hose adapter for 1/8" tubing, for VICI caps	A59251
Cellulose filter, 0.2 µm, 4 mm diameter for VICI Safety Air Inlet Valve, fit any VICI cap or VICI safety cap	A59252
Cellulose filter, 0.2 µm, 15 mm diameter for VICI Safety Air Inlet Valve, fit any VICI cap or VICI safety cap	A59253
VICI barbed hose adapter for 8 mm ID tubing, for VICI caps	A59254
For basic solutions in IC, fit any VICI cap or VICI safety cap	A59255
AZURA® Tubing kit with cap and solvent filter (A3375, stainless steel, 10 µm), suitable for all analytical HPLC systems	A9650

Sample vials



A09340



A09342

Vials

Glass material	Closure type	Nominal diameter*	Volume [ml]	Units per package	Article number
Clear	Screw neck	ND 9	1.5	100	A09302
Clear	Crimp neck	ND 11	1.5	100	A09322
Clear	Snap ring	ND 11	1.5	100	A09342
Amber	Screw neck	ND 9	1.5	100	A09300
Amber	Crimp neck	ND 11	1.5	100	A09320
Amber	Snap ring	ND 11	1.5	100	A09340

Caps



A09324



A09344



A09305



A09304

Closure type	Septum	Pre-slit	Cap material	Units per package	Article number
Screw Neck	PTFE/Silicone	No	Blue Polypropylen	100	A09304
Screw Neck	PTFE/Silicone	Yes	Blue Polypropylen	100	A09305
Crimp Neck	PTFE/Silicone	No	Aluminum	100	A09324
Crimp Neck	PTFE/Silicone	Yes	Aluminum	100	A09325
Snap Ring	PTFE/Silicone	No	Clear Polypropylen	100	A09344
Snap Ring	PTFE/Silicone	Yes	Clear Polypropylen	100	A09345

Vial sets

Glass material	Closure type	Septum	Nominal diameter	Volume	Units per package	Article number
Clear	Crimp neck	Rubber/PTFE	20	10 ml	100 /1 000	A15851 / A1585
Clear	Magnetic screw neck	Rubber/PTFE	18	10 ml	100	A15854



A09362



A09361



A09360

µ-inserts

Volume µl	Bottom	Units per package	Article number
250	Conic with plastic spring	100	A09360
300	Conic	100	A09361
400	Flat	100	A09362



A09339

Accessories

Product	Units per package	Article number
Manual crimper pliers	1	A09339

Laboratory filtration



A09003



A09013



A09033



A09043

Sepapure® Syringe Filters

Sample volume	Diameter [mm]	Pore size [µm]	Membrane material	Housing material	Solvent compatibility	Units per package	Article number
< 10 ml	13	0.22	Nylon	Polypropylen	aqueous and organic	500	A09000
< 10 ml	13	0.45	Nylon	Polypropylen	aqueous and organic	500	A09001
< 50 ml	25	0.22	Nylon	Polypropylen	aqueous and organic	100	A09002
< 50 ml	25	0.45	Nylon	Polypropylen	aqueous and organic	100	A09003
< 50 ml	25	0.45	hPTFE	Polypropylen	aqueous and organic	100	A09013
< 50 ml	25	0.45	CA	Polypropylen	aqueous	100	A09033
< 50 ml	25	0.45	PTFE	Polypropylen	organic	100	A09043
< 100 ml	30	0.22	Nylon	Polypropylen	aqueous and organic	100	A09004
< 100 ml	30	0.45	Nylon	Polypropylen	aqueous and organic	100	A09005



A09101 A09103 A09104

Sepapure® Membrane Disc Filters

Diameter [mm]	Pore size [µm]	Membrane material	Solvent compatibility	Units per package	Article number
13	0.22	Nylon	aqueous and organic	100	A09100
13	0.45	Nylon	aqueous and organic	100	A09101
25	0.22	Nylon	aqueous and organic	100	A09102
25	0.45	Nylon	aqueous and organic	100	A09103
25	0.22	hPTFE	aqueous and organic	100	A09112
25	0.45	hPTFE	aqueous and organic	100	A09113
25	0.22	CA	aqueous	100	A09132
25	0.45	CA	aqueous	100	A09133
25	0.22	PTFE	organic	100	A09142
25	0.45	PTFE	organic	100	A09143
25	0.22	PES	aqueous	100	A09152
25	0.45	PES	aqueous	100	A09153
47	0.22	Nylon	aqueous and organic	100	A09104
47	0.45	Nylon	aqueous and organic	100	A09105



A09211/A09201



A09221



A09092

Filter units and filtration accessories

Filter unit or syringe type	Product	Article number
Glass vacuum filter unit	Set, 1 000 ml, ground glass joint	A09200
	Set, 2 000 ml, ground glass joint	A09201
	Collection flask, 1 000 ml, ground glass joint	A09202
	Collection flask, 2 000 ml, ground glass joint	A09203
	Set, 1 000 ml, rubber joint	A09210
	Set, 2 000 ml, rubber joint	A09211
	Collection flask, 1 000 ml, rubber joint	A09212
	Collection flask, 2 000 ml, rubber joint	A09213
Reusable syringe filter holder	Polypropylen, 13 mm	A09220
	Polypropylen, 25 mm	A09221
	Polypropylen, 47 mm	A09222
Syringe	B.Braun Injekt® Luer Lock 2 ml syringes	A09090
	B.Braun Injekt® Luer Lock 5 ml syringes	A09091
	B.Braun Injekt® Luer Lock 10 ml syringes	A09092
	B.Braun Injekt® Luer Lock 20 ml syringes	A09093



A01242



A0272



A7013



A02330

Osmometry consumables

Pack of 12 ampules NaCl calibrating solution, 300 mOsmol/kg	A01240
Pack of 12 ampules NaCl calibrating solution, 400 mOsmol/kg	A01241-1
Pack of 12 ampules NaCl calibrating solution, 850 mOsmol/kg	A01250
Pack of 12 ampules NaCl calibrating solution, 100 mOsmol/kg	A01242
Pack of 12 ampules NaCl calibrating solution, 2 000 mOsmol/kg	A01248
100 Pack of plastic sample tubes for Semi-Micro Osmometer K-7400S	A02721
500 Pack of plastic sample tubes for Semi-Micro Osmometer K-7400S	A0272
1 000 Pack of plastic sample tubes for the Semi-Micro Osmometer K-7400S	A0720
Cleaning tissue, lint-free, for thermistor cleaning	A02330
Printer paper for the plain paper printer A3711 (60 m roll)	A7013
Ribbon cartridge for the plain paper printer A3711 (black)	A7014

Pipettes and Pipette Tips



A09500/A09501



A09502-A09505/A09509



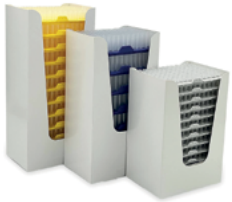
A09506



A09507/A09508

BlueOrchid Manual Single Channel Pipettes

BlueOrchid Pipette 0.1 – 2.5 µl	A09500
BlueOrchid Pipette 0.5 – 10 µl	A09501
BlueOrchid Pipette 2 – 20 µl	A09502
BlueOrchid Pipette 5 – 50 µl	A09503
BlueOrchid Pipette 10 – 100 µl	A09504
BlueOrchid Pipette 20 – 200 µl	A09505
BlueOrchid Pipette 100 – 1 000 µl	A09506
BlueOrchid Pipette 500 – 5 000 µl	A09507
BlueOrchid Pipette 1 000 – 10 000 µl	A09508
BlueOrchid Pipette 150 µl fixed volume	A09509



A09520-A09523

Pipette Tips

Product	Units per package	Packaging	Article number	
Pipette Tips, 0.1 – 10 µl, clear	1000 tips 10 x 96 tips	Bulk bag Reload	A09510 A09520	
Pipette Tips, 0.1 – 10 µl XL/20 µl, clear	1000 tips 10 x 96 tips	Bulk bag Reload	A09511 A09521	
Pipette Tips, 1 – 200 µl, clear	1000 tips 10 x 96 tips	Bulk bag Reload	A09512 A09522	
Pipette Tips, 1 – 200 µl, yellow	1000 tips	Bulk bag	A09513	
Pipette Tips, 1 – 300 µl, clear	1000 tips	Bulk bag	A09514	
Pipette Tips, 100 – 1 000 µl, clear	500 tips 5 x 96 tips	Bulk bag Reload	A09515 A09523	
Pipette Tips, 100 – 1 000 µl, blue	500 tips	Bulk bag	A09516	
Pipette Tips, 100 – 1 250 µl XL, clear	500 tips	Bulk bag	A09517	
Pipette Tips, 1 – 5 ml, clear	250 tips	Bulk bag	A09518	
Pipette Tips, 1 – 10 ml, clear	200 tips	Bulk bag	A09519	



A09524



A09525



A09526/A09527

Rack Boxes

Product	Units per package	Article number
Empty Rack Boxes for 10 µl/10 µl XL Pipette Tips	10	A09524
Empty Rack Boxes for 200 µl/300 µl Pipette Tips	10	A09525
Empty Rack Boxes for 1 000 µl Pipette Tips	8	A09526
Empty Rack Boxes for 1 250 µl Pipette Tips	8	A09527

KNAUER

Standards for Performance Verification (PV)



A PV procedure is recommended for testing newly installed AZURA® systems as well as for regularly monitoring the system performance.

This table gives an overview of the needed PV document, PV standard and separation column for a specific AZURA® system.

Backpressure range	Type of detection	Flow cell path length [mm]	Injection: Sample loop volume [µl]	PV document	Article no. of PV standard	Article no. of HPLC column
UHPLC systems (max. 1 000 bar)	UV, DAD	10	1 – 20	VPV-001: Analytical HPLC, UV detection	A01260-3	10BE181E2F
	UV, DAD	10	21 – 100	VPV-001: Analytical HPLC, UV detection	A01260-2	10BE181E2F
	UV, DAD	50	1 – 20	VPV-001: Analytical HPLC, UV detection	A01260-2	10BE181E2F
	UV, DAD	50	21 – 100	VPV-001: Analytical HPLC, UV detection	A01260-1	10BE181E2F
	FLD	all	1 – 20	VPV-004: Analytical HPLC, FL detection	A01262-2	10BE181E2F
	FLD	all	21 – 100	VPV-004: Analytical HPLC, FL detection	A01262-3	10BE181E2F
	RID	all	all	VPV-002: Analytical HPLC, RI detection	A01261-1	05WE184E2J
HPLC Plus systems (max. 862 bar)	UV, DAD	10	1 – 20	VPV-001: Analytical HPLC, UV detection	A01260-4	15WE181E2J
	UV, DAD	10	21 – 100	VPV-001: Analytical HPLC, UV detection	A01260-3	15WE181E2J
	UV, DAD	50	1 – 20	VPV-001: Analytical HPLC, UV detection	A01260-3	15WE181E2J
	UV, DAD	50	21 – 100	VPV-001: Analytical HPLC, UV detection	A01260-2	15WE181E2J
	FLD	all	1 – 20	VPV-004: Analytical HPLC, FL detection	A01262-1	15WE181E2J
	FLD	all	21 – 100	VPV-004: Analytical HPLC, FL detection	A01262-2	15WE181E2J
	RID	all	all	VPV-002: Analytical HPLC, RI detection	A01261-1	05WE184E2J
	ECD in PAD mode	n/a	20, 100	VPV-106: AZURA systems with ECD	A01132	n/a
	ECD in DC mode	n/a	20, 100	VPV-106: AZURA systems with ECD	A01273-2 A01273-3	n/a
	UV (normal phase)	10, 50	10, 20, 100	VPV-009: AZURA HPLC systems in normal phase mode	n/a	15WE000E2J
	RID (normal phase)	all	all	VPV-009: AZURA HPLC systems in normal phase mode	n/a	15WE000E2J
Preparative HPLC systems	UV, DAD	≤ 2	all	VPV-007: Preparative HPLC, UV detection	A01264-1	05JE181E2J
	UV, DAD	> 2	all	VPV-007: Preparative HPLC, UV detection	A01264-2	05JE181E2J
	RID	all	all	VPV-008: Preparative HPLC, RI detection	A01265-1	05IE184E2J
FPLC systems	UV, DAD	all	all	VPV-003: AZURA FPLC systems	A01261-1	05WE184E2J

Software & PC Hardware

NEW LNP VERSION

Mobile Control for Windows

With the hand-held Mobile Control and Mobile Control Chrom software you have your devices and systems at your fingertips. Remotely control and monitor your devices and enjoy the touchscreen-optimized user interface. Choose Mobile Control as an easy-to-use and cost-effective control software!

Mobile Control Display provides full access to devices. Change device settings, set operating parameters, automate device control or check the system status. Mobile Control features all functionalities of a device display.

Mobile Control Data features data acquisition of pump and detector traces in addition to full device control.

Mobile Control FRC features a fraction collection option for simple preparative applications
Mobile Control LNP features a predefined method structure and ready-to-use workflow for easy formulation.

Only pay for what you use: Mobile Control features basic functions to operate devices and systems. The software can operate dedicated applications which do not require a highly developed and cost-intensive Chromatographic Data System (CDS).

Save space: Mobile Control runs on a tablet. Especially in labs with little space avoiding a desktop PC with keyboard and monitor can be a decisive factor. The touch-optimized user interface allows device control using just your fingers.

Save time: Mobile Control convinces due to an intuitive user interface and a clearly structured menu function. The training period is minimal in comparison to a complex CDS.

Free updates: With every release new features are available in Mobile Control. Download the current version for free.

Free trial: To evaluate if Mobile Control holds up to your expectations, you can download the software and test the free trail option. Perfect for those who'd like to try before they buy.

Customized software design: Mobile Control is made by KNAUER and can be adapted to the requirements of our OEM partners.



This software supports a wide range of instruments:
www.knauer.net/softwarecontrol

For PC hardware & periphery, **see p. 135**

Specifications

Software name	**Mobile Control Display** without data acquisition **Mobile Control Data** with data acquisition **Mobile Control FRC** with data acquisition and fraction collection option **Mobile Control LNP** with data acquisition and formulation user interface
Operating system	Windows 10, Windows 11
Software version	Display, Data, FRC: Mobile Control v6.0x, Data Viewer v6.0x LNP: Mobile Control v6.2.X, Data Viewer v6.2.X
Supported instruments	Consider release notes (downloads below)
Field of application	Display software, device control, simple preparative applications with fraction collection

Expandability

Stand-alone	yes
Multi-user environment	yes
Report functions	yes
Special features	with tablet

Free demo version:
www.knauer.net/mobilecontrol

Ordering details

Software

A9607	Mobile Control Display without data acquisition including tablet
A9608	Mobile Control Data with data acquisition including tablet
A96132	Mobile Control FRC with data acquisition and fraction collection option including tablet
A96134	Mobile Control LNP for control of IJM NanoScaler systems including tablet and mount
A9610	Mobile Control Display without data acquisition
A9612	Mobile Control Data with data acquisition
A9613	Mobile Control with data acquisition and column test option
A96131	Mobile Control FRC with data acquisition and fraction collection option
A96133	Mobile Control LNP for control of IJM NanoScaler systems
A9614	Upgrading Mobile Control Display A9610 to Data A9612
A96141	Upgrade Mobile Control Data A9612 to FRC A96131

Accessories

A96182	USB-C-LAN network adapter for tablets
A64809	WiFi router, 8 × LAN GBit RJ-45 ports, 1 × WAN GBit RJ-45 port
A64809INT	WiFi router, 8 × LAN GBit RJ-45 ports, 1 × WAN GBit RJ-45 port, power plug UK, US or AUS
A64811	Single device WLAN router for Mobile Control - 1x RJ45, 10/100 MBit; WLAN
A9617	Mobile Control Mount - flexible tablet mount for tablets
A131261	Docking station for Tablet MS Surface Go

knauerOS®

NEW

Simplicity where it matters, flexibility where it counts.

knauerOS® is a modern, browser-based Chromatography Data System (CDS) for the efficient acquisition, evaluation, and management of HPLC data. Developed to map complex chromatography workflows as simply as possible, knauerOS delivers maximum flexibility, scalability, regulatory compliance, and cost-effectiveness, all within a server-client web architecture.

The challenge: HPLC data is becoming increasingly complex: more detectors, more methods, more users, all while costs and time pressures are rising. Many CDS solutions are too expensive, too complicated, or too inflexible.

The solution: knauerOS makes HPLC data analysis as easy as possible and as flexible as necessary. Less software effort and costs. More clarity in the data. Faster decisions.



This software supports a wide range of instruments:
www.knauer.net/softwarecontrol

For PC hardware & periphery,
see p. 135

Further information:
www.knauer.net/software-knaueros

Advantages at a glance

Easy to use: Intuitive interface, quick to learn, even for changing users.

Unlimited users: Teamwork without licensing hurdles, in the lab or remotely.

Maximum flexibility: Custom calculations, integration rules, and SST criteria, precisely tailored to method and requirements.

Fast data analysis: Efficient evaluation of many chromatograms, even with multiple detector signals.

Compliant & reproducible: Standardized evaluations for QC, GMP, and documented processes (FDA 21 CFR Part 11).

Cost-efficient & scalable: Grows with your lab without increasing complexity, from single workstation to full lab network.

Server-client architecture

knauerOS is hosted on an internal company intranet server and supports external clients via VPN or secure connection with **no controller and no user limits**.

knauerOS Controller (on-site PC): interfaces directly with up to 4 HPLC systems, captures data in real time and transmits to the central server.

knauerOS Server (application backend): hosts the software, handles data processing, storage, analysis, and user management, enables browser-based remote access from any client.

Specifications

Software name	knauerOS®
Architecture	Browser-based, server-client web application; hosted on company intranet
Operating system	Microsoft Windows 11/10 Professional (Server & Controller)
PC Hardware (Server/Controller)	Intel® Core i 8xxx or higher (Intel® N95 for Server); 8 GB RAM; SSD/NVMe ≥ 500 GB; 2 × Ethernet Adapter
Systems per Controller	Up to 4 HPLC systems per Controller with unlimited Controller setup
Multi-user environment	Unlimited user numbers with role-based Permissions (manager, analyst, administrator, etc.)
Network environment	Remote work with unlimited team members
Security & compliance	Full audit trail; GMP compliance (FDA 21 CFR Part 11)
Fields of application	Analytical HPLC
Supported instruments	Full control for a selected range of analytical KNAUER instruments (see instrument list)
Instrument control	Method-based instrument control, Instrument Status and Direct-Control via Monitor dashboard
Sequence	(Preferred use) for start measurement with chromatogram acquisition
Data processing	Data processing is handled on the knauerOS server side, ensuring optimal performance and efficiency
Data import	Import of chromatography data from multiple chromatographic systems into a single chromatography database
Search & filter	Database filtering based on multiple search parameters
Analysis operations	Perform post-analysis calculations (signal ratios, peak derivates, signal filters, detector offset…) with overlay view, data integration, calibration, calculation processing over samples and signals in multi analysis management mode, definable blank subtraction, custom labels and many more.
Custom variables	Pre- and free defined mathematical expressions
SST criteria	Flexible pre- or free defined sample or system SST expressions
Presentation mode	For interactive online visualization of chromatograms and results
Analysis export	Chart via Clipboard, .png, .jpeg, .svg CSV from signals, peaks table and results table

Supported instruments

KNAUER Autosampler AS 6.1L
KNAUER HPLC Pump P 6.1L LPG / HPG
KNAUER Dosing Pump P 2.1S
KNAUER Dosing Pump P 4.1S / P 4.2S
KNAUER Column Thermostat CT 2.1
KNAUER Conductivity Monitor CM 2.1S
KNAUER Multiwavelength UV Detector MWD 2.1L
KNAUER Valve Unit VU 4.1
KNAUER Diode Array Detector DAD 2.1L*****
Fluorescence Detector RF-20A
Light Scattering Detector ELSD
Fraction Collector FOXY R1

***without 3D spectra functionality (coming soon)**

Ordering details:

ASWKN01A	knauerOS Commercial Edition – License for 1 HPLC System (incl. 3-year updates)
ASWKN01B	knauerOS Commercial Annual Edition – License for 1 HPLC System (incl. updates)
ASWKN01C	knauerOS Community Edition – License for 1 HPLC System (incl. updates)
ASWKN01D	knauerOS Demo Version (1 month testing) – License for 1 HPLC System
ASWKN02A	knauerOS Commercial Edition Update (regardless of the number of HPLC systems)
ASWKN02C	knauerOS Community Edition Update (regardless of the number of HPLC systems)
All knauerOS editions are unlimited expandable to build a server-client environment.	
A13110 /A13120	Chromatography Workstation Win 11 Pro incl. Keyboard/Mouse/Monitor 24" (DE / EN)

ClarityChrom®

KNAUER ClarityChrom® is a powerful, yet easy-to-use chromatography software (or chromatography data system, CDS) for instrument control, data acquisition and data processing. ClarityChrom is designed for smaller laboratories. It is an economical solution compared to other more complex chromatography software while still offering FDA 21 CFR Part 11 compliance.

ClarityChrom comes as a complete package with LC control and including autosampler control. It is scalable from 1 up to 4 systems; depending on the desired instruments. The built-in fractionation option as well as the optional extensions as SST for automated system tests, PDA for 3D (UV spectra) data handling, GPC analysis, MS and GC control cover a wide range of the requirements for a CDS on a modern lab. KNAUER additionally offers a more advanced fractionation with the KNAUER FRC control module.

ClarityChrom supports all KNAUER devices that can be controlled by software. Please refer to the instrument support list in the Support section of our website, the download link can be found below. Beside this, devices and systems from more than 45 manufacturers can be controlled. Additionally, data acquisition can also be performed with any detector providing a voltage output by simply connecting a KNAUER IFU 2.1 interface box or any other supported A/D converter.

The system suitability (SST) extension automates the calculation of system suitability parameters for system validation and calculates up to 12 parameters and compares the results with the limits the user has set.

The PDA extension allows to acquire and process 3D data from a photo diode array detector (KNAUER PDA detectors are fully supported). The PDA extension provides peak purity analysis and peak identification by spectral library search in self-made or commercial spectra libraries.

The SEC/GPC extension provides interactive and automated gel permeation chromatography analysis, including recalibration and GPC reporting, as well as simplifies the retrieval of GPC data. The GPC extension allows flow rate and multi-detector delay corrections and includes Narrow, Broad and Broad on Narrow calibrations.

ClarityChrom comes with some basic fractionation functionality. The KNAUER-exclusive KNAUER FRC control module for ClarityChrom adds more drivers of several fraction collectors and supports the peak recognition by level and/or slope as well as fractionation by time. Also more advanced functionality as solvent recycling, manual fractionation and rack view with detailed fraction information and chromatogram links are available. The functionality corresponds exactly to the KNAUER preparative functionality of discontinued ClarityChromPrep.

ClarityChrom offers all the necessary operations for an analytical lab. Moreover, the preparative version adds fractionation options to this feature list and allows more flexibility in the lab. ClarityChrom is the best solution for all laboratories searching for an up-to-date and robust software with support of devices from many manufacturers to be flexible in instrumentation but also meet the requirements for modern laboratories.



This software supports a wide range of instruments:
www.knauer.net/softwarecontrol

For PC hardware & periphery,
see p. 135

Further information:
www.knauer.net/claritychrom

Specifications

Software name	ClarityChrom 10.1
Extensions / Licenses	PDA / 3D UV, System suitability, Fraction collection, SEC/GPC, Mass spectrometry
System architecture	32-bit CDS
Operating system	Windows 11, Windows 10, Windows 8.1, Windows 7, all 32- and 64-bit

Expandability

Stand-alone	Workstation version, max. 4 systems controlled by one computer, max. 3 LC systems, max. 2 systems with PDA or 1 system with MS or special devices per computer
Client/server	No Client/Server functionality
Multi-user environment	Selectable system of user accounts with independently customizable behavior and appearance for individual users
Network environment	Easy offline data sharing (at the file level) among all stations in a local network
Fields of application	Analytical and preparative HPLC, GPC/SEC, GC, MS
Supported instruments	All KNAUER devices are supported, driver for devices from many other manufacturers are available
Instrument connection	Supports RS-232, Ethernet, PCI interface card, A/D-D/A interface
Recommended PC hardware	Pentium 2 GHz, 4 GB RAM, 80 GB free hard disk space, separate graphics card if one PC should control more than one system, USB for dongle, connectors as LAN, RS-232 etc. for device control
Graphics capabilities	Multiple chromatogram view and overlay, PDA view
Integration	27 integration parameters (peak width, threshold, tangent slope ratio etc.) integration parameters programmable in time, automatic re-integration
Calculation types	With/without calibration (int./ext. standard method)
Security and GLP	Installation qualification test of the software; FDA 21 CFR Part 11 conformance, validation with virtual detector
Instrument control	Method-based instrument control, Instrument status display and Direct-Control mode,
Calibration	6 types of calibration curves, up to 20 levels, reference peaks, groups, unlimited number of standards (peaks), LOD, LOQ
Chromatogram operations	Overlay view, custom labels and settings, also applying mathematical operations to chromatograms
Automation	Sequences, automatic launch of selected commands or applications immediately following chromatogram acquisition – Post run, Batch
Presentation of results	Integrated customizable table of results, columns with user defined calculation, summary table, and export in text or database format
Calculations	Custom: 12 predefined mathematical operators, 15 basic and 4 summary functions, special: Kovats indexes for GC, determination of noise/drift, performance calculations
Data import and export	ASCII, AIA, dBase

Additional options/extensions

FRC option	Separate license option; Control of fraction collectors and KNAUER valve drives as fraction collector, fractionation per time/level/slope, rack info with filling level and chromatogram link
PDA option	Separate license option; 3D chromatogram, peak purity analysis, spectrum search in self-made or commercial spectra library
GPC/SEC option	Separate license option; molecular weight determination in size exclusion chromatography with various calibration methods
System suitability test	Separate license option; automates the calculation of system suitability parameters for system validation
Note	Autosampler control included

Ordering details

Software

A1670	ClarityChrom® single instrument license for one time base
A1674	ClarityChrom® offline license for data evaluation
A1671	ClarityChrom® additional instrument license on additional time base
A1676	ClarityChrom® option for PDA data processing
A1677	ClarityChrom® system suitability option
A1678	ClarityChrom® option for GPC data processing
A1679	ClarityChrom® option for MS data processing
A1682	ClarityChrom® KNAUER FRC control module for preparative HPLC
A1681	Upgrade for one system from former version to latest ClarityChrom®
A1683	Upgrade for ClarityChrom® Offline from former version to latest ClarityChrom®
A1687	Upgrade for former ClarityChrom® Prep to latest ClarityChrom® with KNAUER FRC control module
A1690	30-day trial version of ClarityChrom®
A1675	ClarityChrom® university package one offline license

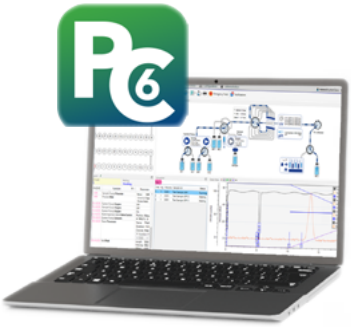
PurityChrom® 6

PurityChrom® 6 is the next generation of KNAUER's purification software.

It is designed to address all separation tasks in bio-purification and preparative HPLC.

PurityChrom 6's animated flow path visualization improves usability and method writing. Methods can be also divided into different steps. The software also enables intelligent and flexible fractionation. Furthermore, different hardware configurations can be managed and controlled. The cherry on top? The software also meets the standards of GAMP 5 and 21 CFR part 11.

The basic license enables the control of up to 2 pump systems, 2 detectors, 1 fraction collector and an unlimited number of valves. To control an extended purification system, please refer to the other PurityChrom licenses. To also get the full GMP documentation including OQ, GAMP 5 and 21 CFR part 11 certificates, please refer to the special GMP license of PurityChrom 6.



This software supports a wide range of instruments:
www.knauer.net/softwarecontrol

For PC hardware & periphery, **see p. 135**

Further information:
www.knauer.net/puritychrom

Specifications

Software

Software name	PurityChrom® 6
Operating system	Windows 10, Windows 11 (English or German only)

Expandability

Stand-alone	License for controlling one system
Multi-user environment	Optional user administration with individual assignment of rights for individual users
Fields of application	FPLC & Prep LC
Instrument connection	Supports RS-232, Ethernet, A/D-D/A interface
Integration	Real-time analysis of peaks, automatic or manual integration and baseline correction
Security and GLP	FDA 21 CFR Part 11, GAMP 5 conformance
Automation	via sequences
Presentation of results	Individual report configuration as pdf or csv
Special features	Method creation based on volume or column volume; hold function; animated visualization; direct control during a run; display of solvent supply; usage of variable values

Additional options/extensions

FRC option	Included
FRC features	Control of fraction collectors and KNAUER valve drives as fractionation valve, fractionation per Time/Level/Slope, rack info with filling level and chromatogram link

Ordering details

License overview

Article no.	License type	Detectors	Pump systems	Fraction collector	Auto-sampler	Valves	Flow-meter	GMP documents	DAD	High flow as pump system
A2680	Basic	2	2	1	0	∞	0	-	-	-
A2681	Full	∞	3	∞	1	∞	∞	-	-	-
A2682	GMP	∞	3	∞	1	∞	∞	+	-	-
A2683	DAD	∞	3	∞	1	∞	∞	-	+	-
A26831	DAD GMP	∞	3	∞	1	∞	∞	+	+	-
A26841	High flow S	∞	2	∞	1	∞	∞	-	-	up to 4 pumps in high flow module
A26843	High flow S GMP	∞	2	∞	1	∞	∞	+	-	up to 4 pumps in high flow module
A26842	High flow L	∞	2	∞	1	∞	∞	-	-	up to 8 pumps in high flow module
A26844	High flow L GMP	∞	2	∞	1	∞	∞	+	-	up to 8 pumps in high flow module
A2685	LNP S GMP	∞	4	∞	1	∞	∞	+	-	-
A2685-1	LNP S	∞	4	∞	1	∞	∞	-	-	-
A2686	LNP L GMP	∞	8	∞	1	∞	∞	+	-	-
A2686-1	LNP L	∞	8	∞	1	∞	∞	-	-	-
A26861	LNP XL GMP	∞	12	∞	1	∞	∞	+	-	-
A26862	LNP XXL GMP	∞	16	∞	1	∞	∞	+	-	-
A2693	Upgrade 2 Autosampler	-	-	-	2	-	-	-	-	-
A2692	Upgrade Remote File Access	Enables MySQL, remote file service and MongoDB for global data management; device control dependent on added license								
A2687-A....	Upgrade PC5 → PC6	Number of controllable devices and article number for upgrade dependent on initial license								

PurityChrom® 5

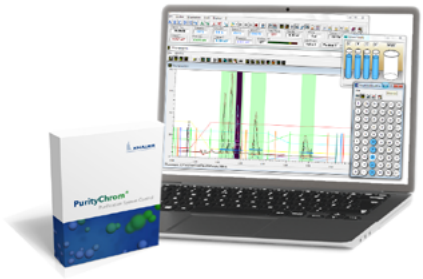
PurityChrom® is a chromatography software especially designed for the area of preparative purifications and FPLC applications. PurityChrom provides a user-friendly and clearly structured interface. The **system visualization** offers a graphical representation of the purification system and allows easy handling even of complex flow processes. Furthermore, each device which is displayed in the fluidic scheme can be manually controlled, giving the opportunity to optimize, change and adapt your conditions during the run.

You have the option to create a **method** based on volume, column volume or time. There is also a possibility to pause or to change the method parameters during the run, which gives you complete control over your chromatography process. In PurityChrom you can define important functions in your method with **variables**. This allows you to write methods that can be adapted more flexibly to a specific sample or column, just before the run with only one click. In combination with the **sequence table** a quick and easy method scouting provides you with the best method for your purification problem in less time.

For **fractionation**, you can use a fractionation valve as well as a fraction collector.

Current guidelines and regulations like 21 CFR part 11 are entirely supported. Please check for more information about supported devices the Release Notes of the latest PurityChrom version. With an unlimited number of **free offline licenses**, you can write methods and evaluate runs on any computer of your choice, without blocking the system.

The **basic version** is limited to 3 data channels and the control of eight devices (excl. autosampler). The **upgrade version** (A2652) supports 8 data channels and an unlimited number of devices including an autosampler. The **3D option** (A2654) allows the support of an diode-array detector and the **MS option** (A2655) the usage of an mass spectrometer.



This software supports a wide range of instruments:
www.knauer.net/softwarecontrol

For PC hardware & periphery, **see p. 135**

Further information:
www.knauer.net/puritychrom

Specifications

Software name	PurityChrom® 5
Operating system	Windows XP, Windows Vista, Windows 7, Windows 8, Windows 10 (English or German only)

Expandability

Stand-alone	License for controlling one system
Fields of application	FPLC and Prep LC
Instrument connection	Supports RS-232, Ethernet, A/D-D/A interface
Recommended PC hardware	CPU/Memory: Pentium III or higher with at least 1 GHz; 32 or 64 bit; at least 512 MB RAM (Windows XP) and 2 GB (Windows Vista and higher) Graphics: Screen with minimal resolution 1 024 × 768 Connectors and slots: USB for license dongle; COM, USB or LAN according to connected instruments
Integration	Real-time analysis of peaks, automatic or manual integration and baseline correction
Security and GLP	FDA 21 CFR Part 11 conformance
Automation	Via sequences and autosampler control files
Presentation of results	Individual report configuration as pdf or csv
Calculations	Column performance calculations according to DAB
Data import and export	Comma Separated Value, AIA/ANDI, ChromStar Slice
Special features	User administration

Additional options/extensions

FRC option	Included
FRC features	Control of fraction collectors and KNAUER valve drives as fractionation valve, fractionation per Time/Level/Slope, rack info with filling level and chromatogram link
PDA option	Special license option; no 3D presentation
Note	For autosampler control the upgrade license is needed

Ordering details

Software

A2650	Basic License for one system
A2652	Extends the Basic License to an unlimited number of controllable devices and 8 data channels, adds autosampler and stacked injections support
A2654	3D option for a diode-array detector (DAD)
A2655	Mass spectrometry (MS) option for supporting the mass spectrometer 4000 MiD®
A2656	PurityChrom® Maintenance and Support including free updates and 5 hours Software support by KNAUER

PurityChrom® MCC / MCC PLUS

PurityChrom® MCC is a special version of our purification software PurityChrom® and is optimized to be used with continuous chromatography systems e.g. SMBC systems. PurityChrom® provides a very user-friendly and clearly structured interface. The system visualization offers a graphical representation and allows easy handling even of complex flow processes. Furthermore each device which is displayed in the fluidic scheme can be manually controlled, giving the opportunity to optimize, change and adapt your conditions also during the run. The new PurityChrom MCC SMB parameter wizard helps you to generate new SMB methods and optimize your parameters while the process is running. With the integrated starting point calculator, you can easily generate you SMB method with the adsorption isotherms of your substances. There is also a possibility to pause your method during a run. The hold function provides you with complete control over your chromatography process.

PurityChrom® MCC Plus is a special software extension enabling monitoring of up to 16 data channels and controlling of up to 8 independent pumps without gradient formation. Accordingly, the software can manage complex, preparative purification systems with an enhanced number of multiple devices. For example, in comparison to other PurityChrom® software packages, the combination of a multi wavelength detector and more than one single UV detector is enabled and up to 8 flow meters can be controlled in one system.



This software supports a wide range of instruments:
www.knauer.net/softwarecontrol

For PC hardware & periphery,
see p. 135

Further information:
www.knauer.net/puritychrom

Specifications

Software name	PurityChrom® MCC / PurityChrom® MCC PLUS
Operating system	Windows 10
Field of application	SMB, prep LC

Expandability

Stand-alone	License for controlling one system
Multi-user environment	Selectable system of user accounts with independently customizable behavior and appearance for individual users
Instrument connection	Supports RS-232, Ethernet, A/D-D/A interface
Recommended PC hardware	CPU/Memory: Pentium III or higher with at least 1 GHz; 32 or 64 bit; at least 512 MB RAM (Windows XP) and 2 GB (Windows Vista and higher) Graphics: Screen with minimal resolution 1 024 × 768 Connectors and Slots: USB for license dongle; COM, USB or LAN according to connected instruments
Security and GLP	FDA 21 CFR Part 11 conformance
Automation	Via control files
Presentation of results	Individual report configuration as pdf or csv
Data import and export	Comma Separated Value, AIA/ANDI
Special features	Controlling of up to 4 independent pumps without gradient formation

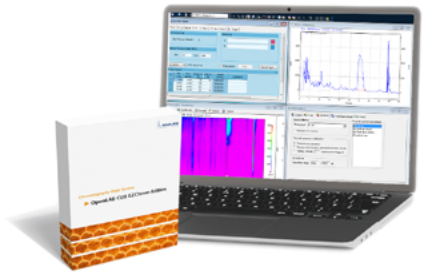
Ordering details

Software

A2659	PurityChrom® MCC: Software solution to control and monitor AZURA® multi column chromatography systems - SMB
A2657	PurityChrom® MCC PLUS: Software solution for complex preparative systems without gradient formation

OpenLab® CDS EZChrom Edition/CDS

OpenLab® CDS EZChrom Edition



OpenLAB® CDS EZChrom Edition is the next generation of chromatography data systems and the successor of ChromGate CDS. OpenLAB® CDS EZChrom Edition provides chromatography data acquisition, processing and control of GC and LC chromatographs and is used in chromatography operations ranging from single user/single instrument to multi-user/multi-instrument laboratories. It provides support of devices from KNAUER and many other manufacturers.

The basic workstation license can only be installed on one PC and allows for control and data acquisition from one system. The license includes System Suitability, Fraction Collector Control and one year Software Maintenance Agreement (SMA).

The system suitability option allows for test if the system is suitable for particular analysis by testing several parameters as resolution, peak asymmetry and theoretical plates.

The KNAUER fraction collector control option includes the drivers of several fraction collectors, including the KNAUER electric valves, and supports fractionation by time, the peak recognition by level and/or slope, also with spectral confirmation. Collet Slices allows for setting a desired volume for each fraction, within the defined fraction vial volume. Also, manual fractionation is supported. The collected fractions will be visualized in the rack view with retention time and volume. If a chromatogram of your separation already exists, the required fractionation commands can be derived directly from the chromatogram with a double mouse-click. The combination of virtual detector and virtual fraction collector allows for optimizing the fractionation settings from an existing chromatogram of your separations without any physically existing device and, therefore, without the loss of solvent or target substance.

OpenLAB® EZChrom Edition and EZChrom Elite are registered trademarks of Agilent Technologies, Inc.

This software supports a wide range of instruments:
www.knauer.net/softwarecontrol

For PC hardware & periphery, **see p. 135**

Specifications

Software name	OpenLAB CDS EZChrom Edition
Extensions / Licenses	Fraction collection, System suitability, PDA / 3D UV
System architecture	32-bit CDS
Operating system	Depends on CDS version. Latest version, supported by KNAUER drivers, is A.04.10. It runs on Windows 11, Windows 10 Prof./Enterprise, 64-bit and Windows 7 Prof., 32- and 64-bit.

Additional options/extensions

FRC option	Always included, for preparative HPLC, adds tools for detector controlled fraction collection, solvent and peak recycling, stacked injection, rack view with information about RT and volume
FRC features	Fractionation can be controlled by time (volume), level, slope including AND/OR combination of these criteria, spectra comparison, local maximum and local minimum, slices, full manual control of fractionation during a run
PDA option	3D chromatogram, peak purity analysis, spectrum search in self-made or commercial spectra library (needs to be converted in OpenLAB® spectral library format)
GPC/SEC option	License is discontinued
System suitability test	License always included, automates the calculation of system suitability parameters for system validation

Ordering details

Software

A2600-1	OpenLAB® CDS EZChrom Edition workstation for one system with SMA and 4x System Suitability
A2610-1	OpenLAB® CDS EZChrom Edition 3D option for UV detectors MW-1, 2550 and 2600
A2611-1	OpenLAB® CDS EZChrom Edition 3D UV Option for DAD DAD6.1L, DAD2.1L, PDA-1, S2850
A2602-1	OpenLAB® CDS EZChrom Edition Instrument Control License
A2614-1	OpenLAB® CDS EZChrom Edition for distributed systems - please ask for desired configuration
A2618-12	OpenLab® CDS EZChrom Edition drivers for Sedex 80/85/90 LT
A2618-13	OpenLab® CDS EZChrom Edition drivers for Sedex 100/LC/FP

OpenLab® CDS

COMING SOON

The latest CDS from Agilent will be available from KNAUER with drivers for our AZURA® (U)HPLC series later this year.

Ordering details:

Software

A2630-1	OpenLab® CDS Workstation
A2632-1	OpenLab® CDS Workstation Plus
A2634-1	OpenLab® CDS non-Agilent Instrument Connection

Chromeleon™ Drivers



Thermo Scientific™ Dionex™ Chromeleon™ is one of the most wide-spread chromatography data systems. Its intuitive handling benefits laboratory workflow and the highly developed algorithms simplify data processing. It offers a broad range of third-party drivers and can be easily used with existing HPLC systems. KNAUER offers drivers for a lot of its devices.

Disclaimer: KNAUER Wissenschaftliche Geräte GmbH is solely responsible for development, testing and support of Thermo Scientific™ Dionex™ Chromeleon™ Chromatography Data System driver software for KNAUER instruments and therefore solely liable for damages associated with the use of this driver software.

Specifications

Further information:
www.knauer.net/chromeleon

Computer requirements	
Operating system	Windows 8.1 Professional, 64-bit;Windows 7 SP1 Professional, Enterprise, 64-bit, (32-bit version is not recommended); Windows Vista SP2 Business, Ultimate, 32-bit (Vista is not recommended)
CPU (recommended)	3 GHz Intel Quad core i7 or better
Memory RAM (recommended)	16 GB
Free Hard Disk Space	250 GB available, for system with PDA detectors
Optical Drive	DVD
Display (recommended)	1 440 × 990, 32-bit color or higher (Minimum 1 280 × 1 024, 32 bit color)
USB Ports	1 USB port 2.0 or higher
Ethernet Port	1 port for router (for system connection)

Ordering details

Drivers

ASWCM73001	KNAUER Instrument Driver Chromeleon™ 7.3.1 and higher, Special Release (selected AZURA® instruments only)
A1783-4	Sedex Driver for Chromeleon™ 7.2/7.3; For Sedex 85LT / 90LT; Instrument Controller Class 3 necessary
A1783-5	Sedex Driver for Chromeleon™ 7.2/7.3; For Sedex FP / LC / 100LT; Instrument Controller Class 3 necessary

Requires Chromeleon™ 7.3.1 and higher, Instrument Controller Class 3.

Additional KNAUER AZURA® driver for Chromeleon™ 7.3.1 and higher will be available soon. COMING SOON

PC Hardware & periphery

Desktop PCs

Instrument Controller with 24″ monitor, English edition Windows 11 Prof. 64-bit English, Intel® Core™ i7, 8 GB RAM, 256 GB SSD, two network cards	A13121
Instrument Controller with 24″ monitor, German edition Windows 11 Prof. 64-bit German, Intel® Core™ i7, 8 GB RAM, 256 SSD, two network cards	A13111
Instrument Controller for PurityChrom® and ClarityChrom® with 24″ monitor, English edition Windows 11 Prof. 64-bit English, Intel® Core™ i5, 8 GB RAM, 256 GB SSD, two network cards	A13120
Instrument Controller for PurityChrom® and ClarityChrom® with 24″ monitor, German edition Windows 11 Prof. 64-bit German, Intel® Core™ i5, 8 GB RAM, 256 SSD, two network cards	A13110
Laptop for PurityChrom® and ClarityChrom®, Windows 11, min. Intel® Core™ i3, 8 GB RAM, 256 GB SSD, German edition	A13112



Configuration on request

Tell us your requirements and we will figure out the matching CDS workstation. We offer the complete CDS installation and promise you a smooth operation.	A13130

Network devices

WiFi router, 8x LAN GBit RJ-45 ports, 1 × WAN GBit RJ-45 port	A64809
WiFi router, 8x LAN GBit RJ-45 ports, 1 × WAN GBit RJ-45 port, power plug UK, US or AUS	A64809INT
8-port LAN GBit Switch NetGear GS108GE, 8 × RJ-45, GBit, Auto MDI-X	A3119
8-port LAN GBit Switch NetGear GS108GE, 8 × RJ-45, GBit, Auto MDI-X, power plug UK, US or AUS	A3119INT
16-port LAN Gigabit Ethernet Switch NetGear GS116GE, 16 × RJ-45, 10/100/1 000 MBit, Auto MDI-X	A3129
16-port LAN Gigabit Ethernet Switch NetGear GS116GE, 16 × RJ-45, 10/100/1 000 MBit, Auto MDI-X, power plug UK, US or AUS	A3129INT



A64809

IT accessories

VSCOM USB 4 COM 4 x RS-232 DE9 on USB	A3114
AZURA® Interface Box IFU 2.1 LAN, A/D converter, 4 channels	AZB00XA
Ethernet Eventbox for 12 digital inputs and outputs each; only supported under PurityChrom® & PurityChrom MCC Plus®	AZB01
Input cable for Ethernet Eventbox (5 m, M3 plug, open ends with wire end ferrules)	AZB01-01
Output cable for Ethernet Eventbox (3 m, hollow plug, open ends with wire end ferrules)	AZB01-02
Cable for connection of an air sensor to an Ethernet Eventbox (2 m, 2-pole and 3-pole plug)	AZB01-03
RS-232 f/f cable 9-pol nullmodem	A0895
RS-232 m/f cable 9-pol	A0884
APC Smart UPS 1500 VA, uninterruptible power supply for up to 8 devices	A3121



AZB00XA



A3121

Power cables

Power cable for Europe, 2 m, with rubber connector type C13, 230 V	M1642
Power cable for Switzerland, 2 m, with rubber connector type C13, 230 V	M1597
Power cable for UK, 2.5 m, with rubber connector type C13, 230 V	M1278
Power cable for USA, 2 m, with rubber connector type C13, 115 V	M1651
Power cable, 1.5 m, with rubber connector for UPS APC Smart connector	M2561
Power Supply: Distribution Box 24 V for 6 devices like air sensor, external pressure sensor, IFU 2.1 LAN	AZS80SA
EU power cable with 4 cold-device plugs and cover caps	A12345



AZS80SA

KNAUER Services

Application Services



With profound application knowledge of analytical and preparative HLPC and FLPC, our team is at your service around the world.

Our experts are pleased to receive your inquiries and requests and will offer attractive customized solutions.

HPLC method development

Qualify, quantify or purify

Do you plan to separate substances by HPLC in order to qualify, quantify or even purify without spending too much time in developing a suitable method? We offer an application and method development service and support you to select a suitable system for your lab.

According to your specifications we prepare an efficient HPLC or FPLC method including advice for an appropriate sample preparation.

HPLC method transfer & optimization

For optimized quality and speed

Do you intend to perform your analyses faster, more efficient and cost effective? We are happy to support you with our profound expertise and experience in liquid chromatography. The team assists in transferring LC applications and methods.

1. Method transfer

We investigate the transfer of your method to one of our HPLC systems. Especially complex separations can cause trouble when transferring them to a different system.

We ensure continuous and consistent quality after the transfer.

2. Method optimization

Using ultra-pure solvents in HPLC can increase the expenses of an analysis substantially. A shift from classic HPLC columns to smaller inner diameters and smaller particle size could cut costs enormously since considerably less solvent is required. We optimize and transfer your LC analyses in order to obtain identical, or even better and faster results, reduce eluent consumption and operating costs.

Column Screening Services

Chiral column screening and/or method development and optimization

As most chiral separations are not predictable, KNAUER offers a screening service to find the best suiting Eurospher II Chiral column for your chiral separation task.

- Column screening with all available Eurospher II Chiral columns
- Optional: Method optimization
- Results will be reported completely and send as a report
- Method parameters and column specifications will be handed over directly

Eurokat column screening for the analysis of carbohydrates

Not sure which column separates your saccharides best? We offer a screening service for Eurokat columns that are recommended for the separation of sugars and all types of carbohydrates.

- Column screening with all available Eurokat columns
- Optional: Method optimization
- Results will be reported completely and send as a report
- Method parameters and column specifications will be handed over directly



Note: Details and requirements must be discussed previously with KNAUER´s application specialists.

KNAUER Academy

KNAUER has been successfully leading courses for many years for its customers, dealers and sales staff. Our main goal is to familiarize every participant with the latest chromatographic technologies in small groups with practical examples. We offer HPLC courses for beginners and advanced users. In individual courses, participants can receive specialized knowledge, e.g. in UHPLC, FPLC or preparative HPLC. Take part in one of the regularly offered courses or book an individual training on special topics.

Workshops at KNAUER in Berlin or on site

Ordering information online or upon request: Tel. +49 30 8097270, E-mail **academy@knauer.net**

Research

Scientific research generates new results and knowledge for industry and society. Currently, KNAUER is involved in different research projects. Obviously, we mainly focus on activities where we can efficiently contribute with our expertise in HPLC technology.

With our research commitments, we intend to generate new knowledge in the field of chromatography as well as learn even more about our own products.

Are you looking for a competent partner in scientific research projects? Do not hesitate to contact us: **academy@knauer.net**

Compliance

Qualification

Note: Standard procedure for IQ and OQ can be handled differently in individual cases for devices.

Installation Qualification (IQ)

The customer may request the IQ document, which is free of charge. In case of a commissioned installation, the Technical Support of KNAUER or a provider authorized by KNAUER, performs this functionality test during the installation.

The IQ is a standardized document including:

- Confirmation of flawless condition at delivery
- Check if the delivery is complete
- Certification on the functionality of the device

Operation Qualification (OQ)

The Operation Qualification includes an extensive functionality test according to KNAUER standard OQ documents. The Operation Qualification is a standardized document. It is not part of the delivery, please contact the Technical Support in case of request.

The OQ includes the following:

- Definition of customer requirements and acceptance terms
- Documentation on device specifications
- Device functionality check at installation site

Test intervals: To make sure that the device operates within the specified range, the device should be tested regularly. The test intervals depend on the use of the device.

Execution: The test can be carried out either by the Technical Support of KNAUER or by a provider authorized by KNAUER (for a fee).



Instrument	IQ Document
All instruments	VIQ-Installation-Qualification

Instrument / Software	OQ Doc.
AZURA® Assistant ASM 2.1L, ASM 2.2L	VOQ-ASM
AZURA® AS 6.1L, AS 3950, PLATINblue AS-1	VOQ-AS
AZURA® CM 2.1S	VOQCM21SA
AZURA® CT 2.1 Column Thermostat	VOQCT21
AZURA® DAD 6.1L, DAD 2.1L, MWD 2.1L	VOQ-DAD
AZURA® RID 2.1L, Smartline S2300	VOQ-RID-2.1L
AZURA® UVD 2.1S, UVD 2.1L	VOQ-Detectors
Flow cells	VOQ-Flowcells
Fraction collectors	VOQ-FRC
Osmometer K-7400	VOQ-K7400
Osmometer K-7400S	VOQ-K7400S
Pumps AZURA®, Smartline, BlueShadow, Platinblue	VOQ-Pumps
PurityChrom® 5 (for Purity Chrom® 6: article no. A2682 includes OQ)	VOQ-PUC
RF20A/RF20Axs	VOQ-RF20
System OQ for analytical systems	VOQ-Sys-01
Valves	VOQ-Valves
Impingement Jets Mixing Systems	VOQ-IJM

Performance Verification (PV)

Definition: The document Performance Verification (PV) is part of the quality management system of KNAUER. The Performance Verification includes a qualification test of an AZURA® LC system and needs to be purchased from the manufacturer. The PV is a standardized KNAUER document and includes:

- Documentation on device specifications
- All necessary method parameters to perform the PV

Goals: The system runs reliably within the documented specifications, and the PV is a summary of the results with comments and evaluations.

Target group: The test can be carried out either by the Technical Support of KNAUER, from a provider authorized by KNAUER or by the customer.

System	Document
AZURA® analytical systems with UV detector used in reversed phase mode	VPV-001-AZURA-UV
AZURA® analytical systems with RI detector used in reversed phase mode	VPV-002-AZURA-RID
AZURA® FPLC systems	VPV-003-AZURA-FPLC
AZURA® analytical systems with FLD detector used in reversed phase mode	VPV-004-AZURA-FLD
AZURA® SMB Lab and Pilot systems	VPV-005-AZURA-SMB
AZURA® preparative systems with UV detector used in reversed phase mode	VPV-007-AZURA-Prep
AZURA® preparative systems with RI detector used in reversed phase mode	VPV-008-AZURA-Prep-RID
AZURA® systems with UV or RI detector used in normal phase mode	VPV-009-AZURA-HPLC-RI-UV-normal-phase
AZURA® systems with ECD detector and flow cell with GC or Au working electrode.	VPV-106-ECD

Material certification

Upon request customized material certification for all wetted parts with varying degrees of complexity from manufacturer statement (only material) to full documentation (e.g. material certification 3.1, FDA compliance statements).

Note: Retrospective material certification is not possible.

FAT / SAT

The factory acceptance test (FAT) refers to the functional test that is performed upon completion of the manufacturing process to prove the equipment has the same specification and functionality that indicated in the data sheet, specification and purchase order. We are experienced in establishing such test procedures with you before your equipment is shipped.

The acceptance of the equipment at your site (site acceptance test, SAT) is also possible: A technician comes to you and ensures that everything works to your utmost satisfaction. In addition, we can integrate the equipment into the existing production environment, if necessary.



Capillary labeling

Complex HPLC systems with a myriad of valves and variable flow paths can be somewhat confusing. We offer professional capillary labeling upon request, to aid end-users in everyday use.

Support

We are committed to provide the best quality support with experienced staff and technical expertise. All standard user instructions, helpful video tutorials, and a structured section of frequently asked questions is freely accessible on our web page **www.knauer.net**.

If you need further support, our friendly Support team is happy to help you via e-mail, phone or Team Viewer. They will work with you personally until all issues are resolved.

Contact

Do you have questions about the installation or the operation of your instrument or software?

Support in Germany
(Austria & Switzerland on case-to-case basis):

Phone: +49 30 809727-111 (workdays 9-17h CET)

E-mail: **support@knauer.net**

International support:
www.knauer.net/local-distributors

Worldwide Technical Services

Our highest goal is to keep your laboratory work as effective and productive as possible. Therefore, we not only pay attention to the highest quality in the development and production of our components and instruments, but also stand by your side after the purchase. With our wide range of services, we are ready to meet any demands to your full satisfaction.

KNAUER offers worldwide quality service of all products, purchased from KNAUER or our authorized partners. All KNAUER Service technicians have completed a specialized service training in the KNAUER headquarter in Berlin, Germany. They are ready to help on site ensuring efficient operation and minimized downtime.

Installation & Instruction

Our experienced KNAUER Service technicians can ensure the proper set-up of your instruments. Get in contact whether you want to use a single device, install a complete system or update your chromatography data system.

KNAUER installations always include introduction in proper handling of the devices as well as tips for self-maintenance and imparting of necessary software knowledge.

On request you may add an IQ, OQ, PV or PQ for compliance (see page 138).

Maintenance

Preventive maintenance has proven to be very successful in ensuring the highest availability of HPLC equipment. Unforeseeable failures of individual system components are thus almost impossible, production processes and laboratory capacities can be planned safely.

We offer maintenance services customized to your needs. You may either ship your instruments to the nearest KNAUER Service facility or contact your local dealer for on-site service of an authorized KNAUER Service technician.

Repair

KNAUER still repairs and maintains the following product lines: the current AZURA®, the former Smartline and PLATINblue devices and - to our best abilities - the Wellchrom equipment which was introduced in the 1990s.

If you discover any malfunction of your device, don't worry, we will repair it for you! Please contact your local dealer for shipment matters or ask for an on-site visit of our skilled KNAUER service technicians.

Development Services

Software development

How does your software limit you?

Many devices rely on some kind of software to run and interact with you, either internal software (firmware) or drivers and application software on your PC.

Development of firmware for HPLC devices like

- UHPLC and HPLC pumps
- UV, PDA, RI, detectors
- Autosamplers
- Valves
- Column ovens
- Fraction collectors



Development of device drivers for

- knauerOS®
- OpenLAB® CDS
- Chromeleon™
- HyStar
- ClarityChrom® (Clarity based)

KNAUER software support for firmware, drivers and software solutions

To provide the most useful tools for your daily work, our team of software engineers combines its expertise in developing firmware, instrument control drivers, as well as application software. KNAUER also has a long experience in customizing instrument operation and in developing drivers for various OEM customers.

Let us know about your software challenges - we will program a solution!

Hardware development

KNAUER has a long experience in customizing scientific equipment according to your needs. With on-site hardware designers, mechanical production and assembly, we can provide tailor made products under certain conditions. Contact us for more information.

Storage of instruments and systems

At times equipment must be removed from your laboratory or you are forced to order equipment before your laboratory is up and running. We can offer storage facilities where your equipment can be stored for future use, giving you peace of mind knowing that you are protecting your investment.

Configuration of your PC

We strongly recommend ordering a KNAUER computer with your HPLC system. However, we understand that sometimes certain constraints do not allow this. We offer a PC configuration service of your PC, in order to assure a safe and reliable installation.

Note: We cannot guarantee installation on a non-KNAUER PC.

KNAUER

Power cable overview

Allocation of power plug types to devices

Every device is supplied with a power plug of the AZURA® series (cold-device plug) in the suitable country-specific version (see table 2).

Exception of allocation (table 1)

Device	Power plug type
■ BlueShadow Pump 40P ■ BlueShadow Detector 40D/50D ■ Smartline Degasser (article no. A5328) ■ Osmometer	Smartline series (see table 2)
■ Router ■ Switch	Power plug is supplied. For outside Europe, a suitable adapter is supplied (see table 2).
■ Degasser (article no. AZE03, AZE03-1, AZE02-1)	Power plug is supplied for US, UK, Europe, Australia.
■ Pressure control (article no. AZG10) ■ Pressure sensor (article no. AZG10-1) ■ Airsensor (article no. A70092, A70093, A70082) ■ Interface Box (article no. AZB00XA)	Power distributor (article no. AZS80SA) and accessories kit with 1 × power plug (article no. F1518) is needed. The distributor can provide power for up to six devices. Only one power distributor per system is required. ■ Power plug for China: article no. M3027D ■ Power plug for Australia: article no. M3027C

Overview of country-specific power plugs, routers and switches

If no suitable adapter is available for a specific country, contact the responsible distributor: www.knauer.net/local-distributors

Overview (table 2)

Power plugs/ routers/switches	Article no. USA	Article no. UK	Article no. China	Article no. Switzer-land	Article no. Europe	Article no. Argentina	Article no. Australia
Power plug AZURA® series (cold-device plug)	M1651	M1278	M3381	M1597	M1642	M3233	M3439
Power plug Smartline series	M1279	M1277	-	M1479-1	M1479	-	-
Router (power plug incl.):	A64809INT	A64809INT	-	-	A64809	-	-
MicroTik	M1651	M1278	M3381	M1597	M1642	M3233	M3439
NetGear, DLink	Adapter: M0447V2	Adapter: M0447V1					
Switch (power plug incl.):							
Switch 8 × LAN	A3119INT Adapter: M0447V2	A3119INT Adapter: M0447V1	-	-	A3119	-	-
Switch 16 × LAN	A3129INT Adapter: M0447V2	A3129INT Adapter: M0447V1	-	-	A3129	-	-

Note: For connecting multiple devices, we provide a special power plug for up to four AZURA® devices (Europe), article no. A12345.

Allocation interfaces to devices

Currently, desktop PCs from KNAUER have one serial interface (RS-232, DE9). If more than one device with a serial interface needs to be connected, you have to install an additional serial interface for the computer. Nearly all laptops have no serial interface; here in general a serial interface has to be installed. For a single device, you can use the USB to serial adapter, article no. A3108 (works also with Shimadzu RF-20A/Axs), for more than one device the USB 4COM, article no. A3114.

- Sedex 85 LT
- Shimadzu RF-20A/Axs
- Bronkhorst Flowmeter
- GJC Flowmeter,
- Chiralyser-MP
- GABI* Gamma Spectrometer
- Osmometer (only with software)

Note: If the tablet for Mobile Control is to be connected via LAN and not via WLAN, the USB-to-LAN adapter (article no. A96181) is required.
You can find the driver on the KNAUER website: www.knauer.net/en/usb-lan-adapter

Detail overview of devices by power plug type

Power Plug AZURA® series (cold-device plug)	AZURA® series (cold-device plug)
All devices of AZURA® series	Autosamplers
PCs and monitors	AZURA® Column Thermostat CT 2.1 (article no. ATC00)
Preparative pumps BlueShadow 80P (article no. APD20xx)	Fraction collectors ■ Foxy® R1/R2 (article no. A59100/A59102/A591021) ■ LABOCOL Vario-4000 (article no. A591022/A591024)
External pressure sensor (article no. AZG10-2)	Liquid Handler ■ Liquid Handler LH 2.1 (article no. A5080) ■ Liquid Handler LH 8.1
Detectors ■ RF20A (article no. A59200) ■ RF20AXS, CBM-20A (article no. A59201) ■ GABI Nova ■ HERM LB500, LB514 Flowstar ■ Sedex85LT, Sedex90LT, Sedex100LT, Sedex LC (article no. A0754-x) ■ CHIRALYSER-MP	Micro devices ■ BlueShadow Pump 10P/20P ■ BlueShadow Detector 10D ■ Degasser 20DG (article no. AZE02)

Power plug Smartline series
Analytical Pumps 40P (article no. APC30xx)
UV Detector 40D/50D
Smartline Degasser (article no. A5328)
Osmometers

This document is subject to technical changes.
Find the latest version of the cable overview
(document no. V1662) here:
www.knauer.net/en/cableoverview

KNAUER BlueShadow Pumps and Detectors

Versatile stand alone instruments for your lab and production systems

KNAUER BlueShadow pumps & detectors are the ideal choice for upgrading your existing LC, reaction system or process instruments.

BlueShadow Pumps 40P and 80P



BlueShadow Pump 40P



BlueShadow Pump 80P

Pumps from the BlueShadow line can be integrated into every existing LC system, but they can also be used for high-pressure dosing applications. KNAUER dosing pumps are highly accurate two-piston pumps for applications in the chemical and pharmaceutical industries as well as in research and method development.

They pump and dose aqueous and organic liquids, aggressive media or liquid gases. The metering pumps impress with their high chemical resistance, excellent flow rate precision and low pulsation of the pumped medium in a wide range of applications.

BlueShadow Detectors 40D and 50D



BlueShadow Detector 40D



BlueShadow Detector 50D

Detectors from the BlueShadow line are spectrophotometers that can be used for LC applications, reaction monitoring, and other applications. They are offering excellent technical specifications in a highly flexible and compact design.

The flow cells are easily accessible, can be changed quickly and cover flow rates from 10 µl/min up to 10 l/min. With the unique fiber optics design of the BlueShadow 40D, the flow cell can also be separated from the detector and directly placed in the stream of the product flow.

Further information on BlueShadow devices:
www.knauer.net/dosing-pumps

KNAUER GMP Services

KNAUER Services for Good Manufacturing Practice for biopharmaceutical industry

KNAUER provides equipment for downstream processing in the biopharmaceutical industry such as skids for the formulation of lipid nanoparticles, or chromatographic systems for mRNA purification or continuous chromatography. KNAUER provides a wide range of services to support our customers and to ensure that GMP requirements are met.

KNAUER's GMP services are based on our hardware- and software-solutions; encompassing product safety, quality control and the training of personnel. Risk management, in relation to GMP, is covered by the user:



Product safety:

Documentation on the compliance of materials used for wetted parts is an important requirement for product safety. In the bio-pharmaceutical industry, potentially harmful substances have to be avoided in liquids for clinical, cosmetic or food applications. Therefore, any materials of the liquid flow path that come into contact with the final product have to meet certain criteria. According to our end user's requirements KNAUER can provide compliance with the order (EN 10204-2.1), certificates of compliance on the materials used for wetted parts, and further documentation from the supplier such as 2.1 certificates.

Overview of KNAUER options:

Certificates are available for KNAUER products and selected third party products. Contact sales@knauer.net.

Type of certificate/statement	Unit of quantity	Article number
Declaration of Compliance with order (EN 10204-2.1)	for 1 order	A0000TDCOO
Declaration of Compliance (EN 10204-2.1) with Certificate and compliance of material of wetted parts	for 1 article with less than 5 components	A0000COMS
Declaration of Compliance (EN 10204-2.1) with Certificate and compliance of material of wetted parts	for 1 article with 5 or more components	A0000COM
Declaration of Compliance (EN 10204-2.1) with Certificate and compliance of material of wetted parts for one pump	for 1 article	A0000COMP
Declaration of Compliance (EN 10204-2.1) with Certificate and compliance of material of wetted parts for one pump head	for 1 article	A0000COMPK
Declaration of Compliance (EN 10204-2.1) with Certificate and compliance of material of wetted parts for one valve	for 1 article	A0000COMV
Declaration of Compliance (EN 10204-2.1) with Certificate and compliance of material of wetted parts for one detector	for 1 article	A0000COMD
TSE/BSE Statement: Customized order- and article related	for 1 article	A0000TDTSE
Stepfile per device without functional groups	for 1 article	A0000IDSTE
Documentation on compliance of material of wetted parts: Compliance with the order (EN 10204-2.1); Certificate of compliance on material of wetted parts; Documentation on references (supplier information of material) and wetted parts (certificates such as 2.1)	customized	A0000TD

Conversion tables

Dimensions

mm	inches
0.10	.004″
0.12	.005″
0.15	.006″
0.25	.010″
0.40	.016″
0.50	.020″
0.75	.030″
1.00	.040″
1.50	.060″
2.00	.080″
4.60	.180″
6.00	.236″
6.40	.253″
7.00	.276″
10.00	.400″

inches	mm
1/32″	0.8
1/16″	1.6
1/8″	3.2
1/4″	6.4
3/8″	9.5
1/2″	12.7
1″	25.4

Tubing volume/length

Tubing ID	µl/cm	µl/in
.004″	0.08	0.21
.005″	0.13	0.32
.010″	0.51	1.29
.015″	1.14	2.90
.020″	2.03	5.15
.025″	3.17	8.04
.030″	4.56	11.58
.040″	8.11	20.59
.060″	18.24	46.33
.070″	24.83	63.06
.085″	36.61	92.99

Pressure

MPa	bar	psi
5	50	725
10	100	1450
20	200	2901
30	300	4351
40	400	5802
50	500	7252
60	600	8702
70	700	10153
80	800	11603
90	900	13054
100	1000	14504
110	1100	15954
120	1200	17405
130	1300	18855
140	1400	20306
150	1500	21756
160	1600	23206
170	1700	24657
180	1800	26107
190	1900	27558
200	2000	29008

Temperature

°C	°F	°C	°F	°C	°F
-40	-40	65	149	170	338
-35	-31	70	158	175	347
-30	-22	75	167	180	356
-25	-13	80	176	185	365
-20	-4	85	185	190	374
-15	5	90	194	195	383
-10	14	95	203	200	392
-5	23	100	212	205	401
0	32	105	221	210	410
5	41	110	230	215	419
10	50	115	239	220	428
15	59	120	248	225	437
20	68	125	257	230	446
25	77	130	266	235	455
30	86	135	275	240	464
35	95	140	284	245	473
40	104	145	293	250	482
45	113	150	302	255	491
50	122	155	311	260	500
55	131	160	320	265	509
60	140	165	329	270	518

KNAUER

Terms & Conditions

1. Definition of terms

The following terms and conditions apply to every order received by KNAUER and every delivery of goods. This holds as well in case of contradictory buying conditions of the purchaser. Exceptions are only valid when confirmed by KNAUER in writing. Purchase orders are only binding if confirmed by KNAUER in writing.

2. Payment

Deliveries are due and payable, net, within 30 days of invoice date or in advance. Deductions are not allowed. Foreign deliveries must be paid by irrevocable letter of credit or in advance. All bank and transfer fees must be paid by the customer. The consequences arising out of delay are due to statutory provisions. Payments are due irrespective of an eventual notice of defect, except such defects are evidently justified.

3. Delivery

Delivery dates are not binding unless expressly stated in the contract as binding dates. Delay in delivery requires a written reminder and an adequate additional grace period set by the customer. KNAUER is only liable for claims for damages under the requirements of no. 6.

4. Claims

Condition for any warranty claim is the immediate inspection of the goods upon delivery, and complaint towards and damage assessment together with the carrier, and an immediate written complaint to KNAUER. The complaint must be made within five workdays in case of visible defects or losses.

5. Risk liability

Delivery is made at the customer's own risk. As soon as the goods leave KNAUER's plant the risk of accidental loss, destruction or deterioration passes to the customer.

6. Warranty and damages

6.1. Warranty claims

The warranty begins with receipt of the goods. If commissioning has been ordered, after commissioning. In the case of delayed commissioning, the warranty begins at the latest four weeks after receipt of the goods unless the supplier is responsible for delayed commissioning.

The warranty for osmometers and liquid chromatography instruments is limited to two years, excluding glass breakage, damages due to stoppage and consumable materials such as membranes, light bulbs, columns, bushings, gaskets and valves. KNAUER's liability shall be restricted to the replacement of defective material or repair only. Transportation costs are borne by the customer. In case of failure of replacement or repair the customer may demand a reduction in price or cancellation of the contract with respect to the defective material. The customer has to inspect the goods delivered immediately and shall immediately give written notification of any defects to KNAUER, in case of non-obvious defects within 10 working days after delivery at the very latest.

6.2. Claims for damages

The liability of KNAUER shall be restricted to intentional acts and acts of gross negligence and compensation shall only be due for direct, foreseeable damages. Liability for breach of a material, essential duty of the contract, liability because of personal injury, liability according to the stipulations of the German Law on Product Liability and liability for the lack of the condition of the contract goods guaranteed by KNAUER remain unaffected.

7. Third party rights on industrial or other intellectual property

KNAUER shall not be liable for the infringement of third party rights founded on industrial or other intellectual property caused by the use of the delivered goods. The customer is fully responsible for the products manufactured with the goods. In particular KNAUER is not obliged to indemnify and hold harmless the customer from all claims raised by third parties based on the infringement of their industrial or intellectual property rights by the use of the goods.

8. Property rights

The ownership of the goods shall remain with KNAUER until payment in full for all our claims resulting from our business relation is received. In case of improper treatment of the goods or in case of default KNAUER may demand the return of the delivered goods. This demand entails resignation of the contract only if KNAUER declares it explicitly.

Resellers are allowed to sell the goods to third parties in due course of the business. The customer herewith assigns his resale claims against third parties to KNAUER.

9. Export

Instruments and products delivered by KNAUER may not be exported to a country other than of the customer's headquarters without KNAUER's prior written permission.

10. Place of settlement and court of jurisdiction

The place of performance is Berlin. Proper venue for all claims is the competent local court at KNAUER's principal place of business - Berlin. KNAUER reserves the right to sue the customer at his principal place of business.

This agreement shall be governed by the laws of the Federal Republic of Germany excluding the UN-Convention on the International Sale of Goods (CISG).

KNAUER Wissenschaftliche Geräte GmbH
Hegauer Weg 37 - 38
14163 Berlin, Germany

These terms and conditions apply since June 1, 2016

Notes

KNAUER Brochures



AZURA® Analytical HPLC/UHPLC
(Document no. V7852US)



AZURA® SMB System Solutions
(Document no. V7741US)



AZURA® SEC System Solutions
(Document no. V7721US)



AZURA® Preparative HPLC
(Document no. V7820US)



AZURA® Bio purification
(Document no. V7855US)



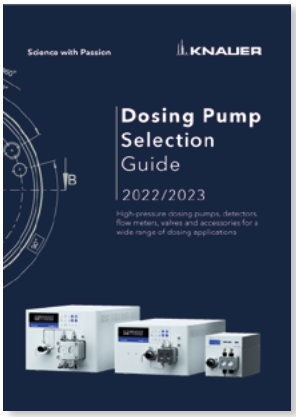
Freezing point osmometry
(Document no. V7716US)



KNAUER LNP Flyer
(Document no. V7720US)

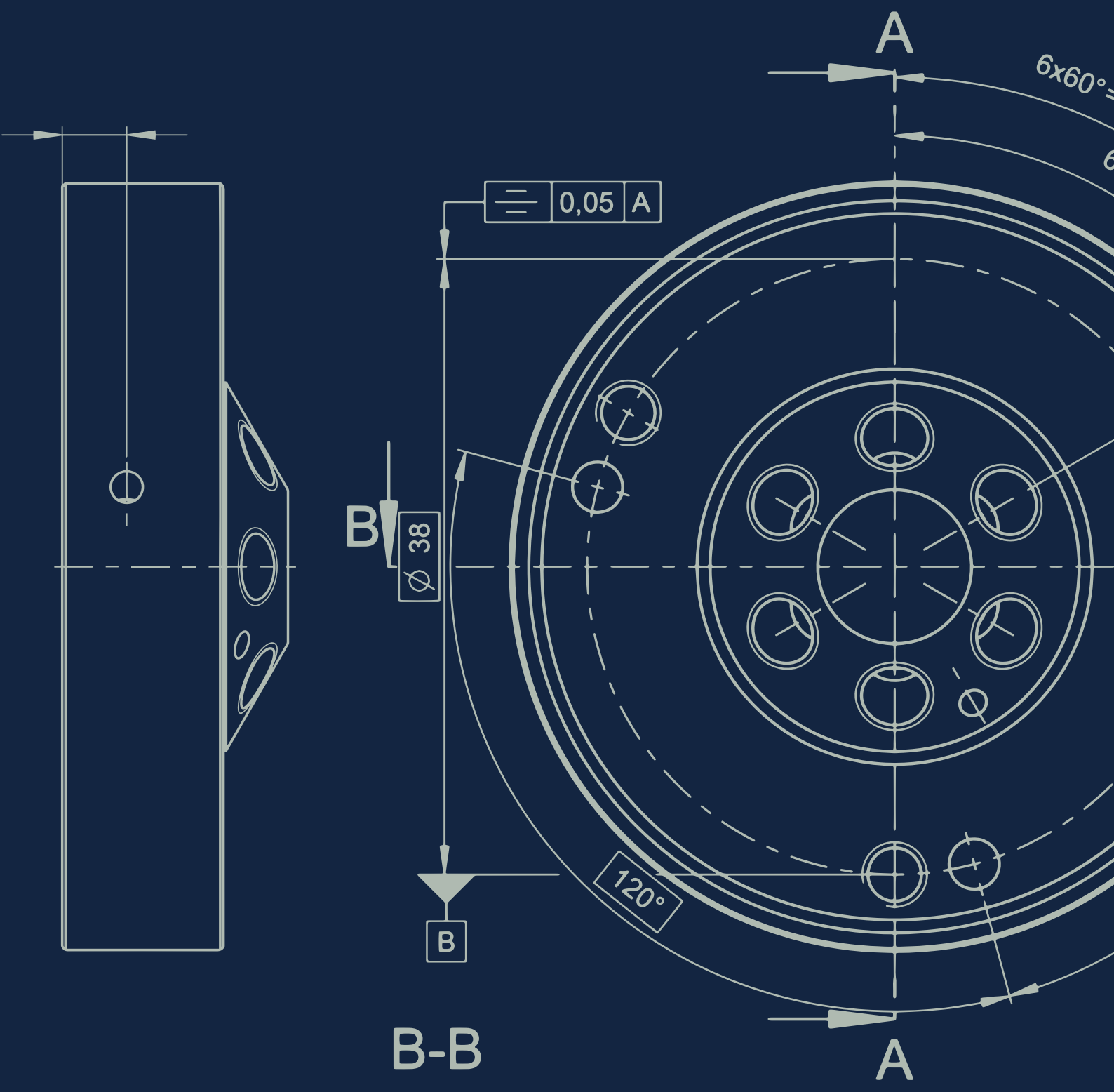


KNAUER OEM Brochure
(Document no. V7712US)



KNAUER
Dosing Pump Selection Guide
(Document no. V7866US)

All KNAUER brochures:
www.knauer.net/brochures



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